



**GREEN
CLIMATE
FUND**

Meeting of the Board
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

GCF/B.37/02/Add.15

2 October 2023

Consideration of funding proposals – Addendum XV

Funding proposal package for FP224

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled "Renewstable Barbados Project" submitted by International Finance Corporation;
- b) No-objection letter(s) issued by the national designated authority(ies) or focal point(s);
- c) Environmental and Social report(s) disclosure;
- d) Independent Technical Advisory Panel's assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled "Renewstable Barbados Project" submitted by International Finance Corporation is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged, or commercially sensitive information of the entity.

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Funding Proposal

Project/Programme title: *Renewstable Barbados Project*
Country(ies): *Barbados*
Accredited Entity: *International Finance Corporation*
Date of first submission: *2023/06/18*
Date of current submission: *2023/09/27*
Version number: *V.010*



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Project	A.2. Public or private sector	Private
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)			
		GCF contribution	Co-financers' contribution
	Mitigation total	Enter number %	Enter number %
	☒ Energy generation and access	100 %	100 %
	☐ Low-emission transport	Enter number %	Enter number %
	☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
	☐ Forestry and land use	Enter number %	Enter number %
	Adaptation total	Enter number %	Enter number %
	☐ Most vulnerable people and communities	Enter number %	Enter number %
	☐ Health and well-being, and food and water security	Enter number %	Enter number %
	☐ Infrastructure and built environment	Enter number %	Enter number %
☐ Ecosystems and ecosystem services	Enter number %	Enter number %	
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	693,000 tCO ₂ e (over a 25-year lifespan)	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	N/A N/A
A.7. Total financing (GCF + co-finance)	169 million USD	A.9. Project size	Medium (Upto USD 250 million)
A.8. Total GCF funding requested	41 million USD		

A.10. Financial instrument(s) requested for the GCF funding	☒ Grant <u>1000000</u> USD ☐ Equity <u>Enter number</u> ☒ Loan 40000000 USD ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>		
A.11. Implementation period	24 months	A.12. Total lifespan	25 years
A.13. Expected date of AE internal approval	11/16/2023	A.14. ESS category	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	Yes ☐ No ☒		
A.20. Executing Entity information	RENEWSTABLE (BARBADOS) INC., a company incorporated under the Companies Act Cap. 308 of the Laws of Barbados (“RSB”). Registration Number: 43324 INTERNATIONAL FINANCE CORPORATION, an international organization established by Articles of Agreement among its member countries including Barbados (“IFC”). Executing Entity (EE) in connection with some Technical Assistance activities. As part of the due diligence carried out by IFC in connection with the pre-investment engagement launched in August 2022, IFC has met with the Executing Entity at various levels and assessed their competence to carry out the project. IFC has also carried out Integrity Due Diligence/Know Your Customer assessment in connection with reputational aspects and potential accusations of corruption, fraud, etc. Based on IFC’s initial due diligence, there were not red flags raised that would prevent IFC from doing business with the EE on reputational or competence grounds.		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

The Renewable Barbados project will be the **first large-scale hydrogen storage solution in the country**. The solar PV power generation with batteries and hydrogen storage will provide clean firm capacity at night, morning and evening peak-hours **providing 6 percent of the country's electricity demand**, avoiding 693,000 tCO₂e GHG emissions over the life of the project and creating 120 jobs at construction and ten permanent jobs.

The government of Barbados (GoB) has set ambitious decarbonization targets for 2030, including a target of achieving a 100 percent renewable energy matrix. The Barbados' Nationally Determined Contributions (NDC) targets GHG emissions reduction of 44 percent compared to its business-as-usual scenario by 2030. At the same time, electricity demand is projected to increase to 1,277 GWh in 2030 under a base scenario, implying a 35 percent increase from its 2019 levels. Given the current power generation matrix that is around 95 percent based on fossil fuel generation¹, steam generation plants (totaling 40MW) are to be decommissioned as soon as possible and gas turbines (totaling around 100MW) that are not suitable for continued operation are expected to be used only as backup and emergency generation. Barbados has abundant solar resources with high irradiation and capacity factors thanks to its location close to the equator. A 10 MWp solar power plant is already operating, but developers of additional solar plants are facing difficulties in obtaining necessary permits to connect to the grid. The permitting process is slowed by concerns about grid stability, which emerged after the construction of the existing solar plant.

Barbados is subject to hurricanes and other regional climate events. These climate events can cause disruptions to the grid and generation facilities and, consequently, loss of power to various areas in the island, particularly those more distant from the existing generation facilities. Another level of resilience challenge relates to the access to and pricing of fossil fuels that currently are needed for most power generation in Barbados. Geopolitical developments such as the recent Russian invasion of Ukraine highlight the vulnerability that this dependency on imported fuels can pose to countries around the world and particularly to small island economies such as Barbados.

The proposed **RenewStable Barbados (RSB)** project will use intermittent solar PV generation and combine it with short-term battery storage and long-term green hydrogen storage to provide a clean generation facility that can be dispatched 24 hours a day. The solar plant is oversized (50 MWp), so that it can make available sufficient capacity (13 MW) to dispatch electricity to the grid during the day, while also generating additional electricity to fully charge a Battery Electricity Storage System (BESS). Once charged, the BESS can cover 15 MW of demand capacity during peak hours and produce sufficient green hydrogen through water electrolysis to then convert back to electricity at a lower capacity of 3 MW throughout the night. This design responds to the requirements of the vertically integrated power company Barbados Light and Power Company (BLPC) and is based on the demand profile for the grid and the dispatch profile of thermal power plants that are being decommissioned. The RSB project will enable the decommissioning of a gas and diesel turbine (GT02) that was originally scheduled for 2019, in which case arguably the GHG emission mitigation could be linked to the emissions of this plant in the order of 55,555 tCO₂/year. However, a more conservative approach has been used taking into consideration the average emission factor of the grid that results in mitigation of at least 27,719 tCO₂e per year, for a total of 693,000 tCO₂e over the 25-year lifetime of the RSB project.

¹ IRENA (https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Central%20America%20and%20the%20Caribbean/Barbados_Central%20America%20and%20the%20Caribbean_RE_SP.pdf)

The RSB project is designed to withstand extreme weather and high wind loads during climate events. The RSB project will be located in the southeastern area of the island, rather than in the west where most of the generation is currently located. This location ensures access to power for people on this part of the island in the event that transmission lines are affected by a severe climatic event. The RSB project is designed to be able to operate in an “island mode” if disconnected from the rest of the generation plants, allowing residents in the vicinity of the project to not only have continued access to electricity, but also to other services that rely on electric power, such as water (pumping stations are close to the RSB site), health services, telecommunications, and other vital services. The modularity of the project’s design means that each of its components (solar generation, BESS and hydrogen storage) can operate independently improving the resilience and flexibility of operation. For instance, if the solar plant is temporarily out of service, the storage capacity of the BESS and storage system can dispatch at 3 MW for up to 40 hours if fully charged.

Finally, the project generates electricity solely from solar panels. This means that the **RSB plant relies on local resources** and not on imported fuels thus making the system more resilient to disruptions in the sourcing and logistics associated with importing fossil fuels. The RSB project will also help reduce the fuel cost volatility which is today fully passed through to the end-users’ monthly bills and fix the production cost of this baseload asset for the country over the next 25 years.

The RSB project provides an innovative solution to address the intermittency challenge of non-conventional renewable energy sources such as solar and wind. This innovative solution is based on technologies that are well known including batteries and electrolyzers/fuel cells. Although the price trend is downward as industrialized production is streamlined and competition increases, the price of these technologies today is still relatively high and only makes sense in specific contexts such as island countries where current HFO/diesel-based generation is very expensive. Retail tariffs in Barbados are very high, at a weighted average rate of approximately USD 0.27 per kilowatt-hour (kWh) well above the LAC average of around USD 0.15 kWh². **The GCF concessional financing, in addition to contributing to make possible this innovative decarbonization solution, will be critical in keeping the average end-user tariff below the current levels and bridge the gap between the Barbados tariffs and LAC average as much as possible.** With a fixed equity return and fixed commercial terms from the IFC and other lenders involved in the transaction, the concessionality of GCF will be fully passed through in the Power Purchase Agreement (PPA) tariff formula with BLPC. GCF concessional financing will also be key in enabling a project that can demonstrate an innovative solution to leverage intermittent renewable energy resources, which will provide increase energy security to Barbados, diversification of its power generation sources, and a potential for replicability in Barbados and other small islands and isolated grids.

NOTE: GIVEN THE TIMING OF THE EXPECTED FINANCIAL CLOSE BY THE END OF 2023, IT IS IMPERATIVE THAT THIS PROJECT BE PRESENTED AT GCF’s OCTOBER 2023 BOARD MEETING IN ORDER FOR THE CONCESSIONALITY COMPONENT TO BE INCLUDED IN THE TARIFF NEGOTIATIONS WITH BARBADOS FAIR-TRADING COMMISSION.

² “Latin America and the Caribbean Energy Price Report”, World Bank/OLADE, 2021.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate Change Problem

Barbados is among the most vulnerable Caribbean nations to climate change.³ The Barbados electricity grid is centralized and vulnerable to hurricanes. Since 2010, hurricanes Tomas (2010), Ernesto (2012), Harvey (2017), and Elsa (2021) and the tropical storms Matthew, Maria, Kirk and Gonzalo (2020) have impacted the island, causing power outages and enormous damage to the people in Barbados. Unlike its neighboring island countries, Barbados does not benefit from a geothermal potential. Power plants burning local biomass would typically be very land-intensive for this SIDS country. Adding to these challenges with the current energy landscape of Barbados is its high dependence on imported fossil fuels, which accounts for 95 percent of the total energy supply.⁴ This arrangement leaves Barbados extremely vulnerable to global oil price fluctuations that directly impact the cost of electricity. The country spends about 7 percent of its GDP on importing expensive fossil fuels. Retail tariffs are high at a weighted average rate of approximately USD 0.27 per kilowatt-hour (kWh), almost double the LAC average of around USD 0.15 per kWh, and outages are common (Energy Transition Initiative, 2020). The cost of fuel purchased is recovered by the utility and directly applied to all customers through the Fuel Adjustment Clause which has represented USD 0.21 kWh on average in 2022.⁵

Climate Targets and Strategies

The government of Barbados (GoB) developed a National Energy Policy (NEP) for 2019-2030, targeting 100 percent renewable energy and carbon-neutral island-state transformation by 2030. In 2019, about one-third of the country's GHG emissions came from its energy sector, according to Climate Watch. Barbados' NDC targets GHG emissions reduction of 44 percent compared to its business-as-usual scenario by 2030. In absolute terms, this represents a decrease of 23 percent compared with the baseline year, 2008. The Integrated Resource and Resiliency Plan for Barbados (IRRP) developed by Mott MacDonald projects electricity demand to increase to 1,277 GWh in 2030 under a base scenario, implying a 3 percent increase from its 2019 levels. At the same time, the IRRP recommends that steam generation plants (totaling 40MW) be decommissioned as soon as possible and that gas turbines (totaling around 100MW) are not suitable for continued operation and expected to be used only as backup and emergency generation. The NEP's ambitious targets are then likely to be met by tapping into the country's proven renewable energy potential, including wind, solar PV, waste, and biomass (primarily sugarcane bagasse). To this purpose, the GoB has already supported investment vehicles such as the **Energy Smart Fund**⁶ that provides financial and technical support to Renewable Energy (RE) and Energy Efficiency (EE) projects in Barbados through subsidized loans, technical assistance grants, and replacement rebates for air conditioning units and the **Public Sector Smart Energy Program**⁷ consisting of (a) retrofitting of public lighting and public buildings with EE technology, and installation of solar PV panels in government buildings, (b) a fleet of government electric vehicles and a prospectus/study for an ocean power plant, and (c) capacity building for the government of Barbados on sustainable energy. These initiatives aim at developing public and private projects focused on climate resilience, energy efficiency, and renewable energy generation, showcasing the increased political interest and will to support renewable energy projects.

³ Stennett-Brown (2019) <https://doi.org/10.1371/journal.pone.0219250.g008>

⁴ Barbados, Energy Profile, IRENA Report 2019.

⁵ <https://www.blpc.com.bb/index.php/customer-care/fuel-clause-adjustment>

⁶ <https://smartenergybarbados.com/energy-division/energy-smart-fund/>

⁷ <https://energy.gov.bb/tag/public-sector-smart-energy-programme/>

Mitigation Benefits

The RSB project is expected to have a significant impact on GHG emissions mitigation with high efficiency and effectiveness. Based on Climate Watch⁸, in 2020, Barbados was estimated to have emitted 3.79 million tonnes of CO₂e or 0.013 percent of global emissions. This represents around 13.21 tonnes CO₂e per capita annually. GHG emissions⁹ from energy generation at 1.07 mTCO₂e are the second sector after waste and followed by industrial processes. All in all, the plant will generate 56GWh/year of electricity, save 27,719 tCO₂/year, contribute with approximately 6 percent of the Barbados grid capacity from renewable sources, and benefit more than 21 thousand persons with up to 25 years lifetime. A diesel plant with similar dispatch profile as the RSB plant is past its useful life and will be decommissioned once the RSB plant comes in line (13MW gas turbine "GT02" running on diesel has been scheduled to be decommissioned since 2019). Under the scenario of displacing a diesel plant, the GHG emission avoidance is up to 55,555 tCO₂/year.

Table 1 – GHG Emissions Calculation

Parameter	Unit	Value	Reference
Renewable generation (daily)	MWh/day	182	8am-5pm 13MW, 7-9pm 13MW; all other times 3MW
Renewable generation (yearly)	GWh/ year	56	
GHG emission factor	gCO ₂ /kWh	484	IFI Grid Emission Factor Database for firm energy provision
GHG emission reductions	tCO ₂ /year	27,719	
Lifespan	years	25	
GHG reductions over a lifespan	tCO ₂	692,981	

Adaptation Co-benefits

The RSB project will result in higher resilience of the power system in Barbados. By shifting from imported heavy fuel oil and diesel to local, clean solar PV resources, the project is improving the energy security and resilience of the electricity system, as well as reducing the price volatility and increasing reliability of power generation in a more decentralized arrangement. The project will probably not reduce regular grid outages, but rather provide a separate source of base-load power generation that does not depend on logistic infrastructure for fuel.

⁸ https://www.climatewatchdata.org/countries/BRB?end_year=2020&start_year=1990

⁹ Note that GHG emissions are measured in carbon dioxide equivalents (CO₂eq). This means non-CO₂ gases are weighted by the amount of warming they cause over a 100-year timescale.

Given the vulnerability of Barbados to extreme precipitation events, such as tropical storms and hurricanes, this project will increase the resilience of the population by reducing dependence on fuel imports, whose delivery depends on the fair weather. In addition, the project increases the resilience of all the population of Barbados, and particularly the residents of its southeast region where the project will be located. During an extreme precipitation event, it is likely that batteries and hydrogen tanks will be charged/filled to capacity and rationed after the event. As long as the batteries or fuel cells are connected to the grid, electricity can be dispatched very quickly; this is particularly important because some thermal plants require a start-up period. The 90MWh hydrogen storage capacity plus the 30 MWh battery storage capacity would allow for a dispatch at a non-peak rate of 3MW capacity of 40 hours, even if the solar plant is temporarily out of service. (This level of damage is unlikely because of the solar plant's climate-resilient design). Furthermore, the project's location in southeastern Barbados is strategically located to increase resilience since it provides grid independence to this area, which currently relies on national grid and generation plants located the west coast. In the event of a hurricane, where transmission lines from the national grid are damaged and power to this area is cut off, the project can operate in "island" mode (especially because the cable layout within the plant and the substation will be underground so that they are more resilient), guaranteeing energy to the residents of this region. Ensuring energy to this region will also help in maintaining its water supply, given that water pumps near the site can use the electricity from the project to operate. Also, if other sources of electricity in the country are damaged because of an extreme precipitation event (e.g. a thermal facility gets flooded or port terminal damaged), and the transmission lines are not damaged, then electricity can also be delivered to the areas of the country being serviced by the damaged facility.

The project itself will be Paris Aligned for power resilience, in line with the Climate Risk assessment that was conducted for the project.

Barriers and risks to climate change

- **Barbados electricity grid is susceptible to the impacts of climate change and highly dependent on imported fossil fuels:**

Barbados' macroeconomic vulnerabilities are aggravated by dependency on fossil fuel imports to meet its energy needs. Almost all energy in the island country is imported, leaving a bill of around 7 percent of GDP and exposing Barbados to adverse effects of oil price volatility. Half of fossil fuel imports go directly to the industry and transportation sectors, and the rest is centrally transformed into electricity, resulting in a polluting energy matrix dominated by oil products (92 percent), with non-fossil fuel product participation declining since 2000 to 3 percent. Unsolicited applications for licenses to generate electricity are allowed under the regulation for up to 20 MW capacity, while two consecutive feed-in-tariff (FiT) programs have allowed and encouraged projects with a size up to 10 MW. Recent additions of solar generation have already created a situation of saturation of certain feeders, exceeding the stabilization capacity of existing spinning reserves. BLPC is reluctant to connect more intermittent generation facilities such as solar or wind, without storage. This situation makes it difficult for Barbados to achieve its goal of carbon-neutrality by 2030 and highlights the need for non-intermittent, renewable solutions such as the proposed project. The plant is expected to provide 13MW during the day and 3MW during the night. With this capacity, the plant will supply 6 percent of the country's current electricity demand and will be designed to sustain hurricanes. The plant does not require feedstock and can run on island mode if needed. A large-scale hybrid solution with mature technologies and long-term energy storage is the quickest, most competitive, and bankable way to deploy locally produced clean firm power.

- **High cost associated with battery storage and green hydrogen equipment:**

Green hydrogen is a technology that, although already mature, is quickly becoming cheaper, but early projects need some concessionality while the price of electrolyzers and the cost of renewable energy continues to drop to levels that would make it fully commercial. The level of concessionality needed is intended to address relatively higher country and technology risk perception resulting in a higher cost of financing at this early stage. Over time, similar projects are expected to be financed on a purely commercial basis, but GCF funding can help demonstrate technical feasibility in the country context, accelerate implementation and provide a successful example to be replicated in Barbados and other islands. Equity contributions for the HDF project in Barbados will be provided by private sponsors, and potentially local credit unions. Part of the debt is expected to be financed by development banks and local banks on a commercial basis, further leveraging the concessional funds from GCF.

- **Current electricity tariff makes the renewable energy solution less viable:**

The proposed solution is competitive when compared with diesel generation, which is the most obvious alternative, but dependent on a fuel that has high and volatile prices, is imported (resulting in lower energy security) and is polluting. The RSB project would provide a natural hedge against a possible rise in international fuel prices, reduce fuel imports and international reserve depletion and improve the country trade balance. Ninety percent of the oil and gas required to operate the plants is imported due to the lack of refining infrastructure. Oil and gas imports impact the competitiveness of Barbados' economy and costs on average 7 percent of GDP every year. The project will help fix the production cost of this baseload asset for the country and reduce the fuel costs volatility currently passed to the end-users monthly bills (100 percent of the fuel cost via the Fuel Clause Adjustment). Initial studies indicate that a solar plus batteries only solution would result in slightly lower pricing than the proposed solution given that batteries only last around four hours of discharge, so a few banks of batteries at the nominal capacity would be needed. Other solutions such as biomass and wind are being explored but are at an earlier stage of development. The proposed project could be viable in the absence of concessional support but at a higher tariff for the electricity sold to the grid by the project. However, this would ultimately impact the end consumers that would have to pay a higher electricity bill. End consumers in Barbados already pay much higher electricity tariff (\$0.27/kWh) than the LAC region (\$0.15/kWh). The GCF funding is therefore not so much a liquidity solution as a critical concessional source of funding which would help keep tariffs at the same or lower level for the people of Barbados (i.e. maximizing the concessionality in an otherwise set financing plan will directly benefit the tariff). The regulatory framework in Barbados allows for a feed-in-tariff¹⁰ for renewable projects, but only for plants with an installed capacity below 10 MW and aimed mainly to intermittent generation (solar and wind). Thus, it would not apply to the proposed project,. The tariff in the PPA for this project is being negotiated with the Barbados Fair Trading Commission (FTC) and involves a mechanism for the project to recover the investment and return based on the up-front capital investment and other factors through a capacity (availability) payment. The concessional loan of GCF allows for the PPA tariff that BLPC would pay the project to be lower. The impact of this lower tariff is then fully transferred to the end user and shared across all customers in Barbados through the BLPC grid. Since Barbados is not ODA-eligible, concessional funding for the proposed project is difficult to come by, making GCF support particularly relevant. The concessional funding is needed at this stage given the relatively high cost of electrolyzers and fuel cells. However, the per-unit price of electrolyzers and fuels cells is expected to fall as the scale of their production increases , the same that happened in the solar PV and wind industries decades ago. Therefore, replication of this model in Barbados or elsewhere is expected to be viable with fewer or no subsidies in the future.

¹⁰ FiT for a 10 MW solar PV plant up to March 2021 would be BD\$21.75 cents/KWh.
(<https://support.blpc.com.bb/support/solutions/articles/42000060831-applying-for-interconnection-to-the-grid>)

Key characteristics and dynamics of the sector or market

The political feasibility of renewable energy in Barbados is unique because the country does not have large petroleum reserves that would provide competing energy from fossil fuels with embedded subsidies. This is a problem that is characteristic of countries in other regions such as North America and the Middle East. The tropical climate of Barbados makes it an ideal location for wind and solar energy. Barbados averages 8.3 hours of sunshine per day and 5.6 kilowatts of solar irradiation per square meter. However, the intermittent nature of these renewable sources of energy can create instability in the power grid. Natural disasters in the past and the COVID-19 pandemic have highlighted the need for a more diverse economy in Barbados. The current generation matrix consists mostly of centralized fossil fuel generation with 54.5 MW LSD Engines, 73.5 MW gas turbines, and 20 MW steam turbine, located in three sites near the coast in the south and southwest of the island. A 10 MW solar power plant in the north is the first utility-sized photo-voltaic solar farm in Barbados. All of this generation is currently owned by Barbados Light and Power Co. (BLPC), a private utility that also is the concessionaire for transmission and distribution in the island. There is also an increasing amount of distributed solar generation that adds up to 25MW and, under a recent change to the regulatory framework (2015), the potential for Independent Power Producers (IPP) to sell to the grid.

BLPC is a regulated, vertically-integrated electric utility on the island of Barbados. It operates the monopoly on generation, transmission, and electricity distribution through a concession that expires in 2028. BLPC is wholly owned and controlled by Emera Group (BBB/Baa3/BBB), an international energy and services company with revenues over USD 5.5 billion that owns utilities and infrastructure assets in Canada, the USA, and the Caribbean. The electricity tariffs in Barbados are regulated by the country's Fair Trading Commission ("FTC") and allow BLPC to recover all prudently incurred fuel costs quickly through a pass through. BLPC plans to continue operating in Barbados for the long run. The company plans to compete with other IPPs in nascent RE market. On this point, the terms of the PPA that is being negotiated are for Capacity Payments, which would mitigate the risk of a preferential dispatch of BLPC power plants over the RSB project.

IFC's current activities in the resilience space include working closely with the Minister of Economic Affairs and Investment of Barbados. As part of the *Roofs to Reefs* program, an integrated public-private investment roadmap for improving the social and environmental circumstances of Barbadian households, IFC is helping the GoB using data to identify and prioritize the pipeline of investment opportunities, focusing on the interventions across all infrastructure sub-sectors that hold the highest potential for supporting the country's [resilience plan](#). A component of this work involves outlining which projects currently underway are at risk of not progressing due to legal and regulatory environmental barriers or are being held up by procedural inefficiencies and which specific changes would be helpful to advance their implementation.

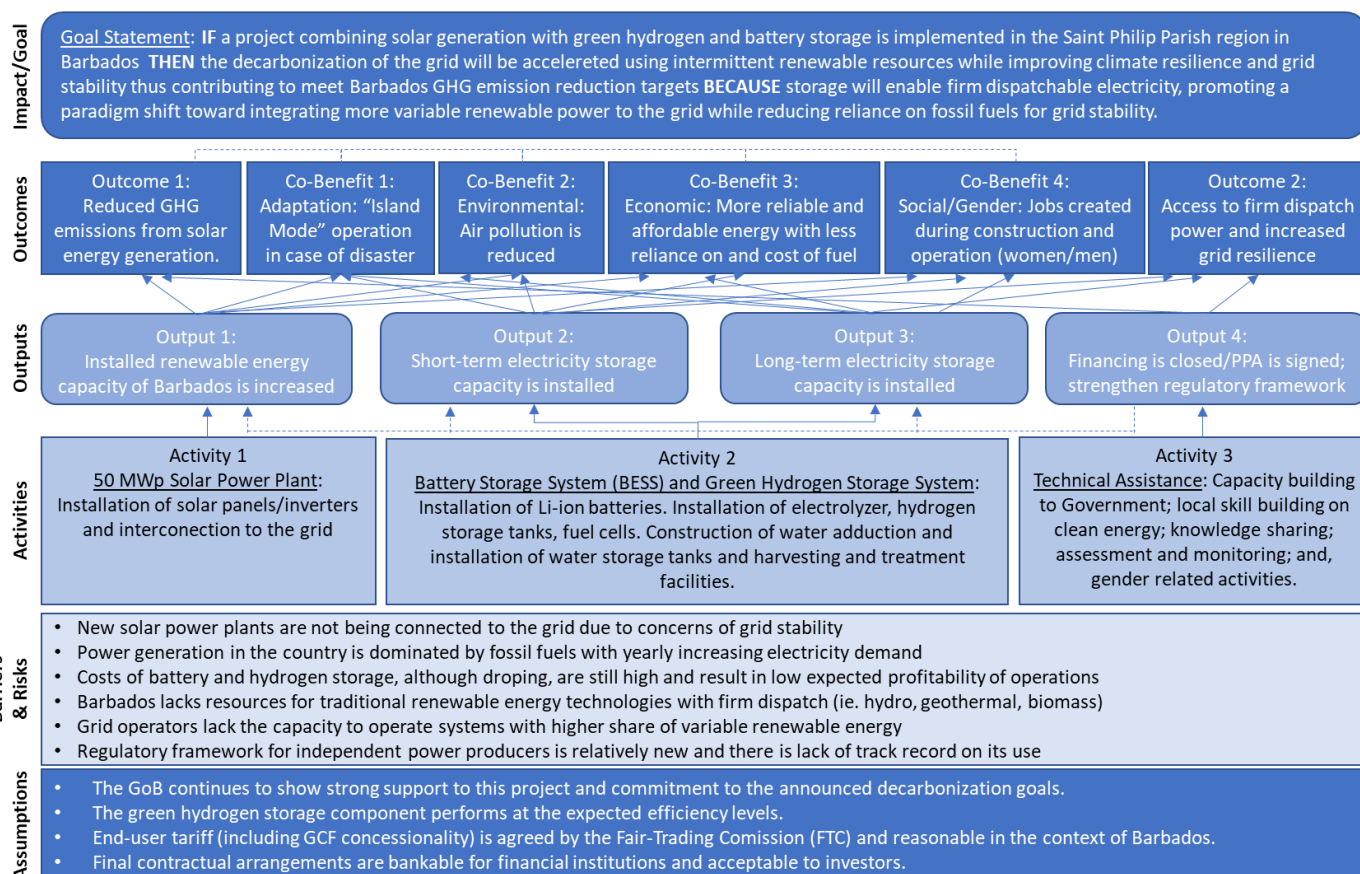
In addition to the IFC intervention above, the World Bank has also approved a USD 100 million financing in January 2023 to support Barbados' low-carbon economic development and resilience to climate change. The **Barbados Green and Resilient Recovery Development Policy Loan** has two main pillars. One of these pillars aims at facilitating Barbados' low-carbon and resilient infrastructure development through the implementation of new standards for agency-level disaster management plans, institutional reviews of national emergency management agencies, as well as increased renewable energy capacity and support for climate change adaptation and mitigation efforts. These engagements nicely complement the work to be supported by this GCF proposed project.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Barbados has very ambitious decarbonization targets, including those aimed at improving its power grid. Going from around 95 percent fossil fuel electricity generation to the target of 100 percent renewable energy requires a paradigm shift. Barbados has plentiful solar and wind resources to generate clean energy. However, these resources of renewable energy are inherently intermittent. The grid has started incorporating solar generation from a 10MW grid-connected plant and distributed generation for residential, commercial and industrial users. But this limited amount of generation is already starting to create instability in the grid.

The proposed project presents a solution that will provide installed capacity for solar power generation, thus avoiding GHG emissions and reducing national reliance on fossil fuels. The RSB project will allow the electricity to be stored and dispatched at times when it is needed so that customers benefit from firm power day and night and the grid stability is not negatively affected. This renewable energy-based solution will promote a paradigm shift and set the basis for an enabling regulatory framework for similar independent power producers.

Figure 1: Theory of Change Diagram



Firm power is added to the grid using solar resources and GHG emissions are being reduced.

(Outcome 1: *Reduced GHG emissions from solar energy generation*) Thanks to the 50 MWp of solar PV generation capacity built under "Output1: *Installed renewable energy capacity of Barbados is increased* " that would be able to generate 66GWh per year using clean solar resources, thus mitigating GHG emissions of more than 690,000 tonCO₂e over the 25-year lifespan of the Project.

"Activity 1. 50 MWp Solar Power Plant: *Installation of solar panels/inverters and interconnection to the grid*" is a critical input to Output 1 that will provide electricity to the grid during the day with a peak capacity of 13 MW. This activity involves site preparation, civil works, procurement of solar panels and inverters, installment and interconnection of equipment (including to storage and control systems), interconnection line to the substation, commissioning and testing. The additional capacity is then used to (a) produce green hydrogen through the electrolyzer that is stored in tanks and (b) recharge the batteries of the BESS. The assumptions that support this activity include that (i) the GoB continues to show strong support to this

project and commitment to the announced decarbonization goals; and, (ii) end-user tariff (including GCF concessionality) is agreed by the Fair-Trading Commission (FTC) and reasonable in the context of Barbados.

The main barriers related to this activity refer to the fact that (i) new solar power plants are not being connected to the grid due to concerns of grid instability; (ii) power generation in the country is dominated by fossil fuels with yearly increasing electricity demand; (iii) Barbados lacks resources for traditional renewable energy technologies with firm dispatch (i.e. hydro, geothermal, biomass); (iv) grid operators lack the capacity to operate systems with higher share of variable renewable energy and, (v) regulatory framework for independent power producers is relatively new and there is lack of track record.

Electrical energy is being stored to be used when needed. (Outcome 2: *Access to firm dispatch power and increased grid resilience*) by providing (i) green hydrogen long-term storage that will make available 3 MW of capacity throughout the night “Output 2: *Short-term electricity storage capacity is installed*”, and (ii) 15 MW capacity for 2 hours of short-term storage (30 MWh) through a BESS to address demand during peak-hours and respond quickly to variation of demand or changes in solar generation supply supporting “Output 3: *Long-term electricity storage capacity is installed*”, which combined together and with the solar PV plant provide a complete solution that is clean, relies on local solar resources, and promotes a more distributed generation matrix resulting in higher grid stability, sustainability and resilience.

Activity “2. Battery Storage System (BESS) and Green Hydrogen Storage System: Installation of Li-ion batteries. Installation of electrolyzer, hydrogen storage tanks, fuel cells. Construction of water adduction and installation of water storage tanks and harvesting and treatment facilities,” enables Output 2 and Output 3 to materialize. For this activity, some sub-activities include: site preparation and civil works; procurement and installation of batteries, electrolyzer, hydrogen storage tanks, fuel cells, and ancillary equipment; interconnection of equipment (including to solar PV plant and control system); commissioning and testing. These activities are supported in particular by the assumptions that (i) the GoB continues to show strong support to this project and commitment to the announced decarbonization goals; (ii) the green hydrogen storage component performs at the expected efficiency levels; and, (iii) end-user tariff (including GCF concessionality) is agreed by the Fair-Trading Commission (FTC) and reasonable in the context of Barbados.

Barriers and risks related to the storage component of the project include that (i) new solar power plants are not being connected to the grid due to concerns of grid stability; (ii) cost of battery and hydrogen storage, although dropping, is still high and results in low expected profitability of operations; and, (iii) Barbados lacks resources for traditional renewable energy technologies with firm dispatch (ie. hydro, geothermal, biomass).

Regulatory framework is being improved. The technical assistance work that will be provided related to capacity building to government and ministerial officials will facilitate the closing of the debt financing for the project and entering into a power purchase agreement with the power distribution company BLPC that is bankable by international standards “Output 4: *Financing is closed/PPA is signed; strengthened regulatory framework*”. Being among the first independent power producers (IPP) in the country, the financing structure and contractual arrangements are expected to serve as a basis for future power plants that are planned as part of the IRRP and to achieve the ambitious decarbonization targets that the GoB has established.

Output 4 is enabled by Activity “4. *Technical Assistance: Capacity building to Government; local skill building on clean energy; knowledge sharing; assessment and monitoring; and, gender related activities*” that is in turn related mainly to the assumption that (i) final contractual arrangements are bankable for

financial institutions and acceptable to investors, but also to the assumption that (ii) the GoB continues to show strong support to this project and commitment to the announced decarbonization goals. The technical assistance will also help facilitate the implementation of the project through training and skill building for local workers (women and men) in clean energy related topics, share the knowledge and lessons learned of the project in Barbados and the Caribbean region, and support the assessment and monitoring of vulnerabilities and gender related actions.

This activity is particularly addressing the identified barriers that (i) the regulatory framework for independent power producers is relatively new and there is lack of track record on its use, and that (ii) grid operators lack the capacity to operate systems with higher share of variable renewable energy.

In addition to the outcomes mentioned above, the project will result in important co-benefits:

Adaptation: In case of a natural disaster, the project will provide an alternate source of power generation in a location of the island where generation is not present at the moment (fossil fuel generation is conveniently located on the western coast where port terminals and logistics infrastructure are nearby) and able to produce electricity from local resources (ie. solar and storage). In the likely event that such a disaster affects transmission and distribution lines connecting to other generation facilities, the RSB project could provide electricity to residential customers nearby as well as as hospitals, water pumps and first response units.

Environmental: Air pollution will be reduced, including avoiding GHG emissions in excess of 690,000 tonCO₂e over 25 years. Particulate emissions are also avoided with this clean solution instead of fossil fuel alternatives.

Economic: A more reliable clean energy source is available and at an affordable energy tariff, fully enabled by the level of concessional financing provided by GCF (i.e the more concessionality, the more affordable the tariffs become, noting the significant gap today between Barbados tariff and the LAC average). The ability to reduce and even eliminate the use of fossil fuels for power generation will result in a significant strengthening of energy security and a reduction of Barbados' reliance on costly imported fuel.

Social /Gender – Job Creation (Women/Men): The project will generate local jobs during construction (around 100) and operation (around 10). Through the Technical Assistance component and the related initiatives for skill development and gender actions, the project will promote the participation of Barbadians in qualified positions in the project with a bias in favor of female employees/interns.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	X	☐	☐	☐	☐	☐	☐	☐
Outcome 2	☒	☐	☐	☐	☐	☐	☐	☐

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Adaptation: "Island Mode" operation in case of disaster	☐	☐	☐	☐	☒	☐
Co-benefit 2: Environmental: Air pollution is reduced	☒	☐	☐	☐	☐	☐
Co-benefit 3: Economic: More reliable and affordable energy with less reliance on and cost of imported fuel	☐	☐	☒	☐	☐	☐
Co-benefit 4: Social/Gender: Jobs created during construction and operation (women/men)	☐	☒	☐	☒	☐	☐

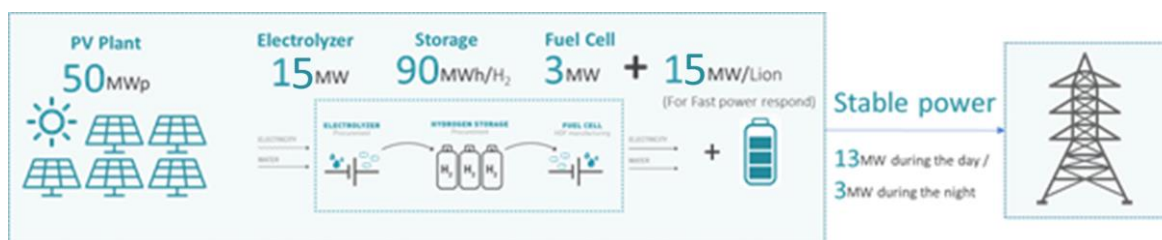
B.3. Project/programme description (max. 2500 words, approximately 5 pages)

The RSB project scope includes the design, financing, construction, operation and maintenance under a PPA scheme of a renewable energy plant in the easternmost end of the island. The project site is in Harrow, Saint Phillip Parish region, 17 km east of Bridgetown, the capital city of Barbados. The project will be based on a 73.4-hectare plot.

The plant consists of a system delivering a stable baseload and dispatchable energy through the following components:

1. 50 MWp solar power plant, as a primary source of electricity when the sun is shining
2. 90 MWh hydrogen storage system produced using electricity and a 15 MW electrolysis system during the day
3. 3 MW hydrogen fuel cell that transforms the stored hydrogen into firm electricity at night and when required.
4. 15MW/30MWh Li-Ion battery storage system that provides peak power from 7 pm to 9 pm and, in combination with the hydrogen storage, ensures the stability of the plant power service.

Figure 2: RSB Project Diagram



The plant will generate clean firm power at a level of 13 MW baseload (i) during daytime (8 am to 5 pm) and (ii) during the two-hour block from 7 pm to 9 pm when Barbados' energy demand and energy costs are the highest. The plant will also generate 3 MW firm power during the rest of the off-peak, including overnight hours.

The ultimate capacity of the renewable power plant is 56GWh/year of electricity, saving 27.7 thousand tons of CO₂e/year (693,000 tCO₂e avoided emissions over lifespan), contributing with approximately 6 percent of the Barbados grid capacity from renewable sources, and benefitting more than 21,000 people during its expected 25-year lifetime.

One key input for the electrolysis is water. The RSB project will procure water from the Barbados Water Authority (BWA). The estimated water demand of the plant is around 65 m³/day, of which 60 m³/day for the hydrogen production and 5 m³/day for other uses. The project includes five-day onsite storage of water for reliability purposes. Unlike higher locations in the western coast of the island (such as St. Lucy), the area where the project will be located has no issues with water stress. BWA has confirmed existing production of around 11,000 to 19,000 m³/day sourced from four different wells. There is plenty of capacity available to supply the project, as well as the ability to increase production to 26,000 m³/day. The RSB project could contribute to the resilience of the water system (and other critical facilities) by providing a new independent source of electricity for water pumping that is generated from solar resources and has storage capability. RSB plant will have an annual commercial contract with BWA and, although a separate supply guarantee cannot be granted, the project is applying two potential plans to ensure uninterrupted supply: Functional Plan (the regular water supply through a pipeline to the project as an industrial customer) and Emergency Plan (supply

of sufficient water to keep the plant running by means of cistern trucks in case the pipeline is off service due to an emergency situation.)

Project activities

Component 1: Solar PV Base Load Generation Plant

Activity 1: Installation of a new source of decentralized renewable energy generation

The 50 MWp solar photovoltaic (PV) plant would enable the country to produce more energy locally, and lower supply risk or political dependence. Its decentralized power generation is designed to sustain hurricanes and does not require feedstock to run. The solar PV plant is an intermittent renewable source of energy. Its output can partially be predicted in advance based on data analysis. When the solar plant is generating, part of the power output is dispatched to the grid in a stable way, while the balance is split to the electrolyzer and the battery following a linear optimization algorithm. The charging capacity of the combined storage (hydrogen and batteries) maximizes the use of solar PV production. An Energy Management System (EMS) will be an integral part of the project that will act as the brain of the power plant and manage the optimal energy dispatch between the different subcomponents of the hybrid power plant at any given time. When the solar PV output is not sufficient to deliver the required power, the EMS will command the battery or the fuel cell to provide power.

The plant can run on an island mode if necessary to provide electricity to those affected by any potential disaster and, with the support of GCF concessional funding, would have a competitive set price of electricity over 25 years for renewable and dispatchable baseload power that can also replace baseload assets and peaking gas turbine. The project would thus reduce the fiscal burden of imported fossil fuel and help Barbados achieve its renewable energy target. Barbados spends approximately USD 350 million on imported energy each year and relies on trade to meet its energy needs. Consequently, a momentary disruption or trade imbalance can lead to uncertainty in access to energy as prices fluctuate on the world market. GOB's NEP for 2019-2030 targeting 100 percent renewable energy is expected to cost around USD 4 billion and yield USD 3.9 billion per year in economic benefits over the 2020s as the country reduces its import of fossil fuels.

Although not a core component of the project, a sheep farming initiative is being planned as a social enterprise. It responds in part to a requirement from the planning conditions to grant the approval to RSB, which is related to the land's current use designation for agribusiness activities. The operation is intended to be outsourced to a sheep farming entity that will be allowed to use the plant site for grazing. This operation will provide vegetation maintenance services which would otherwise need to be performed mechanically, thus reducing the total cost of maintenance for the plant. There is no revenue stream considered from the sheep farm in the financial model. The project will donate the sheep pens, feed silos, water harvesting facilities, some necessary equipment and manure management facilities, and provide support in development of value-added products and services which may offer supplemental revenue streams to the farm. RSB would enter into a lease agreement with the farm entity (with or without nominal payment) for the use of the land and donated assets.

The project's mixed land use contributes to Barbados' food security goals and helps create the circular economy envisioned in the Roofs to Reefs program, by providing grass and other vegetation for native Black Belly sheep to graze between the solar panels, while enjoying the shade provided by the panels. It is worth

noting that there is already an established benchmark of a similar arrangement at what is currently the largest solar facility in Barbados located in Trents, St. Lucy, with a capacity of 10MW owned and operated by BLPC.

Activity 2: Installation of a short-term battery energy storage system (BESS) and a long-term green hydrogen storage system

There are many technologies on the BESS market, but the current leading technology and most suitable for RSB is the Li-ion BESS. For this project, the 15MW/30MWh Li-ion BESS is complementary to hydrogen, being used as a short-term energy storage and for power regulation. Several suppliers on the market offer preassembled enclosures solution for an easy plug-and-play integration that integrates their own firefighting and safety systems in the enclosure. Equipment typically has a lifetime of 15 years assuming one charge/discharge cycle per day, after which a full replacement would be needed. Round-trip efficiency for Li-ion batteries is around 85 percent to 90 percent, and degradation is estimated at 70 percent capacity by the end of the useful life.

The hydrogen storage system is comprised of three major components: electrolyzer, storage tanks, and fuel cell.

Electrolyzer: Water electrolysis is an electrochemical reaction known since the 19th century. Different technologies of electrolyzer exist on the market: alkaline, proton exchange membrane (PEM) and solid oxide. The main difference between the technologies is the electrolyte used. Two technologies are suitable for the RSB project: alkaline and PEM. The electrolyzer for the project is expected to be a 15 MW electrolyzer operating during daytime when the solar PV plant is generating. Suppliers offer solutions which can be containerized or which are pre-assembled and integrated in a building. Electrolyzers are estimated to have a lifetime of 80,000 hours (around 20 years if operating at 10 hours per day) after which the degradation should result in a 10 to 15 percent inefficiency. The energy consumption of the electrolyzer is around 55 to 60 KWh per Kg of hydrogen produced.

Hydrogen Tanks: Hydrogen storage will take place in large vessels (50 to 110 m³) at “low-pressure” (around 30 bar). These steel vessels, especially designed for hydrogen storage, are of a technology widely used in the chemical and petrochemical industry. The storage of hydrogen in gaseous form in large vessels is the simpler and safer solution. The total capacity will be equivalent to the hydrogen needed for 90MWh of electricity generation.

Fuel Cells: As with the electrolyzer, different technologies of fuel cell exist on the market: alkaline, PEM and solid oxide. The only technology suitable for RBS project is PEM. The fuel cell is a strategic brick because of the very few suppliers capable of building a multi-MW fuel cell. HDF Industry will be the supplier of the fuel cell, which is containerized for an easy plug-and-play integration into the plant. HDF has already installed a one MW fuel cell in Martinique that is currently in operation. Fuel cells are estimated to have a lifetime of 30,000 hours (around 7 years if operating at 12 hours per day) after which the degradation should result in a 15 percent inefficiency and a refurbishing of internal parts will be needed. The energy generation of the fuel cell requires around 60 to 65 kg of hydrogen per 1 MWh of electricity generated.

In addition to the main three components of the green hydrogen system, related balance of plant such as water treatment, compressors, pipelines, civil works, etc. will also be needed.

Component 2: Capacity building and knowledge sharing

Activity 3: Technical Assistance: Capacity building and support with contracts; knowledge sharing

Given the innovative nature of the project and the potential for replicability of similar power plants as well as for other applications of green hydrogen in the decarbonization strategy of Barbados, it is important to complement the implementation of the RSB power plant with a multipronged scope of technical assistance. The project intends to use the technical assistance grant funds for capacity building at various branches of the government of Barbados, for developing technical skills in the labor force, and to share knowledge and promote exchanges in Barbados and the Caribbean region. Some funds will also be allocated for project related assessment, implementation and monitoring in connection with MRV requirements and gender-related activities. The proposed scope of work of this activity is as follows:

1. Government assistance for sustainable energy policy development for large infrastructural projects:
 - a) Legislation development
 - b) Rate setting – regulatory assistance in development of capacity models for integrated power and storage projects:
 - i. Tariff structures and modelling
 - ii. Private sector case study development
 - c) Technical support to regional organizations such as CARICOM in strengthening the regional policy, legislative and tariffs frameworks for energy sector development.
2. Ministerial training in strengthening government institutions' capacity to unlock non-recourse project financing:
 - a) Gaining access to finance from multilateral organizations such as GCF
 - b) Unlocking private sector finance and investment (private sector investing for medium/ large scale energy projects)
 - c) Understanding and overcoming bankability issues for DFIs
 - d) Project and blended financing for acceleration of large-scale energy projects (business financial modelling and investment support)
3. Capacity building and skills development for the clean energy sector to elevate the local skills market in hydrogen and hydrogen power specialties. Scholarships to be offered in:
 - a) Medium and high voltage electrical applications – installation technicians, electrical engineering, PV plant design, etc.
 - b) Technical and legal specialties – transition of experienced energy specialists to hydrogen specialties in production, transport, storage, maintenance, policy, financing, legislation, regulation, etc.
 - c) Renewstable® plant management technicians
 - d) Training in financial management and integrity capabilities for the project
4. Knowledge sharing:
 - a) Study of potential applications of green hydrogen for small island states; success stories of GH2 projects in developed countries harnessing lessons learned and best practices. This study will entail, at a minimum:
 - i. Main drivers for success and failure risks
 - ii. Technical challenges typically faced during construction and operation
 - iii. Safety aspects to be considered regarding workers and neighboring communities
 - iv. Examples of regulatory framework highlighting best practices and potential pitfalls

- v. Approach to green certification and quantification of mitigation and adaptation benefits
- vi. Tariff calculation methodology and impact of GCF concessionality to reduce cost to end user;
- vii. Trends and future expectations
- b) Convening of a Caribbean Green Hydrogen Network.
- c) Launch event to present the results of the study in (a) above and to launch the CGHN in (b) above; identify champions to spearhead next steps
- d) Workshop on lessons learned from the Renewstable® plant installation

5. Assessment and monitoring:

- a) Vulnerability assessments including a results chain and MEL (monitoring, evaluation and learning) plan (MRV – monitoring, reporting and verification - requirements), to better understand how the project is reducing the climate vulnerability of the people of Barbados and develop an appropriate logical framework and results chain Gender-related activities
 - i. Gender baseline and gap assessment study
 - ii. Workshop with GoB and other stakeholders on gender issues and integration benefits
 - iii. Development of public education campaigns/workshops which encourage the participation of girls and young women to aspire to leadership and technical roles in the renewable energy sector. Develop communication strategy with targeted messages for women and girls to go for technical jobs/studies
 - iv. Develop and implement a detailed community-based Grievance/ Complaints Mechanism Plan with a survivor-centered approach, which provides confidentiality for GBVH reporting and addressing GBVH claims; gender and GBVH awareness training for all and specialized training for those responding to GBVH claims.
 - v. Develop stakeholder-specific handbooks/tool kits which are user friendly and increase the awareness and understanding of key stakeholders (male- and female-headed households, business owners and other vulnerable populations) on the benefits and impacts of hydrogen energy production and conduct education awareness sessions on such benefits and impacts.
 - vi. Provide training in rainwater harvesting techniques targeting small farmers, community members, business owners and operators, with a focus on women and other underserved populations
- b) Establish a project information meeting; conduct gender-sensitivity training with the RSB project management team.

Implementation arrangement

RSB is the Executing Entity of the solar PV generation and hydrogen and battery storage plant in Barbados. Financial close is preliminary planned towards the end of 2023 and the planning application and the Independent Power Producer (IPP) license application have already been submitted for approval. The Environmental and Social Impact Assessment (ESIA) was submitted around March/April 2022 and the planning application has been approved. Regarding the Power Purchase Agreement (PPA) the government of Barbados has shared a draft solar PPA, and currently it is being discussed with the local utility. Signing of the PPA is estimated to take place by the end of 2023. A fuller description of the implementation arrangement is presented in section B.4 below.

Key financial and operational risks

- **Technology and performance risks (fuel cell in particular) of this new solution.**

Mitigant: Although the proposed solution is innovative, it integrates technologies that have already been proven for years (fuel cells and electrolyzers), albeit there being still room for scaling mass manufacturing and reducing production costs. The electrolyzer and fuel cells will be procured from a tier-1 electrolyzer manufacturer and HDF Industry under a Ballard's exclusivity co-development, respectively, renowned brands in the nascent green hydrogen industry. The fuel-cell technology used is Proton Exchange Membrane (PEM) technology. The electrolyzer technology can either be pressurized alkaline or pressurized PEM, procured through a competitive process. Both electrolyzer technologies compete today. IFC has conducted an initial due diligence and management approval to engage in this project at the development stage (pre-investment). From this initial due diligence, IFC has not found an impediment to support the proposed technology or technology provider in Barbados. Although the expectation is that such a proven technology will operate as designed, as a worst-case scenario the risk of the technology failing is mainly concentrated on the hydrogen part. IFC's understanding is that failure to deliver the agreed electricity to the grid will result in the sponsors carrying the burden of lower revenues and penalties, but not in the tariff to the end consumer being increased as the tariff will be set at PPA signing. IFC will conduct additional due diligence prior to financial closing as part of the approval process for debt financing.

- **Cost competitiveness with other sources of power generation.**

Mitigant: Despite not having access to the exact dispatch profiles of BLPC required to develop an accurate economic assessment and electricity generation price comparison, the team estimates that the project, with the expected concessionality of GCF, competes with a combination of baseload assets and gas turbines (rather than with intermittent solar or wind generation), but entirely carbon-free, domestically-produced, resilient, innovative, and with local return on investment. Another alternative to the RSB solution could be a system using solar generation and battery-only storage (instead of a combination of green hydrogen and batteries). Although no such solution is being considered in Barbados, HDF prepared an in-house comparative analysis between these two options and IFC commissioned an independent study (by consultant DNV) to verify these results. The levelized cost of energy (LCOE) for the BESS-only solution is similar to the solution using GH2 (with the GH2 solution varying from 21 percent more expensive to 7 percent cheaper, across the different scenarios evaluated). Additionally, it is important to consider that the option with GH2 has several advantages such as:

- a) Diversification: not relying on one technology makes the system more flexible. There is also a lower marginal cost of additional MWh when the GH2 component is included.
- b) Higher autonomy: the hybrid GH2 & BESS system has the ability to store energy for up to two days of dispatch in case of emergency. This capacity can be increased economically by adding storage tank capacity, which is important for the adaptation/resiliency benefit and that is prohibitively costly for a BESS-only system to replicate.
- c) Replicability: being a first mover in the nascent green hydrogen space opens the way for Barbados to replicate the solution in tandem with a cost reduction of the GH2 technology that is expected to be steeper than for the more mature battery technology. It will also provide the opportunity for other potential applications such as industrial decarbonization where GH2 is a promising solution.

It is significant that the RSB project is supported by the government of Barbados and there is no BESS-only solution that is ready to be implemented, meaning that an opportunity cost should also be considered. HDF has spent development costs on this project over the last five years to get to the investment stage. The purpose of the DNV report, therefore, was not necessarily to be the decision factor in choosing between possible solutions, but to be considered as one of the various criteria to determine the viability of the proposed RSB solution, which to IFC continues to have sufficient merits to be supported.

- **Grid interconnection**

Mitigant: BLPC confirmed RSB can dispatch power generated to the nearest substation, the Hampton substation, which has upgrade plans to be ready before 2024 independently of RSB's project. Since the project is expected to start commercial operations in 2025, the team will monitor the upgrade's development and explore alternative arrangements to mitigate this risk further. As additional mitigants to the interconnection risk, a Connection Impact Assessment (required as part of the licensing process) will document all technical obligations for interconnection to be met by both parties, including the utility (BLPC). The Interconnection Agreement currently under negotiation will document these obligations and any penalties to be imposed due to delay. The PPA will impose deemed capacity payments in case of delay in COD for a reason attributable to BLPC.

- **Sponsor, Engineering, Procurement and Construction (EPC) and interface risks**

RSB will assume the construction cost and schedule risk. If there are delays in implementing the Project, RSB might be liable to pay Liquidated Damages (LDs) to BLPC such as in the case the plant does not deliver the agreed capacity or does not meet its daily availability commitments.

Mitigants: RSB expects to implement a full wrap fixed price EPC with an experienced, creditworthy contractor. EPC includes a customary liability cap, security package, LDs, and warranties. Additionally, HDF provides fuel cell technology and solar PVs are to be provided by a supplier with a proven track record. Moreover, HDF is currently recruiting professional expertise in Barbados to facilitate and coordinate between international EPCs and local subcontractors to mitigate EPC risks.

- **Supply chain constraints leading to increased costs and schedule slippage**

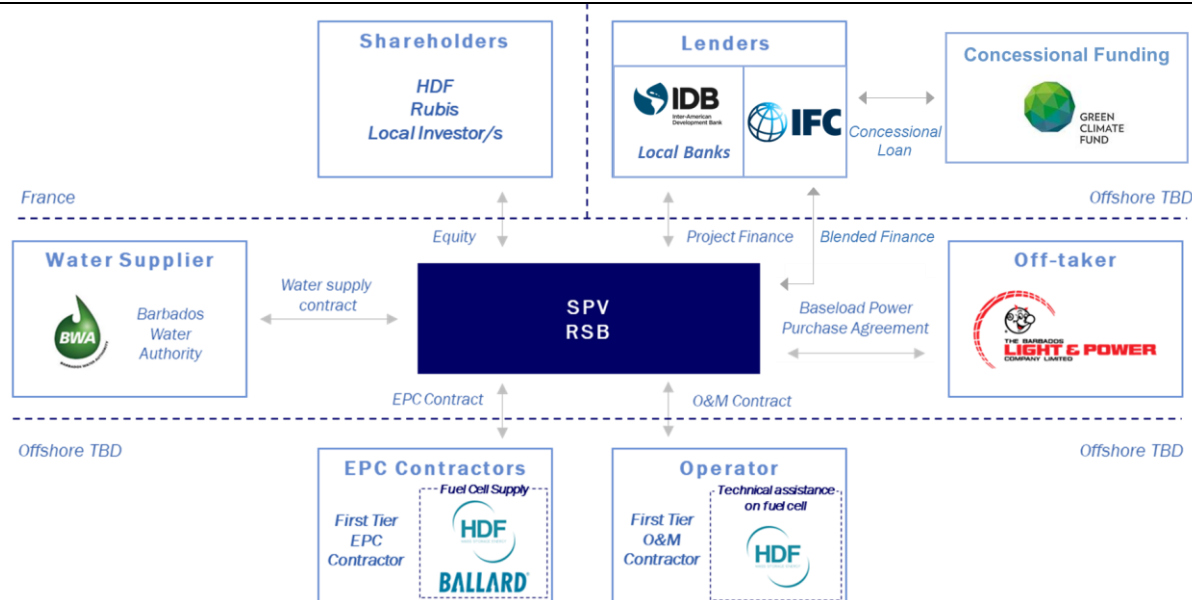
Mitigant: Solar PV and Battery Energy Storage Systems supply chains are currently very constrained, as well as the GH2-related equipment and EPC contractors, but are expected to return to more normal levels in the medium term. HDF is developing similar projects in LAC area, especially in Mexico, and other places in the world, hence a common procurement strategy has been put in place to secure the interest of the suppliers and of the EPCs and find synergies with the different supply chain stakeholders. HDF has already developed relationships with key suppliers on their other similar projects.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Proposed Financing Structure

The figure below illustrates the RSB project's indicative structure and the entities involved.

Figure 3: Stakeholders Structure



Sponsors

Hydrogène de France (HDF) – Registration Number 789 595 956

HDF is a French company listed on the EuroNext Stock Exchange specializing in developing and operating high-capacity large-scale hydrogen-to-power infrastructure projects that provide on-demand electricity from renewable sources energy sources (wind or solar), MW-scale fuel cells, and storage systems. The company was founded in December 2012 and is headquartered in Lormont, France. HDF recently completed its IPO on June 24, 2021. The company raised EUR 132.25 million under the ticker symbol of HDF.

The solutions designed by HDF combine solar and wind power, short-term battery storage, and long-term energy storage using hydrogen to provide clean baseload power and compete with existing polluting diesel generators. The proposed project uses the power-to-power Renewable solution that consists of multi-MW power plants, producing electricity 24h/day (intermittent renewable source and H2 storage). HDF has initiated the proposed project and is a shareholder in RSB. HDF leads all development activities through a development and project management agreement signed with RSB and will be the supplier of the fuel cells. HDF will take over the O&M activities after the two years interim O&M done by the EPC contractor.

HDF Energy is currently developing projects in French Guiana, the Caribbean, Central America, Southeast Asia, European Islands, and Australia. HDF has reported notable progress on the first project deploying this technology in CEOG – French Guiana: a multi-MW electrical power plant using hydrogen to store renewable energy delivering firm power of 10 MW days and evening and 3MW night. The project closed financing and started construction in September 2021 and is expected to be commissioned by 2023. It counts on equity banking from Meridiam and SARA (Rubis Group) and financing interest from international commercial banks (BNP Paribas, SMBC). Given the strong credit rating of French Guiana (France) and EDF as the off-taker, the financing in this case was achieved in commercial terms without concessional financing. The recent partnership with Rubis strengthens HDF's local expertise and networks and adds the means to develop additional projects and fund the construction of their fuel-cell plant in France.

Rubis Caribbean Holding Ltd. – Registration Number 35330

Rubis Inc. is a France-based international company specializing in storing, distributing, and selling petroleum, liquefied petroleum gas (LPG), food, and chemical products. Rubis is the market leader in France, Switzerland, Bermuda, Jamaica, Madagascar, Morocco, French Antilles-Guiana, Senegal, the Channel Islands, and Kenya. Rubis Group (already a 19 percent shareholder in HDF) acquired a 51 percent stake in RSB. Rubis Inc. is the wholly owner of Rubis Caribbean Holding, a majority (51 percent) shareholder in RSB.

Renewstable Barbados (RSB) – Registration Number 43324

RSB will be the Executing Entity implementing the project. The company is registered in Barbados and currently owned at 51 percent by Rubis Caribbean Holding and 49 percent by HDF. The board of directors comprises two board members from Rubis and one from HDF. All the equity needs are already committed by the existing shareholders. HDF is in discussion with local credit unions to acquire up to 30 percent of the project equity at or after financial close (in any case before COD).

RSB is the entity in charge of the development, financing, design, procurement, construction, commissioning, installation, testing, operation, maintenance, insurance, and decommissioning of the plant.

Construction Arrangements:

RSB is currently in the process of selecting an EPC contractor for the solar plant and storage facilities: green hydrogen plant and BESS. The ongoing tender was launched in April 2023 and has already shortlisted four reputable international EPCs. The **EPC contract** is expected to be awarded towards the end of 2023. A separate work package is also being procured for site preparation, potentially from a local or regional firm. Both contracts are intended to be signed in December 2023 following the financial close of the project and construction is expected to be completed in 24 months. RSB, with the support of HDF, will be responsible for procurement of equipment. For the time being, only HDF Responsible Procurement guidelines (annex 10) have been imposed but there is a general requirement to follow IFC standards when applicable. **HDF will be the supplier of the fuel cells** given the limited availability of suppliers of high-power (1.5 MW) fuel cells and the strategic partnership between HDF and renowned fuel-cell company Ballard based on which HDF is building a fuel-cell manufacturing facility in Bordeaux.

RSB is negotiating an **Interconnection Agreement** with the utility (BLPC). BLPC has proposed certain infrastructural upgrades be performed at HDF's cost to facilitate interconnection. Regarding the risk of delay in the interconnection to the grid, the **Connection Impact Assessment** (a requirement of the licensing process) will document all technical obligations for interconnection to be met by both parties, including BLPC.

Operational Arrangements:

The EPC contractor will be also be responsible for operating and maintaining (O&M) the full plant for an interim period of up to two years and will subcontract through **Long-Term Service Agreement** the maintenance of the equipment with their supplier. After that initial period and through a handover process, an O&M Operator (potentially HDF) that will have the human resources, skills and know-how to ensure a reliable operation will take over the O&M activities for the remaining years of the Project. HDF will enter into a **Project Management Contract** with RSB through which, among other things, HDF will carry out all procurement activities (as mentioned above) and supply and provide support regarding the fuel cells, O&M of the plant after the EPC contractor, and in general all development activities.

RSB will also enter into a **Power Purchase Agreement (PPA)** with BLPC, through which RSB will provide BLPC with additional capacity with the previously agreed dispatch profile, and BLPC will pay a capacity (availability) payment.

To be able to operate as a power generator in Barbados, RSB will enter into an **Independent Power Producer (IPP) License** with the GoB.

An agreement to procure the water, **Water Purchase Agreement (WPA)**, will be entered between RSB and the Barbados Water Authority (BWA).

Financing and Subsidiary Arrangements:

In accordance with the Accreditation Master Agreement (AMA) dated 22 November 2017, IFC and GCF will enter into a **Funded Activity Agreement (FAA)** in terms that are being prepared. The FAA will comprise two components: one concessional loan to be blended with the commercial loans financing the construction and implementation of the RSB plant, and a technical assistance grant component. The concessional loan will mirror the terms of the commercial financing except for the concessional pricing that will be fully transferred to the end-user tariff.

RSB is a special purpose vehicle to which the shareholders, HDF and Rubis, have already committed the equity contribution needed for the project budgeted at US\$43 million. The expectation is that the equity component (either as direct equity or as a fully-subordinated shareholders' loan or a combination of both) will be injected first or pari-passu with the debt financing.

Loan Agreements will be signed between RSB and various lenders. The loan agreements are expected to include a standard security package that could include pledge of shares, accounts and assets as well as economic rights under the key Project contracts and insurance policies.

Technical Assistance:

The Technical Assistance grant component of the FAA will support the following scope of work that is detailed more in annex 25:

1. Government assistance for Sustainable Energy policy development for large infrastructural projects
2. Ministerial training in strengthening Government institutions capacity to unlock non-recourse project financing
3. Capacity building & Skills Development for the clean energy sector to elevate the local skills market in hydrogen & hydrogen power specialties (incl. scholarships)
4. Knowledge sharing
5. Assessment and monitoring
6. Gender-related activities.

Funds will be transferred from GCF to IFC in accordance with the FAA that will be entered between the two institutions. Capable consultants will be engaged for these activities. Some of the TA activities are expected to be commissioned by the IFC advisory unit following World Bank Group procurement procedures and guidelines for the selection and contracting of the consultants (competitive selection is preferred in all cases and mandatory for contracts USD 50K and above). There may be some activities that the RSB may hire directly in coordination with IFC. For capacity building activities, participants will be selected in consultation with the GoB and the EE to ensure that the skills are developed for the people that will benefit the most and apply them more effectively in their respective positions. Candidates for the Knowledge Sharing activities

(including for the launch of the Caribbean Green Hydrogen Network) will be selected from the most appropriate public and private sector stakeholders in the region, to target potential champions that can spearhead a green hydrogen economy across these small island developing states (SIDS).

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

NOTE: GIVEN THE TIMING OF THE EXPECTED FINANCIAL CLOSE BY THE END OF 2023, IT WOULD BE IMPERATIVE THAT THIS PROJECT IS PRESENTED AT GCF'S OCTOBER 2023 BOARD MEETING SO THAT THE CONCESSIONALITY COMPONENT CAN BE INCLUDED IN THE TARIFF NEGOTIATIONS WITH THE FAIR TRADE COMMISSION.

The project would enable the development of the **first large-scale hydrogen storage solution in Barbados** and in the Caribbean and inform future IPPs and PPAs for other private sector initiatives that can be scalable and mitigate more innovative vulnerabilities of high dependence on fossil fuel energy generation. Concessional funding is critical for early projects such as this one to be viable, but it is expected that similar projects can be financed on commercial basis as the cost of renewable energy, electrolyzers, fuel cells, and batteries continue to drop going forward. GCF support is being sought not so much for the liquidity as for the concessionality of the financing it provides. Barbados being non-eligible to traditional blended finance sources, GCF is uniquely positioned to bring a cheaper financing than the combined commercial loans of IFC, other multilateral and local lenders, and thus will directly impact the PPA tariff as explained above (i.e. maximizing the concessionality in an otherwise set financing plan will directly benefit the tariff).

GCF funding would be supporting an innovative solution for decarbonization of the grid. Although the electrolysis and fuel-cell technologies are not new, only now the cost of renewable energy and the production cost of electrolyzers and fuel cells are approaching a commercial level that could compete with fossil fuel alternatives. Expected continued trends of cost reduction for both hydrogen equipment and renewable energy generation are likely to result in solutions like this one not needing concessional funding in the medium-term. But to get there, projects like this one are needed to keep the momentum and promote "power to hydrogen to power" project.

A successful experience with this project could be replicated in other island countries or isolated grids. It is critical that early, well-conceived projects such as the proposed solution are supported to deliver a positive result that can then be replicated, not only in Barbados, but in other Caribbean islands and beyond. The possibility of harnessing intermittent renewable energy and, through short- and long-term storage, being able to provide firm dispatchable energy without the need of fossil fuels is sure to have abundant application opportunities. The Technical Assistance component of this proposal includes a study and workshop on lessons learned once the plant has been built to leverage this experience and serve as a catalytic factor for future initiatives using these technologies. Training programs will be required to develop the local workforce to support hydrogen industries. Specialty skills in plant operation, renewable project financing, legislation and contracting will be required, skills that would then be immediately transferrable regionally and worldwide. In this regard, RSB, with the support of GCF TA, is looking at opportunities to partner with local universities and set up a Training Center for H2 services that will be used to form local student and train the potential workforce of the power plant and other H2 project in the region. Already in 2021, HDF executed a letter of intent with the University of the West Indies in which the UWI recognized HDF as an important vehicle for regional research and knowledge transfer and expressed its willingness to work with HDF to explore potential synergies for partnership and collaboration on the Project and regionally.

Through the proposed project, GCF would be able to support the diversification of energy generation sources for Barbados. With a grid that relies heavily on fossil fuels, the proposed solution is not only a way to decarbonize the generation matrix, but also to diversify the energy resources and geographic location of the power generation facilities. Diversification of generation sources and

decentralization of power plants (that no longer need to be close to ports that import fossil fuels) provides additional resilience to the energy sector in the country.

sing solar energy for base load power dispatch will support Barbados to increase its energy security. By shifting the paradigm that fossil fuel is needed for base load generation and grid stability to a solution that is fully renewable and, at the same time, dispatchable at any time and without grid stability issues, GCF would be helping Barbados in reducing imported, expensive and polluting fossil fuels for electricity generation. This will help mitigate a risk price volatility and access to supply that is always present, but that was stressed by the recent conflict between Russia and Ukraine. The Project would provide a natural hedge against a possible rise in international fuel prices, reduce fuel imports and international reserve depletion and improve the country's trade balance. Ninety percent of the oil and gas required to operate the plants is imported due to the lack of refining infrastructure. It impacts the competitiveness of the economy and costs on average 7 percent of GDP every year.

GCF support will be critical in de-risking part of the project's development in the Barbadian environment, thus maximizing chances of reaching financial closure. The concessionality of the GCF contribution will be fully channeled to reduce the tariff to the end user. The recipients would benefit from GCF support to be able to provide a tariff that is competitive with the current electricity generation alternatives in Barbados which are based on fossil fuels which enjoy embedded subsidies. The current cost of energy storage makes this clean and locally-sourced solution expensive to the end customer. As the cost of electrolyzers, fuel cells, renewable energy, etc. continue to decrease, future projects of this type are expected to eventually become competitive without the need for concessional funding. The HDF solution makes sense in island contexts such as Barbados where existing electricity prices are high and driven by diesel/HFO generation assets.

Competitiveness in small islands, such as in the Caribbean or the Pacific, and in similar isolated grids (ie. off-grid villages in Sub-Saharan Africa) is likely to be reached faster, given that diesel gensets and other alternatives are also expensive. In fact, HDF was able to reach financial close for a project in French Guiana (CEOG – <https://en.ceog.fr/home>) that uses the same technology and is very similar to the proposed project in Barbados, except for the country risk (although located in the same geographic region, French Guiana is effectively part of France). The project closed financing and started construction in September 2021 and is expected to be commissioned by 2025. It includes equity funding from Meridiam and SARA (Rubis Group) and financing interest from international commercial banks (BNP Paribas, SMBC). The project cost is similar to the project in French Guiana, however financing cost is higher for Barbados given that French Guiana is part of France (lower country risk) and the off-taker is EDF (state-owned off-taker from France). A successful project in Barbados could potentially open the way for a multi-country platform engagement to reproduce these projects in the Caribbean, LAC, and beyond (IFC is engaged in early discussions about similar systems in countries like Chile, PNG, and Pacific Islands).

B.6. Exit strategy (max. 500 words, approximately 1 page)

The project is expected to be sustainable in the long run. The principal investment of the GCF concessional loan is expected to be recovered from project revenues. Operating costs and revenues are stable and predictable providing higher certainty that the project will be sustainable over its lifecycle. Private investors plan to own, operate and maintain the project in the long run to recover their investment through dividend distributions in the expectation of making a reasonable risk-adjusted return. At the end of the Project life, the plant can be refurbished for continued operation or decommissioned. Lenders will be providing long-term financing alongside GCF concessional loan that is expected to be serviced fully from the Project's cash flows. Looking beyond this project, similar solutions are expected to be implemented in

other locations that may also need concessional funding initially (Barbados is among the most-developed small islands in the Caribbean), and eventually graduate to become a competitive solution financed on commercial terms.

Paradigm shift

Green hydrogen is a technology that is already mature and is on a trajectory to becoming cheaper. Early projects need some concessionality while the price of electrolyzers and the cost of renewable energy continues to drop to levels that would make the technology fully commercial. The level of concessionality needed is expected to be relatively low for this project and intended to address relatively higher country and technology risk perception resulting in a higher cost of financing at this early stage. Over time, similar projects are expected to be financed on a purely commercial basis, and GCF funding can (i) help accelerate implementation and (ii) provide a successful demonstration project to be replicated in Barbados and other places. The project will help develop knowledge and expertise about green hydrogen in Barbados that will facilitate its use in other applications such as industrial, etc., making Barbados a leader in this sector in the sub-region. Equity contributions for the project in Barbados will be provided by private sponsors and the debt is expected to be financed by development banks on a commercial basis, further leveraging the concessional funds from GCF. This project with a solar PV and lithium-ion BESS has the novelty of an H2 long-term storage allowing it to be a clean baseload asset for the country. The HDF solution makes sense in island contexts such as Barbados where existing electricity prices are high and driven by diesel/HFO generation assets. It places this project apart from the various other green hydrogen initiatives being pursued throughout LAC and Europe.

The project will provide catalytic gap financing to support a solution in LAC that shows significant scale-up potential and can demonstrate an economically viable green alternative to imported diesel. The production of cost-competitive green hydrogen is a function of three critical factors: (i) delivered cost of green hydrogen (including cost of electricity, CAPEX, scale, electrolyzers), (ii) policy/legislative environment (including carbon taxes and caps), and (iii) the cost of alternative solutions for a given application. This proposal offers GCF an opportunity to support a paradigm change from fossil fuels to green hydrogen, promote the development of a new green hydrogen ecosystem and help Barbados achieve their net-zero emission targets.

The project is expected to have a significant demonstration effect, to show that on small island states with high power costs, a technical solution combining solar generation, BESS and green hydrogen storage can provide baseload power, thus facilitating a paradigm shift towards further decarbonization of grids on these islands. With the expansion of the industry to include hydrogen and hydrogen technologies, training programs will be required to develop the local workforce to support hydrogen industries. Specialty skills in plant operation, renewable project financing, legislation and contracting will be required, skills that would then be immediately transferrable regionally and worldwide. Moreover, using green hydrogen for power storage will be an anchor for the exploration of other applications of green hydrogen beyond this project's scope in decarbonizing hard-to-abate sectors such as shipping and other transport modes. In this regard, HDF is looking at opportunities to partner with local universities and set up a Training Center for H2 services that will be used to serve local students and train the potential workforce of the power plant and other H2 project in the region. In 2021, HDF executed a letter of intent with the University of the West Indies in which the UWI recognized HDF as an important vehicle for regional research and knowledge transfer and expressed its willingness to work with HDF to explore potential synergies for partnership and collaboration on the Project and regionally.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	41			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	40	18 years	2 years	2.25 %		
(ii) Grants	1					
(b) Co-financing information	Total amount			Currency		
	128			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
RSB	<u>Equity</u>	<u>43</u>	<u>million USD (\$)</u>	25 years 0 years	NA%	<u>junior</u>
IFC	<u>Senior Loans</u>	<u>8.5</u>	<u>million USD (\$)</u>	18 years 2 years	SOFR+4%	<u>senior</u>
Other MDB	<u>Senior Loans</u>	<u>8.5</u>	<u>million USD (\$)</u>	18 years 2 years	SOFR+4%	<u>senior</u>
Local Banks	<u>Senior Loans</u>	<u>68</u>	<u>Local currency</u>	18 years 2 years	3.75%	<u>senior</u>
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	<u>169</u>			<u>million USD (\$)</u>		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	The sources above are sufficient to cover the budgeted project costs. No sovereign guarantee is expected for this financing. Commitment from this financing source is expected to occur concurrently with DFIs by the end of 2023.					
C.2. Financing by component						

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Component 1: Solar PV Base Load Generation Plant	<u>Output 1: Installed renewable energy capacity of Barbados is increased</u>	<u>52.7</u>	<u>12.5</u>	<u>Senior Loans</u>	<u>13.6</u> <u>26.6</u>	<u>Equity</u> Senior Loans	<u>RSB</u> <u>IFC/Other MDB/Local Banks</u>
	<u>Output 2 & 3: Short-term electricity storage capacity is installed & Long-term electricity storage capacity is installed</u>	<u>115.3</u>	<u>27.5</u>	<u>Senior Loans</u>	<u>29.4</u> <u>58.4</u>	<u>EquitySenior Loans</u>	<u>RSB</u> <u>IFC/Other MDB/Local Banks</u>
Component 2: Capacity building and knowledge sharing	<u>Output 4: Financing is closed/PPA is signed: strengthen regulatory framework</u>	<u>1</u>	<u>1</u>	<u>Grants</u>	<u>Enter amount</u>	<u>Choose an item.</u>	<u>Click here to enter text.</u>
Indicative total cost (USD)		<u>169</u>	<u>41</u>		<u>128</u>		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐
The Technical Assistance/capacity building component of the Project will supplement the Project by building the capacity of the government of Barbados and relevant actors, to support and drive green hydrogen and battery storage projects both in Barbados and in the LAC region. The Project is a pilot that can demonstrate scalability of low-emission energy for the Barbados national grid and other fossil fuel-based power grids in the region. Beyond decarbonization of the power grid addressed with this initiative, the proposed project would be Barbados first venture into the green hydrogen field that can be helpful to decarbonize hard-to-abate sectors such as shipping and other transport and industrial applications. The Technical Assistance grant of approximately USD 1 million is planned to be allocated for i) institutional capacity building of the National Designated Authority of Barbados and Ministry of Energy and Business in novel technology such as green hydrogen powered fuel cell energy storage and battery energy storage system ("BESS") via multistakeholder forum and training dedicated for efforts to improve the enabling environment for future green hydrogen and BESS projects in the country; and ii) knowledge generation and sharing with neighboring SIDS countries to develop and tailor similar business cases in those countries.	

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential (max. 500 words, approximately 1 page)

Mitigation Impact:

- Expected tonnes of carbon dioxide equivalent (tCO₂eq) to be reduced or avoided (PMF-M Core 1): The Project will reduce or avoid 27,719 tCO₂e per year, for a total of 693,000 tCO₂e over the lifetime of the asset. This conservative emission avoidance estimation is based on displacing typical fossil fuel power generation in the national grid. Although there is no formal conditionality implicit in the Project for the decommissioning of the gas turbine "GT02" running on diesel, it is expected that with the RSB project coming in line, GT02 can be decommissioned. Note that GT02 was scheduled to be decommissioned by 2019 and is still operating because there is no alternative generation that has been developed with a similar dispatch profile. Based on the characteristic of the GT02 facility, the emission avoidance would be up to 56,700 tCO₂/year, once the Project replaces GT02.
- Degree to which activity avoids lock-in of long-lived, high emission infrastructure: The Project does not rely on fossil fuel but removes barriers for higher penetration of renewable energy in the power grid, and it is therefore considered to have no carbon lock-in risks.
- Degree to which the program/project supports the scaling up of low-emission energy in the affected region by addressing key barriers: The Technical Assistance/capacity building component of the Project will supplement the Project by building the capacity of the government of Barbados and relevant actors, to support and drive green hydrogen and battery storage projects both in Barbados and in the LAC region. The Project is a pilot that can demonstrate scalability of low-emission energy for the Barbados national grid and for other fossil fuel-based power grids in the region.
- Expected number of MW of low-emission energy capacity installed, generated and/or rehabilitated: 50 MWp solar power plant, as a primary source of electricity, with the following supporting equipment to achieve required dispatchability profile: 90 MWh H₂ storage system that converts electricity into hydrogen using a 15 MW electrolysis system during the day, 3 MW hydrogen fuel cell that transforms the stored hydrogen into firm electricity at night, 15MW/30MWh Li-Ion battery storage system that provides the end of the day (7 pm to 9 pm) peak power and, in combination with the hydrogen storage, ensures the stability of the plant power service.

Adaptation Impact:

- Expected total number of direct and indirect beneficiaries: The Project will directly benefit approximately 21,000 people. This is calculated as the incremental energy generated by the plant (56GWh/year) by the average electricity consumption per capita (approximately 7.3 KWh/person per day). The project will indirectly benefit the entire population of Barbados (287,000 people) since all of the population of Barbados is connected to the energy grid and the project will increase the resilience of this grid, particularly during extreme precipitation events, tropical storms or hurricanes.
- Degree to which the activity avoids lock-in of long-lived, climate-vulnerable infrastructure: This project was designed based on the recommendations from a climate risk assessment using the latest climate models and scenarios. For example, the plant's structural design will withstand higher extreme weather and wind loads and include drainage design to higher flood return periods, increased resilience against low water availability, electrical infrastructure designed for higher future temperatures and design suitable for increased dust levels. More information on the climate risk assessment can be found in the annexes.
- Expected reduction in vulnerability by enhancing adaptive capacity and resilience for populations affected by the proposed activity, focusing particularly on the most vulnerable population groups and applying a gender-sensitive approach: The Project is expected to reduce the vulnerability of the population of Barbados by enabling faster restoration of electricity to the local region and the country, which will shorten recovered time post disaster and help to avoid economic catastrophe.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Potential for scaling up and replication

The proposed project has a very high potential for engendering a paradigm shift given its prospects for scaling up and replicability in Barbados and in other small island economies or in off-grid, remote areas. Solar PV generation has become one of the lowest cost solutions for renewable energy production. One of the main drawbacks of solar energy relates to its intermittency, being operational only during the day and with significant variations when passing clouds interfere with sun irradiation. The instability that intermittent sources of power generation like wind or solar produce on the grid can severely limit the penetration of renewable energy capacity in the generation mix. The Project's proposed solution combines a solar plant with green hydrogen and battery storage to provide a clean, stable, base-load dispatch, thus preventing disturbances to the grid.

Potential for knowledge sharing and learning

Being one of the first projects in the LAC region - and the world - using green hydrogen as an energy vector to store energy for base-load dispatch, the Project will offer significant opportunity for knowledge sharing and be a critical learning experience for (i) private sector including the Project sponsors who, pending the success of this Project with the support of the GCF, are aiming to build a pipeline around the world, (ii) for IFC as a potential lender and knowledge source for future green hydrogen projects, and (iii) for GCF to demonstrate the viability of the technology and replicability paving the way to support similar projects following a programmatic approach.

Contribution to the creation of an enabling environment

The Technical Assistance/capacity building component of the Project will supplement the Project by building the capacity of the government of Barbados and relevant actors to support and drive green hydrogen and battery storage projects both in Barbados and beyond. Through capacity building activities, the Project will focus on filling 'gaps' to strengthen the enabling environment for private sector investments in green hydrogen and battery solutions, focusing especially on the needs of small island states. Specifically, the Project will ensure that there is awareness of the up-to-date legislation and that necessary mechanisms are in place bolstering private sector engagement in the energy sector.

Contribution to the regulatory framework and policies

The Project will provide the opportunity to adjust the regulatory framework (i.e. tariff, PPA, etc.) and safety standards to enable the creation and development of a green hydrogen ecosystem. After a new PPA legislation was put in place in 2015, the Project will be among first IPPs in Barbados, thus contributing to establish a more diversified structure in the electricity sector. The Project will serve as a proof of concept for a technical solution and business model that can be scaled to contribute to decarbonization of the power generation sector in Barbados and various countries. Beyond this specific application, the proposed solution will be complementary to other uses of hydrogen as a decarbonization enabler and energy vector for hard-to-abate industries such as transport, fertilizers, chemicals, steel, and synthetic fuels.

Overall contribution to climate-resilient development pathways

The Project is fully aligned with the ambitious objectives of Barbados to increase the share of electricity produced from renewable sources. The Project is fully aligned with the GoB's ambitious objectives to increase the share of electricity produced from renewable sources and the proposed Project is also aligned with key programs being pursued by the government of Barbados, such as the Roofs to Reefs Program

(R2RP), supported by GCF readiness funds, that provides the overarching framework for an integrated approach to addressing the negative impacts of climate change. Additionally, through a NAMA prioritization analysis, the Barbados' Second National Communication Under the United Nations Framework Convention on Climate Change identifies solar generation and energy storage as new technology options for contributing to climate change mitigation and to reduce Barbados' dependence on fossil fuels. Thus the Project has a strong paradigm shift potential and will impact the country's trade balance and limit the exposure to a possible rise in international oil prices and international reserves depletion. Also, the Project will enable the country to produce more energy locally and reduce supply risk and political dependence.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Financing Structure. IFC's additionality results from providing a financing solution that is not available in the local market. Non-recourse project finance is practically nonexistent in Barbados and the tenors available in the market are well below the 15-years that IFC expects to offer to the client. In the past three years only one USD-denominated syndicated loan was issued by a company in Barbados (extractives sector) with a face value of 100 million USD and a one-year maturity. In that same period there were no USD corporate bond issuances at a corresponding rating of B+/- . This context of limited availability of financing is a challenge for the private sector in Barbados. Private investment for example, was 15 percent of GDP in 2022 which is a mild improvement from the 13.9 percent average of the previous 5 years but remains below the 17.7 and 17.8 percent registered in LAC and EMDEs¹¹.

Non-commercial Risk mitigation. IFC's participation gives comfort to the client by mitigating non-commercial project risk derived from the untested nature of solar and green hydrogen generation plus hydrogen and battery storage in Barbados. IFC will contribute to facilitating interactions with the Government to enable the project implementation. The project has been subject to a lengthy permitting process and unforeseen landowner disputes, resulting in a one-year delay on the development schedule. The current development timeline relies heavily on the engagement with the Ministry of Energy, Small Businesses, and Entrepreneurship for the IPP License and the PPA negotiation.

Knowledge, innovation, and capacity building. IFC has established a strategic partnership with the client through an Upstream advisory engagement to develop the project. The engagement has been fundamental to move the project from prefeasibility to near implementation stage. IFC's Upstream engagement is currently active in several areas including assessing solar resources, geotechnical and hydrological characteristics, tax and customs.

Standard Setting. IFC is also working with the client to develop a comprehensive ESAP that improves the project's Environmental and Social Risk Management policies and procedures, making sure that it is aligned with IFC Performance Standards. In addition, IFC has a rigorous approach to managing blended finance to ensure there is a strong rationale for blended finance in every project and each project applies the DFI Enhanced Blended Finance Principles including minimum concessionality and IFC's enhanced disclosure which also includes an estimation of the subsidy at the transaction-level. This includes review of every investment project by a Blended Finance Committee independent from IFC Operations or the Blended Finance Director and supervision by a blended finance team responsible for the application of concessional blended finance and adherence to the DFI Blended Finance Principles spearheaded by IFC and adopted by all major DFIs. In 2019, IFC announced it would hold itself to the highest standards of transparency when deploying concessional resources: IFC now discloses in its [Summary of Investment Information](#), the subsidy levels for each proposed project along with justification for why it is necessary. IFC is the only DFI or blended finance implementer taking this step to date.

¹¹ Emerging markets and developing economies

Affordability/Environmental. The project is expected to generate 56GWh/year of electricity and directly benefit approximately 21,000 people. The plant will enable the replacement of a 13MW gas turbine thus contributing to a reduction of the country's reliance in fossil fuels (over 95 percent currently) and the resulting high exposure to fuel price volatility. Levelized cost of electricity has gone up from 163 USD/Mwh in 2020 to 261 USD/Mwh in 2022 due to increase in oil prices. This volatility is directly reflected in the end consumers' bills since it is passed down to the end consumer via the Fuel Clause Adjustments mechanism.

The plant's two-hours dispatchable block at 13MW effectively replaces a 13MW gas turbine on the grid (GT02) which was scheduled to be decommissioned in 2019 and is still operating. The associated emissions savings per year are preliminarily estimated in 56.7 thousand tons of CO₂.

Sustainability. The project's demonstration effect can be a central component of the country's decarbonization strategy for the power sector. The Project directly supports Barbados's National Energy Policy (NEP), which targets full renewable energy and carbon-neutral island-state transformation by 2030. The Renewstable concept combining solar PV and H₂ generation with H₂ and Li storage represents a significant innovation to provide stable baseload and dispatchable energy to the grid. This configuration enables increased utilization of solar generation in the country's power generation matrix, thus mitigating climate change effects of the sector. Barbados' generation matrix relies heavily on fossil fuel generated power. In 2016, the country had 249 MW of utility scale installed capacity of which 239 MW was powered by oil and diesel plants. The role of solar power in the provision of electricity has remained marginal because of its limited ability to dispatch based on changes in demand. The project's configuration combining generation with daily storage contributes to addressing this limitation, thus effectively diversifying sources of generation, and ultimately reducing the system's reliance on thermal sources. Looking beyond the scope of this project, the experience of producing green hydrogen from local renewable resources can help Barbados in using green hydrogen to decarbonize sectors other than power, such as transport.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

The Barbados electricity grid is centralized and vulnerable to hurricanes. Since 2010, hurricanes Tomas (2010), Ernesto (2012), Harvey (2017), and Elsa (2021) and the tropical storms Matthew, Maria, Kirk and Gonzalo (2020) have impacted the island, causing power outages and enormous damage to the people in Barbados, particularly in areas that are distant from the current generation centers. The ability to locate the Project away from the ports through which fossil fuels are imported provides the local communities around the Project with more reliable electricity supply in case of climate events that may impact the transmission infrastructure.

Barbados is already among the countries with highest electricity tariffs in the region. On the other hand, Barbados has very ambitious targets for decarbonization and low potential for base-load renewable sources such as geothermal, hydro or biomass. To avoid stability issues with its grid, Barbados will have to rely on technologies that are in a downward cost trend, but still at a relatively expensive price level such as battery storage or green hydrogen. The higher cost and inherent higher risk of these technologies compounded by the small size of the Barbados economy makes commercial financing of power solutions of this nature very unlikely to materialize at this stage. The support of GCF alongside development financial institutions like IFC comes to fill a need to help materialize a project that will be a reference for future similar projects that eventually are expected to be financed by commercial banks without the need of concessional funds once the decreasing pricing trends reach the appropriate level.

The Project will also provide a natural hedge against rises in international fuel prices, reduce fuel imports and international reserve depletion, and improve the country trade balance. Ninety percent of the oil and gas required to operate the plants is imported due to the lack of refining infrastructure.

D.5. Country ownership (max. 500 words, approximately 1 page)

Country ownership

As mentioned above, this private sector project is fully aligned with the GoB's ambitious objectives to increase the share of electricity produced from renewable sources. The proposed project is also aligned with key programs being pursued by the government of Barbados, such as the **Roofs to Reefs Program (R2RP)**, supported by GCF readiness funds, that provides the overarching framework for an integrated approach to addressing the negative impacts of climate change. Additionally, through a Nationally Appropriate Mitigation Action (NAMA) prioritization analysis, the Barbados' **Second National Communication Under the United Nations Framework Convention on Climate Change** identifies solar generation and energy storage as new technology options for contributing to climate change mitigation and to reduce Barbados' dependence on fossil fuels. Additionally, the project would strongly impact the country's trade balance and limit the exposure to a possible rise in international oil prices and international reserve depletion. Also, it would enable the country to produce more energy locally and reduce supply risk and political dependence. **The government of Barbados is fully supportive of the proposed project and issued the No-Objection Letter (NOL) early in the process.**

Demand for energy remains strong and the Integrated Resource and Resiliency Plan for Barbados (IRRP) projects electricity demand to increase to 1,277 GWh by 2030 under a base scenario, implying a 35 percent increase from its 2019 levels. At the same time, the IRRP recommends that Spring Garden steam generation plans fueled by HFO (totaling 40MW) be retired in 2023 and that gas turbines (totaling around 100MW) are not suitable for continued use operation and are expected to be used only as backup and emergency generation. The IRRP mentions green hydrogen as an option for decarbonization through long-term storage of electricity. Although this technology was excluded from the analysis for being deemed not mature enough and too expensive, the green hydrogen landscape has evolved significantly in the two years since the IRRP was published. GCF support would help address the high cost of the technology at this early stage. The authors of the IRRP recommend that the report is updated regularly to provide the opportunity to consider alternate technologies as circumstances evolve.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☒ Reduced emissions (mitigation)
☐ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	The current installed capacity of solar PV is 10 MWp of utility-scale solar PV generation capacity plus 25 MWp of solar PV distributed generation ¹² in Barbados. The country has no BESS system nor hydrogen storage and fuel cell system. The Ministry of Energy is launching a national energy storage policy to	<u>Low</u>	(i) 50 MWp of additional solar PV generation capacity (ii) 90 MWh hydrogen storage system (iii) 3 MW hydrogen fuel cell (iv) 15MW/30MWh Li-ion battery storage system	The RSB project will contribute to a paradigm shift in renewable energy and storage in Barbados by scaling up 50 MWp of solar PV generation capacity as a primary source of electricity when the sun is shining, by developing a 90 MWh hydrogen storage system that converts electricity into hydrogen using a 15 MW electrolysis system during the day, installing a 3 MW hydrogen fuel cell that transforms the stored hydrogen into firm electricity at night, and installing a 15MW/30MWh Li-ion battery storage system that provides the end of the day (7 pm to 9 pm) peak power and, in combination with the hydrogen storage, ensures the stability of the plant power service.

¹² BLPC (<https://www.blpc.com.bb/index.php/company/how-we-serve-you/how-we-generate-electricity>)

	incentivize renewable energy developers to build a storage system.			Green hydrogen and battery storage in this project will enable firm dispatchable electricity, promoting a paradigm shift toward integrating more variable renewable power to the grid while reducing reliance on fossil fuels for grid stability.
Replicability	No hydrogen storage, BESS, and fuel cell have yet been demonstrated in Barbados, so replication is not yet possible.	<u>Low</u>	If the installment of solar PV and hydrogen storage systems can be demonstrated, the Project can be replicated across neighboring small island economies and in other off-grid, remote areas.	The project's intervention has never been tested before, so lessons learned through this work will directly inform any efforts to replicate the work in other areas of Barbados and neighboring small island countries. Via technical assistance of knowledge sharing, Caribbean Green Hydrogen Network will be established to share best practices and lessons learned from this project.
Sustainability	Current capacity of the government of Barbados and other small island states with regards to creating green hydrogen solutions to generate and store energy is limited.	<u>Low</u>	The capacity of the government of Barbados is increased and sustained beyond the project's implementation period.	Via technical assistance, the capacity of the Barbados government will be increased via focused trainings and other capacity building activities such as legislation development, rate setting, technical support to regional organizations such as CARICOM in strengthening the regional policy, legislative and tariffs frameworks for energy sector development to create enabling environment and minimize

				regulatory barriers for future green hydrogen projects.		
E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
<u>MRA1 Energy generation and access</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Annual ex-ante and ex-post analyses, electricity generation data from the project Power generated from solar resource (GWh) and multiplied by the Firm Energy Combined Margin Grid Emission Factor IFI grid emission factor¹³ (Conducted by an independent verification company at the mid and final term evaluation, and end of the project)	0	0	Target value at the end of implementation period: 0 Target value at end of total lifespan: 693,000 tCO2e	Total lifespan target for GHG reduction is expected over 25-year asset lifetime of the Project, which is 692,981 tCO2e. During the project period of 18 years, it is expected to be 498,946 tCO2e. The annual emission reduction is expected to be 27,719 tCO2e. Externalities: The PPA is based on capacity payments and BLPC is free to dispatch as they wish up to the maximum agreed. BLPC may decide not to dispatch the plant, which doesn't have an impact in revenues, but may have

¹³ https://unfccc.int/sites/default/files/resource/Harmonized_IFI_Default_Grid_Factors_2021_v3.2_0.xlsx

						an impact on emissions mitigation.
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.3: Installed renewable energy capacity</u>	Annual ex-post analysis by IFC team (commissioning report) Lenders' engineer verification that commissioning is completed and the plant has reached COD	0	0	50MW	Successful construction of solar PV plant Assumes interconnection to the grid is permitted and connecting infrastructure is ready
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.4: Renewable energy generated</u>	Annual ex-post analysis by IFC team Power generated from solar resource (GWh) to be measured at the end of the month through calibrated meter at battery limit of the project concurrently by RSB and BLPC	0	0	66,000 MWh/year	Clean power dispatched during the day over one year

<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.2: Installed energy storage capacity</u>	Annual ex-post analysis by IFC team Verification of hydrogen storage capacity at commissioning	0	0	90 MWh	Green Hydrogen long-term storage capacity of 3MW throughout the night
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.2: Installed energy storage capacity</u>	Annual ex-post analysis by IFC team Verification of battery storage capacity at commissioning	0	0	30 MWh	Battery Storage System with capacity of 15MW throughout the peak hours
<u>MRA1 Energy generation and access</u>	<u>Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions</u>	Project financial report; asset value information report from asset owners	0	0	USD 120 million	Based on CAPEX (excluding Development Costs, Additional Investment Costs, Contingencies, Financing Costs, Interest During Construction and Working Capital)

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
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<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	The country has no BESS system nor hydrogen storage and fuel cell system. The Ministry of Energy is launching a national energy storage policy to incentivize renewable energy developers to build a storage system.	<u>low</u>	Strengthened institutional and regulatory frameworks allow for an increase of renewable energy in the generation matrix and help the country decarbonization targets.	Through the TA activity, the project will strengthen governmental capacity to develop Sustainable Energy Policy for large infrastructure projects.	<u>National level (one country)</u>
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	There is a lack of knowledge of storage technology, good practices and standards.	<u>low</u>	In-depth knowledge of battery and green hydrogen storage technology is developed, extracting lessons are learned and good practices, methodologies and standards are adapted to the local context.	The project's practice of storage technology, good practice, standards, and lessons learned will be shared with target audiences including other government officials, energy companies in the country and in the region.	<u>Multi-countries</u>

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1 Reduced GHG emissions from solar energy generation	Estimated GHG emissions avoided (tCO ₂ e)	GWh generated from solar resource and calculation of emissions of a gas turbine operating	0	0	693,000 tCO ₂ e	The final target is defined as the ex-ante GHG reduction of 690+KtonCO ₂ e expected

		on diesel with the same dispatch profile				over 25-year lifetime of the Project.
Output 1 Installed renewable energy capacity of Barbados is increased	Generation Capacity installed (MW)	Commissioning certificate issued by grid operator.	0	0	50 MW	Tested capacity in line with nameplate capacity
	Annual availability (%)	Grid operator confirmation	0	0		Plant to be dispatched by the grid operator based on demand and operating needs
	Maximum annual production of the Plant(GWh/year)	Monthly joint energy or meter reports issued by the grid operator	0	0	[56] GWh/year	Assumes no curtailing
Output 4 Project financing is closed and a sound PPA is signed	Financing closing	Signing of loan agreements between lenders and RSB	0	0	[1]	Assuming the PPA is signed
	PPA is signed.	Legal effectiveness of PPA document issued by grid operator (BLPC) or government of Barbados	0	0	[1]	Assuming commercial and legal terms are agreed among the parties
Project/programme co-benefit indicators						
Co-benefit 1: Adaptation: "Island Mode" operation in case of disaster	Direct/indirect beneficiaries (#)	Vulnerability assessment	0	0	21,000	Number of customers affected during an event that continue to have access to power from RSB

Co-benefit 2: Environmental: Air pollution is reduced	Estimated NOx, SOx and PM-10 emissions avoided (Ton)	GWh generated from solar resource and calculation of emissions based on emission factors ¹⁴ of a diesel generator with the same dispatch profile	0	0	NOx: 31,099 SOx: 2,057 PM-10: 2,207	The final target is defined as the ex-ante GHG reduction of NOx: 31,099; SOx: 2,057; and PM-10: 2,207 expected over 25-year lifetime of the Project.
Co-benefit 3: Economic: More reliable and affordable energy with less reliance on and cost of fuel	Electricity tariff reduction (%)	Negotiated tariff with FTC	0	0	[6]%	Final tariff negotiated as compared to tariff yielding the same equity IRR without concessional financing.
Co-benefit 4: Jobs created during construction and operation (women/men)	Number of jobs created (#)	RSB payroll	0	100 of which 15% female (Construction)	10 of which 30% female (Operation)	The TA activities are expected to develop the skills to hire local workers and promote female workforce participation.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1 <u>50 MWp Solar Power Plant</u>	Installation of solar panels/inverters and interconnection to the grid	Sub-activity 1.1 Solar panels/modules Sub-activity 1.2 Inverters Sub-activity 1.3 Interconnection to the grid	50 MWp Solar PV plant installed and commissioned
Activity 2 <u>Battery Storage System (BESS) and Green Hydrogen Storage System</u>	Battery Storage System (BESS)	Sub-activity 2.1 Pre-assembled enclosures (plug-and-play) 15 MW/30MWh	Storage systems installed and commissioned

¹⁴ <https://www.epa.gov/air-emissions-factors-and-quantification/ap-42-fifth-edition-volume-i-chapter-3-stationary-0>

	Green Hydrogen Storage System	Sub-activity 2.2 Electrolyzer (15 MW) Sub-activity 2.3 Hydrogen storage tanks (90 MWh) Sub-activity 2.4 Fuel cells (3 MW)	
Activity 3 <u>Technical Assistance</u>	Capacity building to government; local skill building on clean energy; knowledge sharing; assessment and monitoring; and, Gender-related activities.	Sub-activity 3.1 Government assistance for Sustainable Energy Sub-activity 3.2 Ministerial training on non-recourse project financing Sub-activity 3.3 Capacity building & skills development Sub-activity 3.4 Knowledge sharing Sub-activity 3.5 Assessment and monitoring	Government officials who have completed capacity building Ministerial officials who have completed capacity building Skilled labor trained Study on GH2 applications in SIDS Launch of Caribbean GH2 network Workshop on lessons learned from RSB delivered Vulnerability assessment conducted Gender action plan implemented

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

IFC is responsible for the supervision of the project, including obtaining and reporting results to GCF. Relevant indicators in Annex 11 will be tracked and reported by IFC. For mitigation result measurement, the operational reporting will include the renewable energy (GWh/year) generated by the plant and dispatched to the grid. This data is a key variable not only for mitigation measurement purposes, but also for other operating metrics. Energy generation will be measured using calibrated meters located at the battery limit of the power plant and verified by BLPC as counterpart in the power transaction. The evaluation of the annual GHG emissions reduction will be estimated by multiplying the renewable energy generation times the latest IFI Grid Emission Factor for Barbados. IFC methodology for measurement of development impact will be followed. IFC will provide GCF with relevant, non-confidential project information. The services of a qualified external third party may be retained to undertake an evaluation of the project's activities.

The interim and final independent evaluations will be conducted in accordance with the GCF's evaluation criteria as per the GCF Evaluation Policy. Accordingly, these independent evaluations will embed an assessment of the Project's contributions to a paradigm shift and enabling environment using a three-point scale rating. IFC will notify GCF should the project appear unlikely to achieve expected results and/or fail at its objectives due to extraordinary circumstances (natural disaster, political unrest, etc.). A mid-term evaluation is planned on 3Q2026 and a final-term evaluation in 2030 that will allow for lessons learned from the project implementation to be applied at an earlier stage, rather than at the end of the project

lifespan in 2050. Complementing the specific GCF monitoring requirements and independent evaluations mentioned above, IFC's loan agreement to be signed with the EE contains provisions for regular reporting and IFC will monitor and review these reports including: quarterly financial statements, quarterly construction/operations report, annual audited statements, compliance with financial covenants, E&S annual monitoring report, notice of accidents, material adverse effect, litigation, default, insurance, etc. Operations reports will include energy dispatched from this clean source, which would then be used to calculate the emissions reduction.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Select</u>
Description		
Technology and performance risks (fuel cell in particular) of this new solution		

Mitigation Measure(s)

The proposed solution is new and innovative. It integrates technologies that have already been proven for years (BESS, fuel cells and electrolyzers), albeit there is still room for scaling mass manufacturing and reducing production costs. This project's BESS will be lithium ion and is intended to be procured from a top tier manufacturer. The electrolyzers and fuel cells will be procured from a tier-1 supplier and HDF Industry under an exclusive co-development license with Ballard, two renowned brands in the nascent green hydrogen industry, and performance guarantees will be negotiated. The electrolyzer technology can either be pressurized alkaline or pressurized proton exchange membrane (PEM). Both technologies compete today in similar projects (on CEOG, alkaline was used for commercial reasons). The fuel cell technology used is PEM technology. Finally, although these technologies are providing a novel solution, HDF has been operating a 1 MW fuel cell in Martinique since 2019 and reputable international banks and commercial lenders have already invested in a project similar to RSB. Meridiam and SARA provided equity, and BNP Paribas provided debt for HDF's French Guiana CEOG project after conducting technical and contractual due diligence, which uses the same Renewable solution as RSB.

Selected Risk Factor 2

Category	Probability	Impact
<u>Credit</u>	<u>Low</u>	<u>Medium</u>
Description		
Cost competitiveness with other sources of power generation		

Mitigation Measure(s)

Despite lacking access to detailed dispatch profiles of BLPC, the Team estimates that the Project competes with a combination of baseload assets, HFO, diesel, and gas turbines, but is entirely carbon free, domestically produced, dispatchable, resilient, and with local return on investment.

It is also relevant that the proposed project is supported by the GoB and there is no BESS-only solution that is ready to be implemented, meaning that an opportunity cost should also be considered. HDF has spent development costs on this Project over the last five years to get to investment stage.

Selected Risk Factor 3

Category	Probability	Impact
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<u>Credit</u>	<u>Medium</u>	<u>Medium</u>
Description		
Sponsor, Engineering, Procurement and Construction (EPC) and interface risks. RSB will assume the construction cost and schedule risk. If there are delays in implementing the Project, RSB might be liable to pay LDs to BLPC.		
Mitigation Measure(s)		
RSB expects to implement a full wrap fixed price EPC with an experienced, creditworthy contractor. EPC includes a customary liability cap, security package, liquidated damages, and warranties. Additionally, HDF provides fuel cell technology and solar PVs to be provided by the supplier with a proven track record. Moreover, RSB plans to achieve financial closure before the 2023 reelection, which would require a competitive EPC contract tender in 2023. RSB has recruited talent in Barbados to facilitate and coordinate between international EPCs and local subcontractors to mitigate EPC risks.		
Selected Risk Factor 4		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
Ability of the plant to meet performance requirements over its 25-year life.		
Mitigation Measure(s)		
During their operational life, the performance of the components will degrade. Degradation warranties will be sought for all components (Solar PV, BESS, Electrolyzers, and fuel cells). As the lifetime of some parts is less than 25 years, the cost of timely replacement will be included in the operational costs. During the negotiation of the PPA, the performance targets and penalties should be reasonably aligned with the plant's capabilities.		
Selected Risk Factor 5		
Category	Probability	Impact
<u>Prohibited practices</u>	<u>Medium</u>	<u>Medium</u>
Description		
E&S Capacity and Management System: IFC will need to ensure that sufficient client capacity and commitment exists to manage E&S risks and impacts consistently with good international practice while maintaining early, continuous, open, and transparent engagement with stakeholders, avoiding potential conflict or opposition.		
Mitigation Measure(s)		
IFC will work with HDF to screen out and de-risk the Project of sensitive E&S issues during project design and planning phases of development, ensuring the Project works towards consistency with the IFC Performance Standards in the latter stages of the project cycle. IFC will work with the client to develop an ESMS and work on its implementation accordingly. In addition, through the E&S capacity-building advisory, IFC will work with HDF to identify the gaps in the ESMS using IFC ESMS Toolkit and provide a detailed Improvement Plan outlining precise actions, timeline, and staff resources needed to overcome		

these gaps. An ESIA was developed by an external consultant to outline risks and was reviewed by IFC for its thoroughness and rigor.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

This project is provisionally a Category B project: IFC E&S team conducted a preliminary assessment of key E&S risks and potential impact.

Preliminary key E&S risks include: i) Environmental and Social Impact Assessment (ESIA) in line with IFC's PSs prior to construction; ii) Client and contractor ESMS and organizational capacity for construction and operation; iii) management of labor and working conditions including contractor/third party management, migrant workers, Occupational Health and Safety (OHS), and Gender Based Violence (GBV); iv) supply chain risk management related to solar panels; v) pollution prevention; resource efficiency and management; waste and hazardous waste management; and storage safety of green hydrogen; vi) stakeholder engagement and grievance mechanisms; and vii) community health, safety and security.

RepRisk data showed no incidents in relation to the Sponsors of the project in relation to reputational risks on E&S.

IBAT assessment showed that the project is located in a modified habitat but less than one kilometer from an Important Bird and Biodiversity Area (IBA). A Biodiversity Specialist will join the project to understand if there are any material risks to the IBA (due to water abstraction) or to avifauna (due to the powerline).

The project is part of upstream portfolio (46174) with several E&S studies under revision and E&S trainings undertaken. Training in relation to IFC's E&S Performance Standards was given to all project team members as well as senior management in January 2023 by IFC. An additional training on how to develop an Environmental and Social Management System (ESMS) appropriately will be conducted in July 2023 by IFC. Additional studies that have been completed by the client for IFC include:

- The Environmental and Social Impact Assessment (ESIA)
 - o The ESIA was reviewed by IFC, and updates were requested including potential alternative options to water to be incorporated as a separate study. The current water procurement through the Barbados Water Authority is viable and provides sufficient water with the specifications and terms required by the project. Any option identified by the study on alternative water sources can be taken into consideration to improve the water procurement arrangement, if appropriate.
 - o In consideration of the GCF definition of "Indigenous People," available at paragraphs 13-16 of the GCF Indigenous Peoples Policy, there are no indigenous people in the areas targeted by the project.
 - o HDF held a public consultation meeting in December 2022 on the ESIA with the stakeholders and community members of the project to explain the project components and all potential impacts. IFC was present at the stakeholder public consultation meeting. The meeting was recorded and made available online on YouTube for full public disclosure. Stakeholder consultation was done in accordance with IFC's Sustainability Framework.
- Life Cycle Assessment study
 - o The LCA assesses the impacts of major components such as solar panels, batteries, etc. throughout the entire supply chain. This is done to ensure supply chain risks are also taken

into consideration during procurement of equipment such as solar panels to mitigate E&S risks of forced and child labor.

- Drainage Study
 - o The study looks at the scope of drainage topographically based on 50-year storms which are becoming more frequent. The study explores potential route directions for stormwater, including mitigation options such as suck wells in various locations to avoid impact with project site, as well as neighboring communities. This study also explores options for rainwater harvesting techniques to re-incorporate rainwater into the project.
- Climate Change Risk Assessment
 - o This study looks at a detailed climate change risk assessment including impacts from hurricanes, storms and their frequency. It also examines water allocation and distribution across the country and its impact from climate change over the years. As the project site is inland and not coastal, it depends on water from the national grid which may potentially be impacted from climate change as the island of Barbados is already water scarce.
 - o The sponsors contemplate a specific training in SEAH that has also been included in the updated Draft Final Gender Action Plan and will also be included as part of the risk assessment in the updated ESMP. The aim is that all the RSB project staff including EPC contractors will benefit from specific training in these areas (SEAH and SGBV) as well as within the project's updated Standard Operating Procedures that will now include a gender focus/lens.

The client has shown willingness to improve and implement IFC's E&S Performance Standards as well as updating and revising the ESIA and necessary E&S studies for the project. These studies will be provided in Annex 6.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

The Project plans to play a key role not only in the development of the surrounding communities and vulnerable population, but specifically in the empowerment of women. As detailed in the Gender Action Plan, the Project plans to focus on implementing programs to help improve gender equality and to ensure that female and male headed households have access to a more resilient energy supply. In addition to residential customers, the gender action plan also targets small farmers, business owners/operators, temporary field workers and other vulnerable populations.

The Gender Action Plan (annex 8a) developed by the Project, informed by the Gender Assessment (annex 8), details specific objectives, outcomes, and activities that will be followed to improve gender equality in the Project and in the area around the RSB site. In implementing successful community engagement, an important goal of the Project will be to empower local women through increased awareness and understanding of the RSB project and its contribution to enhance energy resilience.

As detailed in annex 8, the Project aims to respond to gender issues by implementing at least five gender responsive interventions (outputs) throughout the life of the RSB project (Outputs 1, 2, 3, 4 and 7).

These outputs will aim at addressing gender issues as follows:

- Output 1: Gender-specific gaps and requirements are identified and incorporated into the Renewstable Barbados Project

- Output 1.1 - Ensure findings of the Gender Assessment are incorporated into the Project's Results Management Framework
- Output 1.2 - Ensure that the project's Environmental & Social Management Plan (ESMP) promotes gender-sensitive actions through the project's life cycle.
- Output 1.3 - Ensure updates to the project's Standard Operating Procedures (SOPs) and (For example; EPC Contracts, Labour Management Plans, etc) address the requirements of gender as well as sexual exploitation, abuse and sexual harassment (SEAH), sexual and gender-based violence (SGBV).
- Output 1.4 - Conduct training with the RSB Project staff on the updated SOPs that include gender considerations.
- Output 1.6 - Organize experience sharing workshops with key government agencies and partners. By engaging government and non-government stakeholders and sharing the importance of integrating gender considerations into renewable energy projects opportunities may be created to influence policy development and decision-making processes.
- Output 2: Enhance capacity of the RSB project staff (including all project-related contractors and employees) to integrate gender considerations/ requirements in renewable energy projects
Activities Indicators and Targets Timelines
 - Output 2.1 - Establish a MOU between HDF Caribbean and tertiary educational institutions to support the training, skill development and capacity building of all RSB Project staff in gender sensitivity, SEAH and SGBV training
 - Output 2.2 - Develop a project-specific SEAH and SGBV Training Program to reflect Gender Sensitivity in Workspaces for all RSB Project employees (including contractors, construction workers and other related employees)
 - Output 2.3 - Develop online self-paced mandatory SEAH and SGBV training to reflect Gender Sensitivity in Workspaces for all RSB Project employees (including contractors and employees)
- Output 3: Increase participation of women performing technical roles in the renewable energy sector
 - Output 3.1 - Develop in collaboration with relevant training institutions N/CVQ in hydrogen energy production, storage and use
 - Output 3.2 - Design in collaboration with relevant training institutions the RSB-Young Professional Energy Sector Internship Program with specific emphasis on encouraging female participation in the sector via technical roles.
 - Output 3.3 - Implement in collaboration with relevant training institutions the above internship program (USD 16,200 stipend co-financing)
 - Output 3.4 - Collaborate with the Ministries of Education and Energy to host public education campaigns/workshops which encourage the participation of girls and young women to aspire to leadership and technical roles in the renewable energy sector.
- Output 4: Conduct gender sensitive stakeholder consultations and awareness-raising programs to ensure public knowledge and buy-in on the use of the RSB hydrogen energy solution and potential impacts.
 - Output 4.1 - Conduct gender sensitive stakeholder consultations
 - Output 4.2 - Update the RSB Project webpage to host communication outputs including a Grievance Complaints Section related to the RSB Project
 - Output 4.3 - Develop a Community and Social Engagement Benefits Sharing Plan which takes on board gender considerations in the development of a community-based program to benefit residents within St. Philip
- Output 5: Increase capacity of the RSB Project Management Team to respond to complaints from community members and other stakeholders

- Output 5.1 - Develop and implement a detailed community-based Grievance Redress Mechanism (GRM) Plan that is (a) easily accessible by all stakeholder groups including women, (b) legitimate, (c) transparent, (d) culturally appropriate, and (e) confidential.
- Output 6: Female and male heads of households, small business owners and other vulnerable populations have an increased awareness and understanding of the use and application of the RSB project in enhancing energy resilience
 - Output 6.1 - Develop stakeholder specific handbooks/tool kits on the benefits and impacts of hydrogen energy production.
 - Output 6.2 - Conduct education awareness sessions with key stakeholders (including women) on the benefits and impacts of hydrogen energy production using the handbooks and toolkits developed.
- Output 7: Female and male heads of households, small farmers, business owners and other vulnerable populations have an increased understanding of how the demineralize water produced by the hydrogen energy plant or rainwater harvesting can be used for irrigation and other ancillary usage
 - Output 7.1 - Undertake sensitization and awareness-raising sessions targeting community members, small farmers and business owners and operators and other vulnerable populations
 - Output 7.2 - Provide training in the rainwater harvesting techniques targeting small farmers, community members, business owners and operators and other vulnerable populations
- Output 8: Small farmers and community members have an increased understanding of how hydrogen energy production can support sustainable agriculture including sustainable farming techniques
 - Output 8.1 - Undertake education and awareness-raising programs with small farmers and industry stakeholders in the agricultural sector on the synergies between renewable energy and the sustainable agricultural sector
- Output 9: Enabling a gender-responsive project management and governance environment for the duration of the Project
 - Output 9.1 - Establish an Inclusive and Participatory Steering Committee (PSC) that includes the Institute for Gender and Development Studies (IDGS) at the UWI Cave Hill Campus or Bureau of Gender Affairs (BGA)
 - Output 9.2 - Conduct gender-sensitivity training with the RSB Project Management Team to ensure gender dimensions are effectively considered throughout the duration of the project. Topics to be included in the training include gender-responsive behavior and interactions, avoiding sexual harassment, and conflict-resolution skills.
 - Output 9.3 - Present annually to the PSC to report on the status and implementation of the Gender Action Plan (GAP)

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

As a lender to the Project, IFC will conduct a due diligence assessment prior to submitting the proposal for approval to IFC's management and Board of Directors. This due diligence will cover financial and credit aspects, but also other non-financial matters such as technical, environmental, social, etc. IFC will also follow its standard procedures to ensure that conditions of disbursement have been met prior to disbursing the funds. IFC also expects to be the mandated lead arranger (MLA) in addition to the AE for the GCF concessional funds, and as such will coordinate with all lenders for a pari-passu disbursement as well as for supervision and monitoring after disbursement. The loan agreements will contain requirements for regular reporting from the client. This reporting may contain among others quarterly financial statements, project progress reports (during construction), annual audits/annual reports, insurance policies,

environmental and social annual monitoring reports (including environmental and social action plan status report), gender action plan implementation status, management letter, compliance certificate, etc. Site visits may also be part of the supervision activities of the Project both during construction and operations. As is customary with all IFC loans, the agreement will include provisions requiring the borrower not to incur in Sanctionable Practices (ie. any Corrupt Practice, Fraudulent Practice, Coercive Practice, Collusive Practice, or Obstructive Practice, in accordance with the *Anti-Corruption Guidelines for IFC Transactions*). and to cooperate in good faith with IFC if there is a concern that a violation has occurred.

Procurement of major components of the Project are detailed in annex 10. The sponsors will manage the procurement process and plan to follow a competitive tender in most cases. One of the sponsors, HDF, is building a fuel-cell manufacturing facility in Bordeaux under license of renowned fuel-cell manufacturer Ballard. Fuel cells are therefore supplied by HDF Energy. An EPC contractor will be engaged to lead the implementation of the project including procurement of long-term service agreements for main equipment such as electrolyzer and batteries. Even though procurement of most of the main components of the Project will follow a competitive tender, IFC usually does not require that procurement follows World Bank Group procedures or even that a competitive tender takes place, when dealing with private sector players (unlike when working with state-owned enterprises or local governments) as long as the transaction is at arms-length. Part of the due diligence will involve making sure that the procurement of the fuel cells is at arms-length and that pricing is reasonable compared to similar fuel cells in the market.

G.4. Disclosure of funding proposal

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☐ Annex 2 Feasibility study - and a market study, if applicable
- ☐ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☐ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☐ Annex 5 Implementation timetable including key project/program milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☒ Environmental and Social Impact Assessment (ESIA) or
 - ☒ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People’s Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/program-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☐ Annex 10 Procurement plan [\(template provided\)](#)
- ☐ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☐ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☐ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/program information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment

- | | | |
|-------------------------------------|----------|--|
| ☐ | Annex 21 | Operations manual (Operations and maintenance) |
| ☐ | Annex 22 | Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects) |
| ☐ | Annex 23 | Drainage Study |
| ☒ | Annex 24 | Climate Change Risk Assessment |
| ☒ | Annex 25 | Technical Assistance scope |



GOVERNMENT OF BARBADOS

The Hon. Ryan Straughn, M.P.

MINISTER IN THE MINISTRY OF FINANCE & ECONOMIC AFFAIRS

Ministry of Finance, Economic Affairs and Investment

Government Headquarters,

Bay Street, St. Michael, BB11156, Barbados, W.I.

Our Ref.: 7045/79/1 Vol. 5

Tel.: (246)535-5300

Fax: (246)535-5629

Date: March 1, 2023

Mr. Yannick Glemarec
Executive Director
The Green Climate Fund
Songdo Business District Incheon
REPUBLIC OF KOREA

Dear Mr. Glemarec,

**Funding Proposal for the Green Climate Fund by
International Finance Corporation, regarding Renewstable
Barbados Project**

We refer to the project titled Renewstable Barbados as included in the funding proposal submitted by the International Finance Corporation to us on February 20, 2023.

The undersigned is the duly authorized representative, Hon. Ryan Straughn, National Designated Authority of Barbados.

Pursuant to the Green Climate Fund (GCF) decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) the Government of Barbados has no-objection to the project as included in the funding proposal;

- (b) the project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Barbados; and
- (c) in accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours sincerely,

A handwritten signature in black ink, appearing to read 'Ryan Straughn', with a long horizontal flourish extending to the right.

HON. RYAN STRAUGHN, M.P.

Minister

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Renewstable Barbados Project
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Private
Accredited entity	International Finance Corporation (IFC)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Harrow, Saint Phillip Parish region in Barbados
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Monday, September 18, 2023
Language(s) of disclosure	English
Explanation on language	English is the official language of Barbados.
Link to disclosure	https://www.renewstable-barbados.com/files/ugd/22dce1_ef2c386467dc4fd78fdcc15c17177f01.pdf
Other link(s)	IFC website: https://www.ifc.org/en/what-we-do/sector-expertise/blended-finance/climate/green-climate-fund Project Planning and Documents: https://www.renewstable-barbados.com/planning ESIA Annexes including ESMP (Annex I): https://www.renewstable-barbados.com/files/ugd/22dce1_73a3a52d72104edba97894ccaf275623.pdf
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the ESIA.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Monday, September 18, 2023
Language(s) of disclosure	English
Explanation on language	English is the official language of Barbados.
Link to disclosure	https://www.renewstable-barbados.com/files/ugd/22dce1_ef2c386467dc4fd78fdcc15c17177f01.pdf
Other link(s)	IFC website: https://www.ifc.org/en/what-we-do/sector-expertise/blended-finance/climate/green-climate-fund

	ESIA Annexes including ESMP (Annex I): https://www.renewstable-barbados.com/files/ugd/22dce1_73a3a52d72104edba97894ccaf275623.pdf
Remarks	An ESMP consistent with the requirements for a Category B project is contained in Annex I to the ESIA
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Tuesday, November 8, 2022
Place	ESIA/ESMP were made available to community for public viewing digitally at RSB website at https://renewstable-barbados.com and physically at (a) Six Roads Public Library, Highway 5, Six Roads, St Philip, Barbados; (b) Bridgetown Public Library, Fairchild St., Bridgetown, Barbados; (c) Planning and Development Department, Warrens Office Complex, Warrens, St Michael, Barbados. Public Consultation advertised in local media and online (https://www.renewstable-barbados.com/copie-de-home) on Wednesday, November 9, 2022. Public Consultation conducted at Princess Margaret Secondary School, Six Roads, St. Philip and virtually via live stream on December 7, 2022.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	Wednesday, November 15, 2023
Date of GCF's Board meeting	Monday, October 23, 2023

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP224

Proposal name:	Renewstable Barbados Project
Accredited entity:	International Finance Corporation
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. Renewstable Barbados is an innovative stable energy provision project based on a 50 Megawatt peak (MWp) solar photovoltaic (PV) system combined with a battery energy storage system (BESS) and green hydrogen storage. This combination will allow continuous and stable electricity provision 24/7 with 13 MW during the day and the evening peak between 7 p.m. and 9 p.m., and 3 MW at other times. The core feature of the project is that the solar PV system will allow for a guaranteed and stable provision of electricity similar to a base-load system. This is achieved through both storage in the BESS to cater for short-term fluctuations during the day and to provide electricity during the evening peak, and through the production and storage of green hydrogen which acts as long-term storage for provision of electricity during the night, or also as reserve power. Such systems are central to achieving high rates of renewables based on intermittent energy sources, in particular for small island states with limited grid interchange options. The project has opted for a hybrid storage system using lithium-ion (Li-ion) batteries combined with hydrogen (H₂) storage for longer periods which is a promising technical solution. The project is of high relevance for countries, especially small island developing States (SIDS), to achieve high or even 100 per cent renewable electricity production and is one of the first large-scale systems of this kind worldwide.

2. The USD 169 million project has a finance source of: (i) USD 41 million from GCF composed of a concessional loan of USD 40 million and a grant of USD 1 million; (ii) USD 43 million of equity from the private project sponsor; (iii) USD 8.5 million loan from the International Finance Corporation (IFC); (iv) USD 8.5 million loan from the Inter-American Development Bank (IDB) and (v) USD 68 million loan from a local commercial bank. The project is private sector-driven and GCF funding leverages an inflow of private capital, including private equity, as well as significant involvement by a local commercial bank into an important climate change mitigation project. The private executing entity, Renewstable Barbados, is composed of Rubis Caribbean Holding (51 per cent) and Hydrogène de France (HDF) (49 per cent). All the equity needs are already committed by the existing shareholders. HDF is in discussion with local credit unions to acquire up to 30 per cent of the project equity at or after financial close. GCF loan proceeds are used to reduce the tariff for energy customers and will not alter the financial internal rate on equity for the project sponsor. GCF grant proceeds are used for capacity-building and knowledge-sharing.

1.2 Impact potential

Scale: High

3. The programme has basically a mitigation impact but, as a side effect, it also increases the resilience of energy provision on the island. The mitigation impact is estimated at around

693,000 tons of carbon dioxide equivalent (t CO₂eq) reduced over the 25-year lifespan of the project.

4. The mitigation outcome is achieved by providing stable carbon-free base-load power to the grid. The project has a capacity factor or energy target compliance of 85 per cent equivalent to a conventional thermal power plant. The 50 MWp solar plant delivers from 8 a.m. to 5 p.m. and again from 7 p.m. to 9 p.m. 13MW and in the remaining hours provides 3 MW of base-load power equivalent to maximum 182 megawatt-hours (MWh) of electricity delivered per day, or on average (using the average availability rate) 57 gigawatt-hours per annum. During the day, the PV plant provides direct electricity, feeds the 15 MW/30 MWh Li-ion battery storage system and produces green H₂ for a 90 MWh hydrogen storage system. The BESS provides the end-of-the-day peak power and, in combination with the hydrogen storage, ensures the stability of the electricity service. In contrast to standard PV systems the proposed project allows for a base load and not an intermittent energy provision. Currently over 90 per cent of electricity provision in Barbados is based on fossil generation and the country has set the high target of achieving 100 per cent renewable electricity (RE) generation by 2030. This project constitutes a critical piece of the puzzle to achieve such a target, not so much for its RE production level but rather for its storage and grid stabilization approach. Storage facilities and base-load systems are a critical component to achieve high levels of RE production.

5. The greenhouse gas (GHG) mitigation impact was calculated using the United Nations Framework Convention on Climate Change registered clean development mechanism methodologies AMS-I.D. ("grid-connected renewable electricity generation"). The source used for the carbon grid factor for Barbados is based on the harmonized International Financial Institutions (IFI) Default Grid Factors 2021 v3.2. for "firm energy provision"¹ which is 484 grams of carbon dioxide equivalent per kilowatt-hour (g CO₂eq/kWh). The projected GHG emission reductions are deemed to have been calculated in a conservative manner.

6. The project is designed to be resilient to extreme weather events and a co-benefit is that the energy provision can be made more resilient to climate change as the system can operate in an "island mode" if disconnected from the rest of the generation plants, allowing for continued access to electricity for residents or critical local services, benefiting more than 21,000 people. The ability to store hydrogen for longer potential outage periods, for example, caused by hurricanes, also increases resilience of the power sector to extreme climate events. The hydrogen storage capacity could be increased in the future in a cost-effective manner by adding more hydrogen storage tanks which would enable maintenance of base electricity provision through a longer period, if required.

7. The stand-alone direct impact of the project is limited. The main impact is indirect in nature, by serving as a lighthouse for SIDS as well as many other countries that want to achieve high or very high levels of renewables in the future. Storage, base-load provision, stable supply and grid stability are core factors to achieve such a target. Barbados, similar to many other SIDS, has abundant solar resources with high irradiation and capacity factors; however, adding increasing numbers of solar plants results in problems of grid stability and base-load power provision. The technology approach chosen by the project is one of various possibilities. Results of the project are critical to design optimal solutions for other countries.² Only very few large-scale base-load and storage projects with solar PV exist today and the medium- and long-term indirect impact of the project is thus considered to be very high.

8. Overall, the Independent Technical Advisory Panel (TAP) considers that the project has a high mitigation impact.

¹ Based on recommendations of the independent Technical Advisory Panel the GHG calculation methodology was updated to firm energy provision reducing the GHG impact from the initially estimated 930,000 tCO₂ to 693,000 tCO₂.

² Solutions for other countries might or might not use the hybrid approach using batteries and H₂ storage.

1.3 Paradigm shift potential

Scale: High

9. Barbados relies currently on over 90 per cent fossil fuel-based electricity generation. The country has the intention to shift to 100 per cent renewables by 2030. No large-scale storage and base-load plant from renewables currently exists in Barbados.
10. The major paradigm shift is not so much towards renewable energy, although this alone is already a major change compared to the current situation. The real paradigm shift is the change from solar PV being an intermittent source of energy that potentially creates grid stability problems, to stable base-load energy provision. The proposed energy storage solution adopts a hybrid, combination of a BESS with H₂ storage. It is the first of its kind in the Caribbean and one of the only plants worldwide at this scale. A large investment will have to be made in RE base-load provision and storage to achieve the RE targets of countries. The project plays a critical role in assessing and choosing optimal technology solutions to enable this paradigm shift.
11. The approach of combining H₂ and a BESS is one possible technological solution. Another option, already implemented in various countries, is to use only a BESS. This system is more energy efficient as producing H₂ and then generating electricity from the H₂, results in larger energy losses than using batteries. At current costs, a hybrid approach is also more expensive compared to adopting only the BESS. However, a hybrid system has other advantages related to flexibility and reduced impact of battery degradation over time. Trialling different technological approaches in energy storage is considered a positive contribution to identifying and choosing optimal storage and base-load systems relative to country requirements and situations in the future. However, information, capacity-building and training should be technology-neutral for storage solutions and should not be exclusive or uniquely prefer the proposed hybrid solution.
12. The project benefits critical experience in green hydrogen production and storage. Green hydrogen can play an important role to decarbonize short- and medium-haul maritime transport, while electro-fuels (e-fuels) such as e-liquefied natural gas or e-methanol can also decarbonize long-haul maritime and aviation services that are critical for islands. However, the independent TAP believes it is important that hydrogen is promoted only in areas where it has an environmental, technical, and financial advantage compared to other solutions. In the case of Barbados, battery electric vehicles which have a factor of three to four times higher efficiency in energy usage compared to H₂ vehicles, are better suited for all road transport modes, while H₂ or e-fuels based on green H₂ are an option for maritime shipping and aviation. The independent TAP wants to avoid GCF grant resources being used for an indiscriminate promotion of hydrogen technology.
13. Knowledge-sharing is critical for a paradigm shift. The project includes measures for capacity-building and development of skills. Information, know-how and capacity-building should however be technology neutral and should not be steered by the investor with vested interests in the H₂ economy.
14. The project is private sector with a private company taking the investment risk. Private sector capital is utilized for renewable energy storage solutions. This is considered as highly positive and with a high replicability potential by showcasing that renewable energy base-load and storage provision can be delivered in a cost-effective manner with private capital.
15. Overall, the independent TAP considers that the project has the potential to attract more private companies and capital to invest in storage and base-load RE provision. Renewable energy storage and base-load provision are critical, and the proposed system creates a paradigm shift. The independent TAP therefore considers that the paradigm shift potential of the project is high.

1.4 Sustainable development potential

Scale: Low to Medium

16. The project is aligned with several Sustainable Development Goals: SDG 3 (good health and well-being); SDG 7 (affordable and clean energy); SDG 8 (decent work and economic growth); and SDG 13 (climate action).
17. The project has a positive impact on reducing local air pollution and noise due to reducing fossil electricity generation. However, the impact is considered to be minor, as fossil generators are not located in areas with high population density and, compared to the air pollution impacts caused by the transport sector, the health and environmental impact of fossil generators is limited.
18. The project has a positive economic impact by using locally available resources and by reducing the need for import of fossil fuels. This increases the sustainability of the economy of Barbados. No impact on job creation is expected as the project will potentially create a similar number of jobs to those lost in fossil-based electricity generation. This is also true for the planned sheep-farming initiative which replaces jobs of former agricultural activities on the land (sugarcane farming).
19. The major potential negative impacts of the project are associated with end-of-life hazardous products (solar PV systems and batteries). These will be taken care of through extended producer responsibilities ensuring the used products are shipped to the country of origin or to a closer recycling facility. Freshwater availability for H₂ production is secured and is at a very low level compared to current usage and water availability rates. The land used for establishment of the solar PV system is currently used for sugarcane farming which is a declining crop industry in Barbados. A sheep-farming initiative is being planned as a social enterprise, thus replacing a declining agribusiness activity with one that is more attractive financially. This avoids potential negative land-use change impacts.³
20. Overall, the sustainable development potential of the project is considered low to medium. It can make important contributions in the area of reducing dependency on fossil fuel imports while the major focus and impact of the project is clearly on mitigating climate change.

1.5 Needs of the recipient

Scale: Medium

21. The Barbados electricity grid is vulnerable to hurricanes. Since 2010, hurricanes Tomas (2010), Ernesto (2012), Harvey (2017) and Elsa (2021) and the tropical storms Matthew, Maria, Kirk, and Gonzalo (2020) have impacted the island, causing power outages and enormous damage to the people of Barbados. The project allows for a more resilient energy system with a more stable and reliable energy supply and can thus reduce the impact of climate events.
22. Comparable to other SIDS, the electricity tariffs are already very high. Barbados wants to decarbonize energy generation but has a low potential for base-load renewable sources such as geothermal,⁴ hydro or biomass. While solar PV technology has a downward cost trend, energy storage is still very costly and would result in additional costs. The higher cost and the risk of hybrid storage technologies make commercial financing very unlikely to materialize at this stage. The support of GCF alongside development financial institutions such as IFC and IDB Invest help to establish a project that will be an important reference case for future storage investments in other countries and which will increasingly be financed by commercial banks. The use of concessional GCF funds to reduce energy costs for the population while also venturing into a renewable energy provision combined with storage enables uptake of the technology. The independent TAP therefore considers the needs of the recipient as medium.

³ A similar arrangement is already in place at the largest solar facility in Barbados located in Trens, Saint Lucy.

⁴ See e.g. [Barbados Central America and the Caribbean RE SP.pdf \(irena.org\)](#).

1.6 Country ownership

Scale: high

23. The energy sector is the largest GHG emission source in Barbados. The Barbados National Energy Policy targets achieving a fossil fuel-free electricity sector by 2030. This private sector project is fully aligned with the government's ambitious objective to increase the share of electricity produced from renewable sources. It is also in line with the target to reduce the country's vulnerability to extreme climate events.
24. The no-objection letter of the Government of Barbados has been provided.
25. The project has made a stakeholder assessment and has responded to the issues raised during the assessment in a satisfactory manner.
26. The country ownership is assessed by the independent TAP as high. The project fully aligns with the country's nationally determined contribution and its strategic national plans.

1.7 Efficiency and effectiveness

Scale: Medium

27. An alternative viable storage solution for Barbados would be based on BESS alone. A battery-based solution with current prices is expected to have a lower overall cost due to the higher efficiency of batteries relative to usage of hydrogen. This is confirmed by an external Det Norske Veritas (DNV) study (annex 28). However, the two systems are not entirely comparable due to battery degradation (resulting in a faster reduction of storage capabilities than with a hybrid system) and a lower potential level of medium-term storage in case of extreme events. The proposed system therefore has its merits but care needs to be taken to have a technology-neutral approach for replication and outreach without promoting indiscriminately a hybrid or H₂-based solution.
28. Green hydrogen production and storage opens the door for future usage in difficult-to-abate transport sectors in the country, especially for medium- and long-haul maritime transport. Generating experience with a hybrid system is thus considered to be an economic asset in itself, warranting some additional costs compared to an alternative of only BESS.⁵ The power purchase agreement costs of the project of around USD 250 per MWh have been compared by IFC with other projects worldwide and are deemed as competitive. Private investment is also available for the hybrid system proposed and not for an alternative BESS alone.
29. The concessionality of the GCF loan is intended to reduce the energy tariff and not to increase the profit margins of the project sponsor. This will only be true if capital expenditure costs of the system are not overrated and are market conform. As project sponsor, HDF will supply the fuel cells. The AE needs to ensure that fuel cells provided are cost competitive to avoid siphoning off concessionality of GCF funds.
30. The total investment amount is USD 169 million with USD 41 million from GCF with a co-financing ratio of 76 per cent and a leverage factor exceeding 3.
31. The estimated GCF cost per t CO₂ eq avoided is around USD 59. While this cost may seem high, the project technology is novel and storage costs are still high. Also, due to the project providing baseload and storage, other RE projects can be established in Barbados. Stand-alone RE projects to a certain extent underestimate the cost per tonnes of CO₂ avoided as they do not factor in the costs of grid reinforcement and storage which is required to achieve high degrees of RE provision based on intermittent energy supply. The GCF funding also has important adaptation co-benefits in increasing the resilience of the energy system in Barbados. The

⁵ The cost comparison between a BESS only and a hybrid system depends on various critical parameters and results vary from comparable costs to 15 per cent lower costs for BESS only. As mentioned, the two systems do not have identical output parameters which means that differential costs can also be attributed to differential outputs.

replicability potential of this project is considered to be very high, with future decreasing unit costs. Therefore, the specific project cost is not considered to be excessive.

32. IFC has experience and together with IDB also proven capabilities to make such an endeavour successful and to achieve replicability. HDF has proven experience with hydrogen systems. Core components, such as the BESS, the electrolyzers and fuel cells, will be procured from a tier-1 supplier with performance guarantees.

33. The overall the independent TAP assessment concerning efficiency and effectiveness is medium as the proposed technology has considerable potential to be a future game changer for countries targeting high levels of RE and is crowding in significant additional private finance. On the other hand, other technology solutions could be more cost-effective.

II. Overall remarks from the independent Technical Advisory Panel

34. Based on the assessment above, the independent TAP endorses this funding proposal with conditions and a recommendation.

35. The project is financed to a considerable extent by private sources and implements an innovative base-load energy plant using a solar PV system combined with a hybrid storage solution with a BESS and H₂ component which allow delivery of stable electricity to the grid. This is particularly important for Barbados where the RE potential is primarily solar PV which can only deliver intermittent and unstable energy thus requiring cost-effective storage and grid-stabling solutions.

36. The independent TAP recommends that the project reports data on energy usage and inputs/outputs for the solar energy system, the BESS and the H₂ system. This would enable future optimal design of systems. The claim of the project sponsor that this information is intellectual property is not fully shared. Disclosure of this information would be valuable and would increase the project impact. Non-disclosure of the information should entail a lower concessionality level of GCF funds as this reduces the project impact and paradigm shift potential.

37. The independent TAP recommends that TA funds are used as far as possible in a technology neutral manner including (i) focusing scholarships on energy storage technologies in general and not only on potential usage of hydrogen; (ii) financing and promoting studies of energy storage options for SIDS in general without focusing exclusively on hybrid or H₂ solutions; (iii) not financing directly H₂ platforms but technology open platforms.

38. The independent TAP recommends that no GCF Proceeds under Component 2 of the Funded Activity are used for the training and/or capacity building of RSB's plant technicians.

39. The AE needs to ensure that GCF concessionality is used to reduce energy tariffs and not siphoned off to potentially overpriced components provided by the project sponsor. This has been acknowledged by the IFC without disclosing to the independent TAP the required documentation describing how this will be achieved.

40. Therefore the independent TAP endorses this funding proposal subject to the condition: Prior to the effectiveness of the FAA, delivery by the AE to the Fund, in form and substance satisfactory to the GCF Secretariat, of the procedures setting out the methodology used to ensure cost competitiveness of the fuel cells to be supplied by HDF.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP224)

Proposal name:	Renewstable Barbados Project
Accredited entity:	International Finance Corporation (IFC)
Country/(ies):	Barbados
Project/Programme size:	Medium

Impact potential

We thank the panel for this assessment. We agree that the impact potential of the Project is high. The input from the Independent Technical Advisory Panel (TAP) regarding the GHG emissions calculations were very useful and relevant to revise the emission factor based on the firm energy provision of the proposed plant.

We concur that there will be significant indirect impacts as a result of the Project in terms of replicability and providing a demonstrative example to other Small Island Developing States (SIDS). The stand-alone, direct impact of the Project is also worth emphasizing: the plant has the potential to generate about 6% of the energy dispatched to the Barbadian network using local, renewable energy and without a negative impact on the grid stability.

Paradigm shift potential

We thank the panel for this assessment. We fully agree with the independent TAP that the paradigm shift goes beyond renewable energy by itself and relates more to using an intermittent source of energy to provide a firm, base-load dispatch that avoids grid stability problems. Regarding the comparative cost against a battery-only solution, while initial indications from an independent study commissioned by the AE pointed to a pricing gap favoring a Battery Energy Storage Systems (BESS)-only solution, further iterations aimed at providing the same flexibility that the hybrid H2 + BESS system yielded an overlapping range of levelized cost of electricity (LCOE) for both alternatives.

We also agree that different technological approaches to energy storage are a positive development and the intent of using GCF concessional funds is not to support the hydrogen technology as a preferred solution, but rather to make the best use of the opportunity to learn from this Project and add this innovative solution to the menu of potential approaches that will help support the paradigm shift desired. As indicated by the panel, private sector participation will be key in the future development of similar solutions and the support provided by GCF to this project will help catalyze future opportunities that will gradually need less and eventually may not need any concessional support to yield an attractive tariff.

Sustainable development potential

We thank the panel for this assessment. IFC's expectation is that the Project will have relevant positive impact in various fronts, while minimizing the negative impact at the end of its life. Moreover, the impact on climate and health by avoiding local air pollution and noise should be considered in the context of an island such as Barbados where the relative impact is more

significant than the absolute effect. As compared to the transport sector, the electricity and heat sector in Barbados emits 37% more, according to:

<https://ourworldindata.org/co2/country/barbados>.

Furthermore, the Project will increase the share of renewable energy in the Barbadian generation mix. The current share of renewables in Barbados is limited by the variable nature of solar PV but the introduction of renewables into the baseload through short- and medium-term storage enables increased renewable energy in the mix. This could be replicated also in the future. Therefore, this is aligned with SDG 7 - to provide affordable, reliable, sustainable, and modern energy for all - with a core target for increasing the share of renewable energy in the mix.

Needs of the recipient

We thank the panel for this assessment. The panel clearly identifies the resiliency aspects of the Project as a key component of the recipient's needs. We would like to also highlight how this Project helps address curtailment and stability issues that the country is already facing even with a relatively small installed capacity of solar PV generation. Given the Government of Barbados' ambitious decarbonization targets, replacing old fossil fuel generation facilities is becoming an issue given the reluctance to install other diesel generators that will be emitting greenhouse gases and other pollutants for a long time. At the same time, the country is facing challenges to increase penetration of intermittent renewable energy without negatively affecting the stability of the grid.

While energy storage is still costly, this is not exclusively related to the financing of electricity storage using green hydrogen. Given the island context of Barbados, in order to increase the penetration of renewables, storage is required. And due to the high fuel costs in Barbados, renewables with storage is cost competitive with diesel plants, which is not the case in other locations.

Country ownership

Thank you for this assessment. We fully agree and have been able to hear firsthand the interest of various public officials in seeing this Project move ahead and close as soon as possible.

Efficiency and effectiveness

We appreciate the independent TAP's assessment. While the roundtrip efficiency of a BESS-only system is higher than a system that combines H2 with batteries, as the panel indicates, the latter provides more flexibility regarding medium-term storage and resilience in case of extreme weather events. In fact, the higher LCOE cost of the hybrid system is closer and even as competitive as the BESS-only system under scenarios where the batteries-only system is sized to replicate the hybrid system, as was evaluated by DNV in a revised version of the comparative study.

It is also important to highlight the location-specific nature of this solution which can work in Barbados due to high fuel costs, but probably not in economies where the electricity tariff is more competitive.

Overall remarks from the independent Technical Advisory Panel:

Thank you for your endorsement to proceed to the GCF Board and we acknowledge your recommendations and condition.

We take note that the panel does not fully share the view that information on input/output of each component of the system (solar PV, Li-ion batteries, and green hydrogen system) is intellectual property and subject to confidentiality. IFC's view is that for the most part, the project impact is achieved regardless of the disclosure of this information and the potential for replicability and paradigm shift are likely subject to other developers with knowledge of the technology or of other technologies (following your comments on maintaining a technology neutral approach) to be able to develop other projects without the need of the level of detail suggested by the panel. Given that the concessionality is fully allocated to reduce the tariff, the rationale of linking the level of concessionality to the provision of information specific to the Project is unclear.

We also take note of the recommendation to keep the TA activities open to a technology-neutral approach. While we are supportive of keeping the TA work on a technology-agnostic basis, we will need to consider the amount of resources available as compared to a broader scope of work, as well as the immediacy of the hydrogen technology from the Project to be leveraged by the TA work.

We acknowledge the recommendation to not use GCF proceeds for the TA work to train or provide capacity building to RSB employees. If there is a restriction on using GCF proceeds for this purpose, we will work with the EE in adjusting the scope of work accordingly.

We understand GCF's interest in ensuring that the concessionality of the funding is fully allocated to reducing the energy tariff. As explained by the AE, the tariff for the life of the Project is determined up-front and takes into consideration the impact of the concessionality in reducing the Project costs, rather than being channeled to benefit the project sponsors. The relevant information has been made available to the independent TAP and GCF subsequent to the submission of the Funding Proposal to the independent TAP.

Thank you for your endorsement. Based on our understanding of the requirement for the condition precedent proposed by the independent TAP, we have already provided the TORs to address this request. We look forward to the consideration of the Renewable Barbados Project at the GCF Board in October 23-25, 2023.

The Green Climate Fund Gender Assessment

RENEWSTABLE BARBADOS (RSB)



Rev	Issue Date	Originated by	Reviewed by	Approved by	Revision
1	28 July 2023	M. Clarke	J. Beckles; L. Theron	H. Frossard	Third Issue--Final

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We are very grateful for Ms. Jacqueline Beckles, Project Manager for Development and Mr. Leo Theron, Project Manager, Hydrogen Projects, both of HDF Energy, France for their technical inputs and review of the draft Gender Assessment for the Renewstable Barbados Project. The Gender Assessment was developed with the aim to identify the social and gender considerations specific to the project and devise ways of streamlining gender into the project proposal.



2 Acronyms

BGA	Bureau of Gender Affairs
CEDAW	Convention on the Elimination of All Forms of Discrimination Against Women
ESIA	Environmental and Social Impact Assessment
EIA	Environmental Impact Assessment
ESMP	Environmental and Social Management Plan
ESF	Environmental and Social Framework
GAP	Gender Action Plan
GBV	Gender -based violence
GGGI	Global Gender Gap Index
GRM	Grievance/Complaints Mechanism
HyPCe	Hydrogen Power Centre
IGDS	Institute for Gender and Development Studies
ILO	International Labour Organization
NDC	Nationally Determined Contribution
SEP	Stakeholder Engagement Plan
SOP	Standard Operating Procedures
RSB	Renewstable Barbados Project
RoW	Rights-of-way
SEAH	Sexual Exploitation Abuse and Harassment
SGBV	Sexual and Gender-based Violence
UWI	University of West Indies



3 Key Gender Terms and Concepts

Gender analysis or assessment: Identifies, analyses and informs action to address inequalities that arise from the different roles of women and men, or the unequal power relations between them, and the consequences of these inequalities on their lives, their health and well-being. The way power is distributed in most societies means that women have less access to and control over resources, and are less likely to take part in decision-making.

Gender and energy nexus: Women and girls disproportionately bear the burden of energy poverty because of gender norms and traditions, which hinder equal access to modern energy services. Women and men have different energy needs and capacities to access energy services and appliances. This is a consequence of societal gender norms, and differences in responsibilities and opportunities. In this context, gender-blind approaches further institutionalise and perpetuate existing inequalities. This is why we need to create a gender-responsive energy sector that enables women and men to equally benefit from energy services and supports their socio-economic potential.¹

Gender-inclusive: This term refers to initiatives, activities or programmes that are open to both women and men but are designed to overcome any barriers to the full participation that one of the two sexes may experience. In most cases, it targets the increased participation of women and girls. Implementing gender-inclusive programmes is a powerful way to promote gender equality and eradicate gender bias.²

Gender-neutral: Something that is not specific to women or men. This can be policies, investments, projects or business models, whereby it can be assumed that there is no different effect on women or men. Often, something is mistaken to be gender-neutral when it is “gender blind”. This means that the policy, programme or business model did not take into account the ways in which it would affect women and men differently. In the end, gender-neutral most commonly means that the needs of men are assumed to be the needs of everyone, which leads to women’s needs being left out³

Gender-based violence (GBV): An umbrella term for any harmful act that is perpetrated against a person’s will and that is based on socially ascribed (gender) differences between females and males. The nature and extent of specific types of GBV vary across cultures, countries and regions. Examples include sexual violence, including sexual exploitation/abuse and forced prostitution, domestic violence, trafficking, forced/early marriage, harmful traditional practices such as female genital mutilation, honour killings and widow inheritance.

Gender mainstreaming: Mainstreaming a gender perspective means assessing the implications for women and men of any planned action as a way to make women’s as well as men’s concerns and experiences an integral

¹ International Network on Gender and Sustainable Energy, 2023. Available at: <https://www.energia.org/what-we-do/why-gender-and-energy/>

² ILO, Gender-inclusive service provision: A quick guide for financial and business development services providers. 2022,

³ Ibid



dimension of all policies and programs in all political, economic, and societal spheres. In this way, women and men may benefit equally and inequality is not perpetuated.⁴

Gender-responsive programming and policies: Intentionally employing gender considerations to affect the design, implementation and results of programmes and policies. Gender-responsive programmes and policies reflect girls' and women's realities and needs, in components such as site selection, project staff, content, monitoring, etc. Gender-responsiveness means paying attention to the unique needs of females and males, valuing their perspectives, respecting their experiences, understanding developmental differences between girls and boys, women and men and ultimately empowering girls and women.

⁴ UN WOMEN, 2022. Available at: <https://www.unwomen.org/en/digital-library/publications/2022/02/handbook-on-gender-mainstreaming-for-gender-equality-results>



4 Introduction

This gender assessment provides a summary overview of the gender equality situation in Barbados specifically within the context of the renewable energy sector. The report summarizes the principal findings regarding the gender and social impact assessment of the Renewstable Barbados project within the fourteen (14)⁵ neighbouring and surrounding communities located nearest to the project site at Harrow Plantation, in the Parish of Saint Philip, Barbados (Please refer to [Annex Table 5](#)). The aim of the gender assessment is to identify the social and gender considerations specific to the project and devise ways of streamlining gender into the project proposal.

The Renewstable® Barbados Project (referred to herein as “the Project” or “RSB”) is an innovative and unique large-scale green energy project which combines the use of solar, hydrogen and lithium-ion battery technologies to construct and operate a hybrid solar photovoltaic (PV) energy facility with hydrogen storage. RSB will deliver non-intermittent, carbon-free, and renewable electrical power to the national grid and will also accommodate a Blackbelly sheep farming facility within the solar power plant and surrounding green areas.

The Project will be located at Harrow Plantation, in the Parish of Saint Philip, Barbados, which is situated on the southeastern end of the island. Harrow Plantation consists of approximately 123.0 hectares (ha) of privately-owned land, of which approximately 73.2 ha will be leased by RSB for the purpose of carrying out the Project. The RSB leasehold area (referred to herein as “the Project Property”) will encompass the following proposed project components and activities:

- Approximately 59.1 ha will be used for the solar PV power plant, including associated rights-of-way (RoWs) and various restrictions, such as drainage reserves. Most of this land (approximately 57.4 ha) will also support solar grazing (i.e., the practice of grazing livestock, in this case Blackbelly sheep, between and beneath the solar panels of the power plant).
- Approximately 1.6 ha will be used to implement energy storage and management systems in support of the solar PV power plant. These systems will be contained within an area referred to as the “Hydrogen Power Centre” (HyPCe).
- Approximately 2.3 ha will be used for buildings and facilities associated with the on-site Blackbelly sheep farm as well as general office and storage facilities.
- Approximately 10.3 ha will be left as green space (i.e., undeveloped land that is covered with grass) to be used as a fodder pasture for grass harvesting and baling.

The approximate geographic coordinates of the centre of the Project Property are as follows: 13.127548, -59.472712. The land within the Project Property is existing agricultural land that is currently used for sugarcane farming augmented by rotational crops (e.g., cotton). Cultivation of existing sugarcane fields and rotational crops within the Project Property will continue until the start of Project construction. On June 16th, 2023 the project received its planning approval and in maintaining the agricultural integrity of the land has incorporated as a dual usage facility a Blackbelly Sheep Farm as key component of the project development. This facility will include on-

⁵ Padmore Village, Circular Road, Marchfield Village Road, Harrow Land, Neil Kirk Gardens, Farm Road, Harrow Road, Crescent Drive, Farm Gardens, Farm Road Terrace, Harrow Park, Green Way/ Farm Gardens, Bushy Park Road, and Niel Kirk Road.



site agricultural facilities such as: sheep pens, a barn for hay storage, a feed silo, water storage tanks, a designated waste skip area on a concrete pad, and a farm building/staff facility.

In-addition to maintaining the project's agricultural integrity, the project initiated prior to its planning approval, community engagement with the St. Philip communities through the conduct of its Environmental and Social Impact Assessment (ESIA), which was carried out by Stantec and included a gender responsive Social Impact Assessment, as part of the study which was undertaken by Dr. Janice Cumberbatch. The results of which are captured in the successive sections of this report. Additional community engagement was facilitated through the hiring of a local community based private sector firm, Haigh Communications Inc (HCI), and as an output of this, the project is developing a Community and Social Engagement Benefits- Sharing Plan derived partially from the data and inputs collected from their social engagement work. It is important to note that the Project will not generate physical resettlement of local communities in any of its components or stages or will it result in the displacement of any minority groups or indigenous populations. However, reference is made to vulnerable populations and or specific gender-based groups (i.e., women, youth and the elderly) in which the Assessment provides an analysis and proposed recommendations for addressing such gender constraints.

5 Methodology

In conducting the Gender Assessment, a mixed methodology approach was utilised which combined the use of qualitative and quantitative research analysis to help identify the gender and social constraints of the RSB project; to best devise ways in which gender could be effectively integrated into the project's proposal and inform the Gender Action Plan. The main qualitative research method used included that of a desk review of all the relevant policy and legislative frameworks specific to the energy sector in Barbados. A further deep dive of the Social Impact Assessment, as part of the ESIA, was also undertaken to provide additional analysis and insights into the project's gender related impacts.

The quantitative data collected from the Social Impact Assessment and the community stakeholder consultations via the community surveys and interviews coordinated was also used as means of gathering sex-disaggregated data on the project. From the data collected a qualitative gender analysis of these results was computed and from this, there were some notable constraints as well as opportunities observed from which several recommendations were proposed as part of the Gender Assessment.

6 Limitations

One of the core limitations in undertaking this Gender Assessment was the timelines. Given that the timelines for the completion of the Assessment had to be fast tracked to meet the financing requirements. The core methodology used to gather and assessed data was that of a desk review. Stakeholder feedback and input was gathered from the data collected under the Social Impact Baseline Study as part of the project's broader ESIA. Additional stakeholder consultations and feedback would have been useful, but were not feasible within the



timeline given for completion. In mitigating this shortfall further stakeholder consultations and feedback are planned during the project's initial inception and throughout the project's implementation.

7 Purpose/ Rationale

As indicated above, the Gender Assessment was undertaken to identify the social and gender considerations specific to the RSB project in order to devise ways of streamlining gender into the project's proposal. The purpose of the Gender Assessment is therefore, to identify the social and gender specific constraints which impact the project's specific stakeholder groups and propose recommendations and actions which can be used to inform the activities captured within the project's Gender Action Plan (GAP).



8 Barbados Country-Specific Gender Context

Barbados is globally recognized as a leading voice for small island developing states (SIDS) specifically in the championing of climate, climate finance and environmental issues. The country has also been consistently ranked amongst the top countries in governance and social development indicators, and amongst the countries with lowest corruption indexes⁶. In 2021, the country formally became a Republic becoming the fifth Caribbean Community (CARICOM) country⁷ to do so. The country's current population is estimated to be 281,976 people. According to the World Population Prospects the country's annual population growth is around 0.18%. In 2021, there were 3,040 births and the population of males totalled 139,000 while females stood at 148,000⁸. The female population at that time was 51.57% compared to 48.43% males⁹. Barbados' female population exceeded that of men by approximately 9,000 females and at the time (i.e., circa 2021) Barbados ranked 30th out of 201 countries/territories in terms of female to male ratio¹⁰.

Since the start of the 20th century Barbados has transitioned from a largely agrarian sugar producing nation-state to a diversified service sector-based economy- where tourism, international business and finance - are among the country's main economic drivers. However, with this transition the traditional gender stereotypes that cast women and femininity as essentially different from and inferior to men and masculinity still remain intact. According to the Barbados Country Gender Assessment, which was carried out by the Caribbean Development Bank in 2016, these gender binaries continue to affect the economic options and behaviours of women and men across the public/ private and professional/ domestic spheres. In recent years, Barbados has however made considerable progress in these areas, specifically in (i) its efforts to enhance women's economic participation and opportunities within society, (ii) educational achievement, (iii) reproduction, health and survival and (iv) political empowerment.

In the most recent Global Gender Gap Index (GGGI), Barbados scored 0.769, ranking it 31st out of 146 countries.¹¹ The GGGI showed that the country ranked highest, at 4th globally, in economic participation and opportunity demonstrating the enhanced gender parity achieved in this area.¹² In the educational attainment sphere, Barbados also performed reasonably, ranking 65th globally but 3 scores up from its 2022, 67th ranking in this area.¹³ A major contributor being the country's high literacy and secondary school enrolment rates, which both ranked 1sts. Even

⁶ Caribbean Development Bank. *Barbados Country Gender Assessment (CGA)*. January 2016.

⁷ Caribbean Community (CARICOM). Barbados Becomes A Republic. November 30th, 2021. <https://caricom.org/barbados-becomes-fourth-caribbean-republic/>

⁸ UN World Population Prospects (2019). Demographics of Barbados. October 31st, 2021. <https://statisticstimes.com/demographics/country/barbados-demographics.php>

⁹ UN World Population Prospects (2019). Demographics of Barbados. October 31st, 2021. <https://statisticstimes.com/demographics/country/barbados-demographics.php>

¹⁰ UN World Population Prospects (2019). Demographics of Barbados. October 31st, 2021. <https://statisticstimes.com/demographics/country/barbados-demographics.php>

¹¹ World Economic Forum. *Global Gender Gap Report 2023*. Geneva: Switzerland. June 2023.

¹² World Economic Forum. *Global Gender Gap Report 2023*. Geneva: Switzerland. June 2023.

¹³ World Economic Forum. *Global Gender Gap Report 2023*. Geneva: Switzerland. June 2023.



though the country has benefitted from stellar female political leadership in the last five (5) years that has been internationally recognized, in terms of political empowerment Barbados only ranked 58th. Altogether, when conceptualizing Barbados global gender performance against the larger grouping of Latin America and Caribbean states, Barbados' ranks 5th overall with the only other English speaking Caribbean state ranking higher being Jamaica at 3rd. This demonstrates the strides the country has made in advancing gender equality and women's empowerment.

The Bureau of Gender Affairs (BGA), a department within the Ministry of People Empowerment and Elderly Affairs, is the entity responsible for integrating gender into all development plans and policies and to achieve gender equity and equality. However, budgets to support gender mainstreaming efforts are limited and are not available in any ministry except the host ministry. Notwithstanding this, Barbados has put in place numerous environmental and energy-related policies including (among others) the *2012 Barbados National Climate Change Policy* and the *Barbados National Energy Policy 2019-2030*. The BGA has not been systematically involved in environmental bodies such as the National Climate Change Committee. This means there has been a dearth of gender mainstreaming and the consideration of gender specific dimensions within many of the country's national development policies and plans.

9 The Gender Policy Framework in Barbados

Barbados has ratified several key international and regional human rights instruments related to gender equality, namely the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) and the Inter-American Convention on the Prevention, Punishment and Eradication of Violence against Women¹⁴. It has also ratified the eight core International Labour Organisation (ILO) Conventions.¹⁵

There is no National Gender Policy currently in place. The BGA, the national agency responsible for gender mainstreaming, areas of work have been shaped by the *Beijing Platform for Action* and *Commonwealth Plan of Action for Gender Equality 2005–2015* and include poverty alleviation but not broader engagement in economic development. Achievements include the development of legislation and support mechanisms relating to gender-based violence and delivery of gender training.¹⁶ Links to some NGOs and the Institute for Gender and Development Studies at the UWI Cave Hill Campus are strong. However, there are few links with the business community and agencies concerned with economic development.

¹⁴ Caribbean Development Bank. *Barbados Country Gender Assessment (CGA)*. January 2016.

¹⁵ Caribbean Development Bank. *Barbados Country Gender Assessment (CGA)*. January 2016.

¹⁶ Caribbean Development Bank. *Barbados Country Gender Assessment (CGA)*. January 2016.



10 The Renewable Energy & Climate Change Policy Sector Landscape in Barbados

The renewable energy and climate change landscape in Barbados is shaped by several policy and legislative frameworks which guide the development of these sectors. In terms of the execution of the Renewstable Barbados project, the ESIA makes reference to twenty-one (21) related policy documents and legislative frameworks associated specifically with the project's development. Captured below in the tabular matrix is an overview of the relevant policies and legislations.

Table 1. Barbados' Relevant Legislative Frameworks and Policy Documents related to the execution of the Renewstable Barbados project

Legislation or Policy	Description
Legislation Associated with Renewable Energy in Barbados	
<i>Electric Light and Power Act, 2013 and associated amendments and Fees Regulations</i>	"An Act to revise the law relating to the supply and use of electricity, to promote the generation of electricity from sources of renewable energy, to enhance the security and reliability of the supply of electricity and to provide for related matters." · Allows for the generation and supply of electricity by independent power producers. · Establishes the requirement for licences to supply, distribute, store, transmit, and dispatch electricity. · Sets out licence application fees. · Allows for the public utility to make interconnections for electricity supply. · Allows for the Minister to set targets for the supply of electricity from sources of renewable energy. · Sets out licensees' powers, obligation, and restrictions in respect of electricity supply (e.g., in relation to the carrying out of works, the placement of electrical lines, etc.).



Legislation or Policy	Description
<i>Fair Trading Commission Act, Cap. 326B</i>	“An Act to provide for the establishment of a Fair-Trading Commission to safeguard the interests of consumers, to regulate utility services supplied by service providers, to monitor and investigate the conduct of service providers and business enterprises, to promote and maintain effective competition in the economy, and for related matters.” · Establishes a Fair-Trading Commission to enforce the <i>Utilities Regulation Act</i>. · Allows for the Commission to set maximum rates and determine standards of service for providers of utility services, including the supply and distribution of electricity. · Empowers the Commission to investigate suspected breaches of the <i>Utilities Regulation Act</i> and take appropriate enforcement action.
<i>Utilities Regulation Act, Cap. 282</i>	“An Act to provide for the regulation of utility services.” · Sets out requirements and principles for setting utility rates. · Establishes a duty to provide adequate utility service. · Allows for joint use of equipment by more than one service provider.
<i>Electricity Act, Cap. 277</i>	“An Act to provide for the inspection and control of electrical works and for other purposes in connection therewith.” · Establishes requirements for inspection of electrical installations prior to the supply of electricity, and for inspection of alterations and extensions to installations. · Establishes requirements for reporting of accidents involving electrical works and circuits.
Legislation Governing Physical Development in Barbados	
<i>Town and Country Planning Act, Cap. 240</i>	“An Act to make provision for (a) the orderly and progressive development of land; (b) the grant of permission to develop land; (c) powers to regulate land use and development; and



Legislation or Policy	Description
	(d) related matters" - Regulates physical development, defined as “the carrying out of building, engineering, mining or other operations in, on, over or under any land, the making of any material change in use of any buildings or other land or the subdivision of land”. - Establishes requirements for environmental impact assessments. - Superseded by the <i>Planning and Development Act, 2019</i>.
<i>Planning and Development Act, 2019 and associated amendments and regulations</i>	“An Act to make provision for (a) the orderly and progressive development of land; (b) the grant of permission to develop land; (c) powers to regulate land use and development; and (d) related matters" - Regulates physical development, defined as “the carrying out of building, engineering, mining or other operations in, on, over or under any land; the making of any material change in use of any building or land; or the subdivision of land”. - Establishes requirements for environmental impact assessments. - Establishes a new Planning and Development Board and a new Planning and Development Department.
<i>Safety and Health and Work Act, 2005</i> <i>Cap. 356</i>	“An Act to make provision (a) for securing the health, safety and welfare of persons at work; (b) for protecting other persons against risks to health and safety in connection with the activities of persons at work; (c) for controlling certain emissions into the environment; (d) to consolidate the law relating to health, safety and (e) for related matters.” Sets out general duties of occupiers and employers as well as duties and requirements in relation to safety, health, employee welfare, medical examinations.



Legislation or Policy	Description
	Includes specific provisions for building operations and works of engineering construction; storage, handling and use of hazardous substances and articles in the workplace; processes involving special risks to safety and health of employees; use and management of pressure vessels.
<i>Health Services Act, 1969, Cap. 44 and associated amendments and regulations</i>	“An Act relating to the promotion and preservation of the health of the inhabitants of Barbados.” - Includes provisions for the abatement of nuisances and the removal or correction of conditions injurious to public health. - Authorises the Minister to compel works in the interest of public health. - The associated <i>Health Services (Nuisance) Regulations</i> allow for the control of nuisance, including any “matter, thing, deposit, or accumulation of liquid or solid matter” deemed unsanitary, injurious, or dangerous to health, or likely to become so; unpermitted discharge of industrial waste or other noxious matter; conditions that promote the proliferation of public health disease vectors. - The associated <i>Health Services (Building) Regulations</i> regulate the construction, extension, or alteration of buildings, with specific reference to drainage, sanitation, and ventilation.
<i>Emergency Management Act, 2006</i>	“An Act to provide for the effective organization and management of disasters and other emergencies in Barbados.” Authorises disaster hazard inspections in the event that the “condition of any premises is reasonably suspected of posing a danger of serious injury to persons outside of the premises in the event of an emergency or an impact of a hazard.”
<i>Trees (Preservation) Act, 1981, Cap. 397</i>	“An Act to provide for the preservation of trees.” - Restricts the removal of trees over a certain size. - Authorises the Chief Town Planner to require planting, re-planting, maintenance, and protection of trees to preserve and enhance the amenity of “any land abutting upon, adjoining or near a public road”.
<i>Marine Pollution Control Act, 1998, Cap. 392A</i>	“An Act to prevent, reduce and control pollution of the marine environment of Barbados from whatever source. “



Legislation or Policy	Description
	- Prohibits release of any pollutant into the environment if such releases violate standards, conditions or requirements specified under the Act or associated regulations. - Scope includes land-based sources of marine pollution.
<i>Policies Associated with Renewable Energy in Barbados</i>	
<i>National Strategic Plan of Barbados 2006–2025</i>	The <i>National Strategic Plan of Barbados 2006–2025</i> was developed to enhance Barbadian society to become prosperous and globally competitive by 2025. One objective within the Plan was to ensure an efficient and reliable energy sector. The objective included programs to expand the supply of renewable energy. Specific targets included meeting 30% of energy requirements from renewable energy by 2025 and substantially increasing the number of renewable energy businesses by 2025 (Government of Barbados 2007).
<i>National Sustainable Development Policy</i>	The <i>National Sustainable Development Policy</i> attempted to encourage an integrated and holistic approach to sustainable development. In addition, the Policy emphasized that measures to support and promote the adoption of renewable energy, energy efficiency, and energy conservation would help Barbados to meet its obligations as a party to the <i>United Nations Framework Convention on Climate Change</i> (Government of Barbados 2004).
<i>Sustainable Energy Framework for Barbados</i>	The Sustainable Energy Framework is an ongoing technical assistance program with the Inter-American Development Bank, the objective of which is to promote renewable energy and energy efficiency to reduce fossil fuel dependency while enhancing energy security and environmental sustainability. The most recent instalment of the program includes a focus on further development and modernization of the electricity framework in Barbados to support adoption of renewable energy technologies (Government of Barbados 2019b).



Legislation or Policy	Description
<i>Draft National Sustainable Energy Policy</i>	Developed as a complementary policy to the Sustainable Energy Framework, the <i>Draft National Sustainable Energy Policy</i> addressed the high importation of fossil fuels. It sought to increase efficiency and sustainability in the energy supply and demand. In addition, the Policy aimed to encourage economically viable utility-scale renewable energy, promote energy cost reduction technologies, reduce fossil fuel dependency, and decrease the impacts of global warming (Government of Barbados 2019c).
<i>Barbados National Energy Policy 2017–2037</i>	The <i>Barbados National Energy Policy 2017–2037</i> aimed to provide a direction for Barbados to transition from a fossil fuel-based economy to mainly renewable energy. The Policy sought to ensure affordable energy security, a sustainable energy sector, and increased renewable energy and energy efficiency (Government of Barbados 2019d).
<i>Barbados National Energy Policy 2019–2030 (BNEP)</i>	The BNEP outlines the transition to a 100% renewable energy and carbon-neutral island by 2030. The policy attempts to ensure the provision of reliable, sustainable climate-friendly energy with zero domestic fossil fuel consumption and expansion of research and development in renewable energy (Government of Barbados 2019a).
<i>Renewable Energy Rider (RER) Program</i>	The former RER Program allowed domestic customers to sell energy back to the electrical grid at 1.6 times the level of the Fuel Clause Adjustment. In 2016, the RER credit rate began to be calculated using a resource cost approach, rather than being tied to the Fuel Clause Adjustment (Howard 2019). In 2019, the RER was replaced by a feed-in tariff, and RER agreements were grandfathered for a period of 20 years (FTC 2019).
<i>Policies Associated with Physical Development in Barbados</i>	



Legislation or Policy	Description
<i>National Groundwater Protection Zoning Policy, 1963 (revised in 1973</i> <i>and</i> <i>2010)</i> <i>and</i> <i>Green Paper on the 2020 Water Protection and Land Use Zoning Policy</i>	The <i>National Groundwater Protection Zoning Policy</i> delineates the island into five zones. Public water supply wells are located in the most restricted Zone 1 areas (Stantec 2022). The Green Paper (i.e., a tentative government consultation document of policy proposals for debate and discussion, which was produced by the Ministry of Energy and Water Resources, the Government of Barbados, and the Barbados Water Authority [BWA]) proposes that the groundwater protection zones introduced in the <i>National Groundwater Protection Zoning Policy</i> be re-configured for the following purposes (Stantec 2022): · To boost groundwater water quality protection due to the presence of more recalcitrant persistent pollutants. · To release more lands in the current Zone 1 protection areas for development. More specifically, the Green Paper states the following (Stantec 2022): · The existing <i>National Groundwater Protection Zoning Policy</i> is over 50 years old and does not protect against persistent chemical pollutants. · The existing Zones 1 to 5 are proposed to be replaced with Zones A to E, with Zone A termed a strict exclusion zone smaller than the existing Zone 1. · Only agriculture Class Two will be permitted within Zone A; this type of agriculture includes “horticulture, fruit growing, seed growing, use of lands for farmers markets, bursary grounds, woodland or forestry”. · Solar PV farms and wind farms – deemed to have low environmental impact – may also be allowed in Zone A areas under specific permitting conditions. · As described in Appendix B, the Project is currently located within a Zone 3 groundwater protection zone as per the existing Policy. If the Policy is revised as proposed in the Green Paper, the Project will be located within a Zone D groundwater protection zone.
<i>Physical Development Plan (PDP) and associated 2003 amendment and draft 2017 amendment</i>	The purpose of the PDP is to promote sustainable growth and development within Barbados by setting out policies and regulations to guide proponents of proposed developments, including



Legislation or Policy	Description
	requirements related to the conduct of an Environmental Impact Assessment (EIA) or ESIA.
<i>Note:</i> ¹ The information regarding policies associated with renewable energy in Barbados was adapted from Evelyn (2020). Other sources are cited throughout the table where applicable.	

Source: The Environmental and Social Impact Assessment for the Renewstable Barbados Hybrid Solar Power Plant with Hydrogen Storage.

Equally important to the identification of the relevant legislative and policy frameworks within the scope of the project's development, is an understanding of how gender intersects or has been integrated into several of these core policies and legislative frameworks. This type of analysis is critical to contextualising whether gender and women's rights have been effectively considered and prioritised within the scope of the renewable energy and climate change policy frameworks. Such an understanding is critical in appreciating the underlying gender inequalities, discrimination and exclusion that may impact women and other vulnerable groups in their access to energy resources and services.

10.1 Gender integration in Barbados National Energy Policy 2019-2030

The Barbados National Energy Policy was approved in November 2019 and aims to achieve 100% renewable energy and carbon neutral transformational goals by 2030, positioning Barbados as the first island-state in the world to do so. Specific goals include the provision of reliable, safe, affordable, sustainable, modern and climate friendly energy services to all, zero domestic consumption of fossil fuels economy wide and to export all hydrocarbons produced both on land and offshore.

Even though the policy includes a specific goal to maximise local participation in distributed renewable energy (RE) generation and storage (democratization of energy), no reference is made to the interlinkages of women/gender and energy and the potential to use energy access to unleash women's economic empowerment, for example by creating opportunities for women in energy value chains.

Furthermore, the policy includes 10 visionary goals for Barbados that reflect the general direction that stakeholders wished the sector to go and what they wanted to see the sector achieve. Even though not specifically mentioning women, Goal 10 calls for the creation of opportunities for all Barbadians including the most vulnerable to the impact of climate change to participate in and benefit from the transformation to 100% Renewable Energy.



10.2 Gender integration in The International Framework and Barbados National Climate Change Policy

From a global perspective, the links between gender equality and climate change mitigation efforts are clearly articulated. The Sustainable Development Goals framework (SDGs) and the connections among goal 5 on gender equality, goal 7 on clean affordable energy, and goal 12 on climate action speak directly to this. However, gender considerations are often not clearly reflected in national policy frameworks. An analysis of 192 national energy frameworks from 137 countries showed that only a third of the frameworks include gender considerations to some extent¹⁷ This is also the case for the Barbados National Climate Change Policy of 2012. Even though the policy aims at establishing a national process for adaptation to climate change, reduce the country's greenhouse gases emissions and design appropriate mechanisms to respond to climate change, no considerations for gender integration in such processes has been made.

10.3 Gender integration in Barbados' updated NDC

Barbados' updated Nationally Determined Contribution (NDC) was communicated to the UNFCCC on 30 July 2021. The latter is a non-binding national plan highlighting climate actions, including climate-related targets, policies and measures, that governments aim to implement in response to climate change and as a contribution to achieve the global targets set out in the Paris Agreement.

Barbados' updated NDC recognizes that climate change disproportionately impacts women, youth and lower income communities. Furthermore, the NDC highlights that three quarters of vulnerable families are female headed households, which is why the Government of Barbados, in addition to social assistance, places strong emphasis on facilitating and supporting women's entrepreneurship and business development as one way of responding to rising levels of unemployment, retrenchment and to assist with the economy's recovery. Incentives for micro and small businesses such as grants, loans and technical assistance for women are explicitly mentioned in the updated NDC.

With regards to the interlinkages of gender and energy, the updated NDC highlights the importance of addressing structural inequities faced by women, young people and those socially and economically disadvantaged in the renewable energy sector. The NDC also makes reference to the Roofs to Reefs Programme (R2RP), that aims to improve the standard of living and quality of life of the most vulnerable segments of the population of Barbados. Under R2RP, protocols and standards for rooftop solar PV installation will be developed and include recommended options for energy storage technology. The NDC further highlights that the government will assess the social and gender impacts, and changes in vulnerability related to the project.

¹⁷ Prebbles and Rojas, 2017.

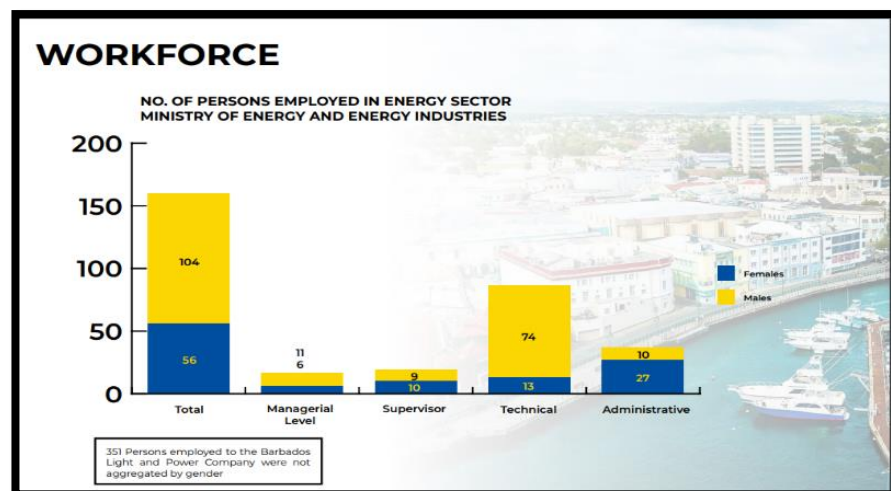


11 Gender Gaps in the Energy & Renewable Energy Sector in Barbados

11.1 Women's workforce participation in the energy sector

The renewable energy sector much like the energy sector is a male dominated field. The International Renewable Energy Agency (IRENA) corroborated this point in its 2019 report titled, *Renewable Energy: A Gender Perspective*, by stating that, "the energy sector both in terms of its conventional fossil fuel and renewable energy sectors remain predominately male dominated fields with women been significantly underrepresented."¹⁸ A look into Barbados' energy sector workforce participation profile, for the data sets collected by the Caribbean Centre for Renewable Energy and Energy Efficiency, clearly highlights that similar to the international context, the Barbados energy sector is male dominated. Kindly see table below:

Table 2. Barbados' Workforce Profile of the Energy Sector



Source: CCREEE. Barbados Energy Report Card 2019.

With statistics showing a total of 104 males in comparison to a total of 56 females being employed within the energy sector, excluding those persons employed with the Barbados Light and Power Company (BL&PC) based on the data collected. Men's participation in the energy sector dominates that of women in every category with the exception of the administrative levels. The impact of which is the lower participation of women in the sector generally and in technical areas more specifically. For example, in the engineering and other STEM related fields. The data captured highlights women's marginalization in the energy sector to specific job roles which in turn limits

¹⁸ International Renewable Energy Agency. *Renewable Energy: A Gender Perspective*. 2019. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2019/Jan/IRENA_Gender_perspective_2019.pdf



their participation and access to senior level and managerial positions within the sector. The knock-on effect is that women are paid less as a result of being in lower paying positions. In addressing this constraint educational and internship programs which place a focus on women and provide access into the sector via non-traditional job roles are critical in helping to make an effective change. The RSB project may consider as one of its recommendations and seek to incorporate under its Gender Action Plan, an internship and/or apprenticeship program which provides opportunities for women specifically, and men also, to gain entry into the sector via technical job roles.

11.2 Women's participation in the energy sector educational programs

Moreover, women's lack of participation in energy related training and educational programmes also served to exacerbate the gender gaps seen in the Barbados' energy sector. In reviewing the profile of educational offerings available via the country's technical and vocational institutions, which often acts as the first point of entry for access into the energy sector in Barbados. The data collected highlights men continued and higher levels of representation across all tertiary and vocational level educational programmes. A look into the profile of courses offered and the participation of women vs men illustrates once more that men significantly dominate women in the area of energy sector training and educational development. Kindly see table below:

Table 3. Tertiary Level Educational Training Programmes on Offer in Barbados in Sustainable Energy

NUMBER AND TYPE OF TERTIARY LEVEL & VOCATIONAL TRAINING SUSTAINABLE ENERGY PROGRAMMES OFFERED							
Name of Education Programme Provider	No. of Persons Enrolled		No of Persons Graduated	Types of Programme			Programme Link
				Vocational Certificate	Apprenticeship	M.Sc.	
Barbados Community College	Male	94		Photovoltaic Installation - Electrical Technician Level 3			http://www.bcc.edu.bb/Divisions/Technology/CertificateinPhotovoltaicDesignPr.aspx
	Female	3					
Samuel Jackman Prescod Institute of Technology	Male		91	Photovoltaic Installation 1			http://sjpi.edu.bb/?page_id=3795
	Female						
	Male		272	Photovoltaic Electrical Installation			http://sjpi.edu.bb/?page_id=3796
	Female						
Barbados Vocational Training Board	Male				Solar Water Heater Technician		https://bvtb.gov.bb/apprenticeship-programme/solar_water_heater_technician/
	Female						http://www.sbcs.edu.tt/programme/windpower/
The University of the West Indies - Cave Hill Campus	Male	23	27			Renewable Energy Management	https://www.cavehill.uwi.edu/gradstudies/programmes/science-and-technology/folder/renewable-energy-management.aspx
	Female	12					

Source: CCREEE. Barbados Energy Report Card 2019.

This data helps to highlight why women's participation in the sector is significantly lower than that of men. Added to this, the underlying assumptions and perceptions around women's participation in certain career fields, further help to drive these misconceptions and gender biases. The International Renewable Energy Agency (IRENA), notes



that these assumptions are often driven by gender biases and mis-perceptions, around “occupations in engineering and technology being seen by women as less socially useful and acceptable when compared to other disciplines such as the medical and biological sciences.”¹⁹ These assumptions help to inform the selection of certain subjects and programmes by women which ultimately determine their final career path. Greater levels of education and sensitization is therefore required is helping to address these misconceptions. Partnerships with the Ministry of Education and the educational institutions offering such programmes working alongside the Ministry of Energy is critical in bringing about such change. RSB can leverage the execution of the Renewstable Barbados project to stimulate discussions and conversations around these issues and where possible establish Memorandum of Understanding (MOU) with these entities to effect change.

11.3 Women’s participation in decision making positions and roles in the energy sector

Consequently, women’s participation or lack thereof within the decision-making levels of the energy sector serve to further erode a focus on women and women’s issues within the context of the energy industry and the renewable energy sector more specifically. Even though women, currently over represent men in a ratio of 2:1, in-terms of the current positions held by the top three (3) critical stakeholders within the renewable energy sector. This includes namely: the Minister with responsibility for Energy, the Chief Executive Officer of the Fair-Trading Commission and the Managing Director of the Barbados Light and Power Company. A further analysis into the composition of the Boards governing these entities, as well as the privately owned and managed renewable energy firms and businesses in operation currently within Barbados, highlights a predominance of male ownership, management and financing.

Women’s limited participation in positions of power translates into an absence and/or very limited focus on women specific issues and challenges within the energy and renewable energy sector. Where women’s issues are not identified and cannot be appropriately articulated they are unable to be addressed as reflected in the current Barbados National Energy Policy 2019-2030. Within the policy context this has significant negative implications, as women ultimately become marginalize and this marginalization reinforces the proliferation of gender barriers and inequalities seen across the energy sector. In addressing these concerns, the RSB project has undertaken this gender assessment to better devise ways into how gender can be appropriately integrated within the scope of this project, which will hopefully serve as a platform for the upscaling of long-term gender mainstreaming actions in the energy sector and the renewable energy sector more specifically by Government.

¹⁹ International Renewable Energy Agency. *Renewable Energy: A Gender Perspective*. 2019. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2019/Jan/IRENA_Gender_perspective_2019.pdf



12 Stakeholder Mapping & Analysis of the Renewstable Barbados Project

In undertaking the stakeholder mapping of the RSB project, a Social Impact Assessment Baseline Study was completed as part of the project's ESIA. The baseline assessment was one of the tools used to identify the project stakeholders; and specifically, the stakeholders immediately impacted by the project in terms of their proximity to the project site. The Baseline Assessment utilised a 1 km radius in the development of a sample size as a baseline, to support the collection of data from the neighbouring community residents as one of the main stakeholder groups.

Captured below is a photo of the project site and a 1km radius surrounding the Harrow Plantation. This radius was used as a sample size in the stakeholder mapping and analysis of the neighbouring community residents as part of the stakeholder analysis.



Figure 1. Study area

In-addition to the residents in nearby communities, other key stakeholder groups were also identified. Using the following support tools developed under the ESIA which included namely:

- Social Impact Assessment Baseline Study
- Environmental and Social Management Plan and
- Stakeholder Engagement Plan



These outputs were all used to inform the stakeholder mapping and analysis captured within this section of the report. Outlined within the project's Environmental and Social Management Plan, is a Stakeholder Engagement Plan (SEP) which identifies and defines the project's stakeholders.

The RSB project defines stakeholders as; persons, groups, or institutions with an interest in the Project or the ability to influence the project outcomes, either positively or negatively. Stakeholders are diverse and may be affected by the Project either directly or indirectly. Captured below is the preliminary stakeholder list which shows each stakeholder's relevance to the project and the extent to which gender is a concerning issue.

Table 4. Preliminary Stakeholder List

Stakeholders	Relationship to the Project	Gender Analysis
RSB	Overall Project Management	Both females and males in this group benefit from the Project. The current information does not suggest a gender bias.
Engineering Consultants	Quality control for the construction	Both females and males in this group benefit from this Project. The current information does not suggest a gender bias.
Contractors and construction companies	Responsible for the Project's construction	The predominance of male workers in the construction industry means that more males than females are likely to benefit from employment opportunities in this Project.
Residents in nearby communities	Could potentially be adversely impacted by the noise, dust and vibration from the construction	Both females and males will be affected by the construction. Of specific concern are the retired residents who may be at home during the construction activities.
Town and Country Development Planning Office and Prime Minister's Office Planning Unit	Regulatory agencies responsible for ensuring the development is consistent with the Physical Development Plan and other national Plans and Laws.	Not applicable



Stakeholders	Relationship to the Project	Gender Analysis
Division of Energy	Regulatory agencies responsible for monitoring and regulating the energy sector, renewable energy and energy conservation.	Not applicable
Environmental Protection Department	Regulatory agencies responsible for ensuring compliance with national or international standards for air quality, noise, etc.	Not applicable
Labour Department	Regulatory agency responsible for ensuring decent work standards on the island.	Not applicable
Unions	Regulatory for protecting workers' rights on the island.	Not applicable
Media	A likely option for residents who are affected to share their complaints will publicly report the resident's complaints. Can provide a channel for sharing information about the Project with the public.	Not applicable

Source: Environmental and Social Impact Assessment for the Renewstable Barbados Hybrid Solar Power Plan with Hydrogen Storage

In-addition to the residents in neighbouring communities, the Stakeholder Engagement Plan (SEP) identified nine (9) other stakeholder groups and in total ten (10) which related directly to the RSB project; and mapped the relationship of each stakeholder group specific to their role in the project. In providing a cursory analysis, four (4) of the stakeholder groups were noted as having gender-related impacts. That is, gender was identified as having a potential positive or negative impact on the project.

Firstly, the RSB project team was identified as one of the project's stakeholder groups. While the cursory gender analysis provided highlights that there is no suggested gender bias and both women and men are likely to benefit as well as be positively impacted by the project, the result of which may be due to the current and positive gender balance in the team composition of the Renewstable Barbados project. Given, that both the Project Manager for Development and the Environmental, Social and Governance & Communications Lead are both females. It is important to note, that the absence of women in large-scale energy infrastructure related projects, specifically in



senior level positions during the project's planning and design phases can contribute to the development of energy projects that do not consider the specific needs and circumstances of women, girls and other diverse groups²⁰. Research shows that this in turn can lead to negative impacts and missed opportunities for individuals and communities whose lives the infrastructure seeks to improve.²¹

Furthermore, engineering consultants were also identified as another critical stakeholder group. While the initial gender analysis, captured within the Stakeholder Engagement Plan notes that there is no suggested gender bias and women and men are likely to both benefit positively, the systemic gendered differences in the educational trajectory of women, highlights the persistent gender gaps in women's education, specific to their participation in science, technology, engineering and mathematics (STEM) related subjects.²² Although STEM fields are widely regarded as critical to national economies only 35% of all students in STEM related fields of study are women.²³ This in turn impacts the ability of women to access jobs in STEM related fields as they lack the necessary training and skills required to perform these job roles. Consideration of these gender barriers and biases should be considered in the implementation of the RSB project and opportunities should be created to increase women's participation in the engineering and engineering-related fields. This can be achieved either through the creation of training opportunities for women in STEM subjects, or via the development of internship or apprenticeship programmes which specifically allocate a percentage of the participation to women.

Contractors and construction companies were also recognized as a key stakeholder group. The analysis provided demonstrates that the predominance of male workers in construction will result in greater employment opportunities for more men than women. Addressing this issue, will require a balance of innovation and practicality in providing the project with the relevant and available skills sets on island to support the successful completion of the project's outcomes while addressing the gender constraints. Inclusion of companies with progressive recruiting, training and human resource policies²⁴ which supports women's inclusion may be a useful tool here. RSB, in the project's execution, should seek to adopt such gender responsive policies within the contracting and procurement frameworks utilized in the recruitment and selection of the contractors responsible for undertaking the project's construction and civil works. However, it should be noted that limitations in the size of local market as well as availability of suitable and qualified female workers may impact the effective recruitment and mobilisation of gender inclusive companies.

The research also shows that an additional factor that may discourage women's employment in the energy sector is a gender-insensitive work environment.²⁵ Construction sites are often cited as high-risk environments for gender-based violence affecting workers and service users and discourage women's involvement, as these sites are often spaces for sexual abuse and exploitation as well as sexual and gender-based violence. These specific gender constraints hamper women's participation and employment in the construction sector. In addressing these constraints, the opportunity exists within the scope of the implementation of the RSB project to integrate training

²⁰ UN Women. *Guidance Note Gender Analysis in Technical Areas: Energy Infrastructure*. New York: December 2022, 6.

²¹ UN Women. *Guidance Note Gender Analysis in Technical Areas: Energy Infrastructure*. New York: December 2022, 6.

²² UNESCO. Institute of Statistics, Science Report, 2023

²³ Ibid

²⁴ UN Women. *Guidance Note Gender Analysis in Technical Areas: Energy Infrastructure*. New York: December 2022, 7.

²⁵ UN Women. *Guidance Note Gender Analysis in Technical Areas: Energy Infrastructure*. New York: December 2022, 7.



in the Gender Action Plan, in the areas of Sexual Exploitation Abuse and Harassment (SEAH) and Sexual and Gender-based Violence (SGBV) as core a mandatory training which must be undertaken by all contractors and construction workers on the project.

As indicated previously, the neighbouring community residents were also noted among the project's core stakeholder groups. From the sample of the 155 households that participated in the Social Impact Baseline Study and the corresponding surveys, the results show that, 70.3% were female headed households, while 10.3% and 19.4% were male and jointly managed households respectively (Please refer to [Annex Table 6](#)).²⁶ The predominance of female head-households within the baseline assessment is reflective of the national trend in Barbados where female-headed households are predominant. This further demonstrates that women are most likely to be negatively impacted by the project outcomes; and given their added economic as well as care responsibilities, as heads of households these impacts are likely to have a multiplier effect. Research has shown that large-scale energy infrastructure projects are mostly likely to negatively impact women. Community and/or benefit sharing programmes, if designed to consider gender, can help to offset some of the likely negative project impacts and advance gender equality in the community.²⁷ The RSB project team will therefore design a Community and Social Engagement Benefits Sharing Plan, which will outline the specific community activities as well as the proposed benefit-sharing among the neighbouring residents and the other stakeholder groups identified. Inclusive within such a plan will be opportunities for knowledge building, training and skills development, infrastructure development via road repairs and the creation of new employment opportunities.

13 Social & Gender Baseline Assessment of the Renewstable Barbados Project

The main objective of the Renewstable Barbados project is to enhance sustainable energy generation in Barbados, by providing a clean, renewable baseload solution that addresses the intermittency issues of solar and wind to enhance energy resilience in the country. The energy generated by the project represents substitution in the demand of approximately 18,680 residential customers, providing them all with clean, resilient and stable electricity. In doing so, the project is anticipated to have specific social as well as gender related impacts. Outlined below are the specific social and gender related impacts across the varying project phases.

13.1 The Social & Gender Impact Assessment of RSB- Project Construction Phase

i) Social and gender impacts related to employment

The proposed project can facilitate employment as well as economic opportunities and benefits for local businesses. It is estimated that there could be a maximum of 150 construction jobs over two years. There could

²⁶ Stantec. *Environmental and Social Impact Assessment for the Renewstable Barbados Hybrid Solar Power Plant with Hydrogen Storage*. May 2022.

²⁷ UN Women. *Guidance Note Gender Analysis in Technical Areas: Energy Infrastructure*. New York: December 2022.



also be opportunities for capacity-building and skills development for local workers, particularly youth, since training will be offered. In addition, collaboration and working relationships with foreign companies involved in the Project's construction could increase the technological capacity of local businesses. These are all benefits that should accrue during the construction phase and require no mitigation measures.

However, the influx of many temporary workers during the construction phase could affect social networks and dynamics in nearby communities and could result in negative interactions with residents. Moreover, care must be taken to establish fair employment and decent work conditions to prevent perceptions of unequal employment benefits and conditions between local workers and equivalent foreign personnel. These potential adverse effects can be readily mitigated and made compliant with local labour laws and regulations. To enhance the employment benefits of the Project, the hiring process will be transparent and allow eligible locals to apply for and have a fair chance of acquiring work at the Project. In addition, consideration will be given to employment opportunities being available to some of the unemployed residents of neighbouring areas. Further mitigation, including gender sensitization training for workers, is detailed later.

The Project could support local businesses that provide supplies for the construction and renewable energy sectors. The construction phase is typically a boost to local entrepreneurs operating small retail businesses near the Project, especially food and beverage, because the construction workers will provide additional customers. This can have a positive impact on the local food and beverage businesses, many of which are led by women. In addition, accommodation and rental car companies could also gain business from foreign personnel working on the Project. The construction phase could therefore boost the economy and have a positive effect on local livelihoods.

The provision of employment is a particularly positive impact given that the Baseline Assessment noted that 30.1% of the other persons living within the sampled households were reported as unemployed (Please refer to [Annex Figure 2](#)). Moreover, the economic opportunities would be a considerable benefit in the current economic context, where the prolonged Covid-19 pandemic has resulted in a decline in tourism arrivals, the closure of many businesses and the loss of several jobs. The benefits require no mitigation measures. To enhance the employment benefits of the Project, the hiring process will be transparent and designed so that eligible locals can apply for and have a fair chance of acquiring work at the Project. In addition, consideration will be given to employment opportunities being available to some of the unemployed residents of neighbouring areas including women.

The project document indicates that approximately twenty (20) full-time-equivalent local jobs, direct and indirect, may be created to support project operation, maintenance, and security requirements and that the hiring process will allow eligible locals to apply for and have a fair chance of acquiring work at the project including some of the unemployed residents of neighbouring areas. However, it is not specified whether women and men will benefit from these opportunities. Recent research suggests that some individuals may gain new jobs in green sectors, but others may not be well-positioned to avail themselves of these opportunities. In the renewable energy sector, for example, women may not automatically be part of new high-tech value chains unless deliberate policy incentives and capacity support measures targeting women are provided.²⁸ It is therefore suggested to conduct a mapping

²⁸ UNDP Just Transition Brief, 2022



of entry points for potential employment to foster the inclusion of local women in the project's operations and where possible support the development of internship or apprenticeship programs as avenues for entry into the sector.

ii) Social and gender impacts related to noise, dust and vibration

The noise, dust, and vibration during the construction of the Project may have adverse effects on the physical and psychological well-being of people in surrounding communities, especially individuals with a sensitivity to dust and noise. In this regard, 27.7% of the residents in the sample households were retired and would likely be at home during the construction activities. In addition, the peaceful and quiet nature of the communities is the primary feature valued by 64.5% of the respondents, most of which are women (Please refer to [Annex Table7](#)) for more details). The potential impacts from the construction vehicles and equipment are an added concern. There is potential for further damage to the roads, already rated as “poor” by 51% of the respondents. In addition, the absence of adequate sidewalks or bicycle paths place these users, many of which are women at increased risk. Especially if, construction drivers do not drive with due care and within the required speed limit. Road restoration and the implementation of traffic management measures and road restoration projects will be essential in helping to address these potential adverse effects and protect health and safety.

iii) Gender related impacts

During the construction of the project, there can be various environmental and social impacts that may affect women and men differently. It is important to recognize that gender roles, norms, and power dynamics can influence how individuals experience and respond to these impacts. Construction projects can bring significant changes to the social fabric of communities. Within the baseline study conducted, 70.3% of the 155 households are headed by women. Women play essential roles in the community, like caretaking and maintaining social cohesion. Noise, dust, and vibration can disrupt these activities and affect the overall well-being of women and children. Additionally, gender norms might restrict women's mobility or participation in decision-making processes related to the construction, limiting their ability to voice concerns and preferences as well as the possible restrictions on mobility in terms of travel/transport to and from their homes.

In addition, the psychosocial effects of construction noise, dust, and vibration can differ between genders. Women may experience stress and anxiety due to concerns about their family's well-being, health, and safety. For men, the pressure to provide for their families and endure harsh working conditions may result in mental health challenges.

With regards to health and safety women and men may have different levels of exposure to noise, dust, and vibration depending on their roles and activities during construction. For example, if men are more likely to be involved in activities near the construction site, they may experience higher exposure levels. This could lead to different health outcomes, with men potentially facing greater risks to their health due to prolonged exposure.



Table 5. Potential benefits and adverse effects that could occur during the construction phase.

Potential Impacts	Phase	Nature	Gendered Impact	Geographic Extent	Duration	Significant	Likelihood
Construction-related jobs	Construction	Positive	Yes	Site specific	Short-term	No mitigation required	Likely
Capacity-building and skills development for local workers	Construction	Positive	Yes	Site specific	Long-term	No mitigation required	Likely
Increased technological capacity of local businesses	Construction	Positive	No	Site specific	Long-term	No mitigation required	Likely
Support for local suppliers in the construction and renewable energy sectors	Construction	Positive	Yes	Local	Short-term	No mitigation required	Likely
Increased retail business, primarily food-related from construction workers	Construction	Positive	Yes	Local	Short-term	No mitigation required	Likely
Business for accommodation and car rental companies	Construction	Positive	No	Island wide	Short-term	No mitigation required	Likely
Noise and vibrations from construction vehicles and equipment	Construction	Negative	Yes	Local	Short-term	Can be mitigated	Likely



Potential Impacts	Phase	Nature	Gendered Impact	Geographic Extent	Duration	Significant	Likelihood
Impaired air quality from dust and other emissions from construction vehicles and equipment	Construction	Negative	Yes	Local	Short-term	Can be mitigated	Likely
Reduced visual amenity due to construction activity	Construction	Negative	Yes	Local	Short-term	Can be mitigated	Likely
Damage to roads from construction vehicles and equipment	Construction	Negative	Yes	Local	Short-term	Can be mitigated	Likely
Health and safety risks to on-site workers	Construction	Negative	Yes	Site	Short-term	Can be mitigated	Unlikely
Health and safety risks to adjacent communities and general public	Construction	Negative	Yes	Local	Short-term	Can be mitigated	Unlikely
Influx of temporary workers affect social dynamics	Construction	Negative	Yes	Local	Short-term	Can be mitigated	Unlikely
Perceptions of inequitable labour conditions	Construction	Negative	Yes	Site	Short-term	Can be mitigated	Unlikely
Delays to construction if any archaeological and cultural heritage features are found within the PDA	Construction	Negative	No	Site	Short-term	Can be mitigated	Uncertain
Accidental spill or leak	Construction and operational phase	Negative	Yes	Local	Long-term or irreversible	Can be mitigated but could be significant if it occurs	Unlikely

Source: Environmental and Social Impact Assessment for the Renewstable Barbados Hybrid Solar Power Plan with Hydrogen Storage



13.2 The Social & Gender Impact Assessment of RSB – Project Operational Phase

i) Social and gender impacts related to employment

The energy facility will provide an estimated twenty (20) permanent jobs when fully operational, and the sheep facility is expected to employ ten (10) people. Employment opportunities in these areas are expected to be created for persons within the community. These benefits require no mitigation. However, the hiring process will be transparent and allow eligible persons to apply for and have a fair chance of acquiring work at the Project. In addition, consideration will be given to employment opportunities being available to some of the unemployed residents of neighbouring areas and in particular women who are often left behind in the execution of large energy sector projects and in agriculture sector.

As indicated previously, the sheep facility is expected to employ up to ten (10) persons directly, thereby potentially providing employment opportunities to the community. The sheep facility can create employment opportunities in various ways for women as well as men, by promoting gender equality and empowerment within the workforce. The planned sheep facility can offer a range of positions that cater to different skill sets for women. Jobs could include shepherders, farm managers, veterinary assistants, administrative roles, marketing, and sales positions. In addition, training and skill development programs to empower women with the necessary knowledge and expertise to excel in various roles within the sheep facility could be provided. This can include training in animal husbandry, farm management, veterinary care, sustainable agriculture practices, and business management. In addition, mentorship programs or leadership initiatives aimed at supporting and encouraging women to pursue leadership roles within the sheep facility could be facilitated.

Table 6. Potential benefits and adverse effects that could occur during the operational phase.

Potential Impacts	Phase	Nature	Gendered Impact	Geographic Extent	Duration	Significant	Likelihood
The generation of clean, renewable energy	Operation	Positive	No	Island-wide	Long-term	No mitigation required	Likely
Employment at the energy facility	Operation	Positive	Yes	Local/parish	Long-term	No mitigation required	Likely



Potential Impacts	Phase	Nature	Gendered Impact	Geographic Extent	Duration	Significant	Likelihood
Employment at the sheep farm	Operation	Positive	Yes	Local/parish	Long-term	No mitigation required	Likely
Improvement in national agricultural productive capacity	Operation	Positive	No	Island-wide	Long-term	No mitigation required	Likely
Reduced visual amenity due to the presence of Project components and perceived negative aesthetic impacts	Operation	Negative	No	Local	Long-term	Can be mitigated	Uncertain
Potential health and safety impacts on workers	Operation	Negative	Yes	Site	Long-term	Can be mitigated	Unlikely
Potential health and safety impacts on adjacent communities	Operation	Negative	Yes	Local	Long-term	Can be mitigated	Unlikely
Decreased water pressure due to Project-related water usage	Operation	Negative	Yes	Local	Intermittent	Can be mitigated	Uncertain
Odours from the sheep farm	Operation	Negative	Yes	Local	Intermittent	Can be mitigated	Likely



Potential Impacts	Phase	Nature	Gendered Impact	Geographic Extent	Duration	Significant	Likelihood
A change in property values	Operation	Unknown	Yes	Local	Long-term	Cannot be determined based on available information	Uncertain
Accidents (non-routine) events from hazardous products	Operation	Negative	No	Site	Long-term	Could be significant if they occur	Unlikely

Source: Environmental and Social Impact Assessment for the Renewable Barbados Hybrid Solar Power Plan with Hydrogen Storage

14 Challenges / Constraints

In completing the Gender Assessment for the RSB project a number of gender constraints and challenges were identified, as corroborated by the Social Impact Assessment Baseline Study and the Stakeholder Engagement Plan. These specific gender related constraints / challenges are captured below in bullet format for ease of reference:

- **Women's lack of inclusion from the project inception (design phase) in the execution of large-scale renewable energy infrastructure projects** – Although the design of the RSB project benefitted from both the early inclusion and participation of women in senior technical-level positions within the project team and in the conduct of the Social Baseline Assessment as part of the ESIA, men's over representation in the construction sectors can mean that during implementation the gender and social biases identified could be largely ignored and not appropriately considered in the project's implementation.
- **Men's over representation/ dominance in the energy and construction sectors** – The energy and construction industries, and similarly the renewable energy sectors, are male dominated fields. On energy infrastructure projects where most of the employment opportunities are generated from the construction sector, men are most likely to be the main beneficiaries as women are less likely to have the training in STEM related fields which provide the skill sets and training needed to perform these job roles. Where women do possess the training and skill sets, they are still significantly under-represented in comparison to men and are most likely to be in junior or administrative positions and/or roles.
- **Project construction sites are gender-insensitive workspaces** – Construction sites, due to entrenched gender norms and stereotypes, are traditionally gender-insensitive workspaces. The prevalence of sexual harassment, gender-based violence and gender discrimination are known to widely exist. In curbing these issues, it is important that the appropriate procedures and protocols are put in place and the appropriate



training provided to increase awareness around these issues and ensure that preventive measures are enacted.

- **Women's lack of participation/ involvement in decision making positions** – A number of factors hinders women's participation in energy sector related infrastructure projects and their ability to also participate in decision making positions within such projects. It is therefore critical that actions which include the adoption of inclusive gender responsive recruitment policies and practices, the scheduling of meetings at times convenient for women and in spaces women feel comfortable as well as the participation of women's base groups for institutions be effectively considered in the project's implementation.
- **Women are more adversely impacted by the project's impacts and are least likely beneficiaries of community and social engagement benefit programmes** – The Social Impact Baseline Assessment Study, showed that women are most likely to be impacted by the RSB project. Research also shows that women are least likely to benefit from community or social benefit programmes as their needs and concerns are often not prioritised or even considered. The first step in addressing this issue is in the collection of sex-disaggregated data which will be used to inform the design and development of community and social benefit programmes for community residents.

15 Recommendations

In addressing the gender constraints/ challenges referenced above the following recommendations are being proposed. These recommendations will be integrated where feasible within the Gender Action Plan to ensure that appropriate steps are taken to reduce the gender vulnerabilities and biases uncovered while enhancing women's participation and empowerment at all levels of the project.

- **Develop a Gender Action Plan** – To support the implementation of gender specific actions which address the gender constraints and challenges identified under the Gender Assessment. It is important that a project specific Gender Action Plan be developed which identifies specific activities, objectives, indicators and outcomes. Principal among these, are actions which seek to support the participation of women at all stages of the project implementation and support their inclusion in high-level decision-making positions such as gender responsive recruitment and hiring practices and the updating of the project's Standard Operating Procedures (SOP) to reflect gender considerations.
- **Design and Implementation in collaboration with relevant training institutions of the RSB Young Professional Green Hydrogen Energy Sector Internship Programme encouraging female participation in the sector via technical roles** – Recognizing that the energy sector is a male dominated field the RSB project team should seek to increase the level of participation of women in the technical related areas of the field through the creation of a paid internship programme. The internship programme would target new entrants as well as recent graduates of any Diploma, Graduate or Postgraduate energy and/or gender related programmes on offer by tertiary level institutions in Barbados. The programme would reserve a percentage of its enrolments to women seeking to pursue a career in this field.



- **Develop Mandatory SEAH and SGBV Training Courses and Update Project SOPs to reflect Gender Sensitivity in Workspaces** – Construction sites are widely known as gender insensitive workspaces with a high occurrence and incidence of sexual harassment, exploitation, gender-based violence and gender discrimination. In addressing these concerns, the RSB project team should seek to enlist the assistance of the University of West Indies Cave Hill Campus, namely Institute for Gender and Development Studies (IGDS) Unit in developing specific training courses and modules in SEAH and SGBV. These courses should become a mandatory requirement for all RSB project staff and construction personnel hired under the Project. Technical support could also be provided in the updating of the Project's specific SOPs to include gender sensitivity with the project's operational policies and procedures.
- **Development and implementation of a community-based Grievance/Complaints Mechanism (GRM)** - Women may face unique challenges and grievances related to the project's activities (given that women are more likely to be adversely impacted than men). Having a well-defined and accessible GRM allows women to raise their concerns and complaints in a safe and supportive environment. It ensures that their grievances are heard, taken seriously, and addressed appropriately, providing them with a pathway to seek redress. An inclusive and community-based GRM can therefore provide a confidential and secure platform for women to voice their concerns without fear of reprisals, ensuring their safety and protection.
- **Organize experience sharing workshops with key government agencies and partners-** By engaging government and non-government stakeholders and sharing the importance of integrating gender considerations into renewable energy projects opportunities may be created to influence policy development and decision-making processes. When decision-makers recognize the value of the RSB gender-responsive renewable energy project, they are more likely to advocate for and support gender-inclusive policies and projects in the energy sector. In addition, the workshops can showcase how women in the RSB project can actively participate in the renewable energy sector, both as beneficiaries and as decision-makers. The potential exists for the RSB project to serve as a model for gender integration into future large scale renewable energy projects.
- **Conduct gender sensitive stakeholder consultations to inform and solicit input from community members into the design, implementation and monitoring phases of the Project** -Gender-sensitive consultations provide an opportunity to include women's voices and perspectives, especially those of women-headed households. This inclusion can empower women by giving them opportunities to share their concerns and needs and where possible have these gender specific considerations be effectively incorporated and/or address in the project's implementation. It also creates opportunities for women to share information on how decisions related to the RSB project that directly affect their lives, livelihoods, and access to resources. At least one inclusive community-based stakeholder consultation should be held once every quarter this total at minimum four (4) stakeholder consultations per year.



- **Education and awareness raising programs and development a community-based handbook-** The handbook and related capacity building sessions for small farmers and community members will provide valuable information about the renewable energy project, hydrogen energy production, and its applications in the energy and agricultural sectors (specifically as it refers to the application of solar and rain water harvesting within the agriculture sector to support sustainable farming practices). The handbook will act as a guide in promoting sustainable practices and reducing greenhouse gas emissions while highlighting the benefits of adopting a gender approach within large scale renewable energy projects. Women involved in agriculture can benefit from learning about these innovative techniques and contribute to more eco-friendly farming practices.
- **Establish an Inclusive and Participatory Project Steering Committee that includes the Institute for Gender and Development Studies (IGDS) –** Women’s voices, needs and concerns are often left out of decision making. This is because of their lack of inclusion and involvement in projects of this nature and at a decision-making level. To this end, the RSB project team should establish a Project Steering Committee (PSC) and ensure in its composition adequate space is made for the inclusion of women. Moreover, in seeking to ensure that the voices of all women are heard, specific to the project across all levels, the PSC should include representation from the UWI-IGDS Unit or other qualified gender or social inclusion consultant. This would help to ensure that decision making is not only inclusive, but gender informed.
- **Develop a Community and Social Engagement Benefits Sharing Plan which takes on board gender considerations in the development of a community-based programme to benefit residents within St. Philip –** Women are often among those most negatively impacted by energy infrastructure projects, yet they are among those least to benefit. The RSB project team should develop a Community and Social Engagement Benefits Sharing Programme which considers gender to benefit the community residents within St. Philip with a specific focus on women and youth. The Plan should outline several community-based projects and activities which are meant to directly benefit the neighbouring and surrounding residents within the St. Philip community.



16 Conclusion

Although, Barbados has and continues to put in place policy measures which aim to bridge the gender gap between females and males, the energy sector, and the renewable energy sector more specifically, is still very much a male dominated field. Men are still among those most represented in the technical related areas of the sector, they (men) hold the highest percentage of the decision-making and senior-level positions and by extension are among those highest paid in comparison to their female counterparts.

In this regard, the Renewstable Barbados Project represents an opportunity for gender considerations to be appropriately integrated within the scope of a large-scale renewable energy infrastructure project in Barbados. In this regard, the RSB project would also become a practical demonstrable model for the integration of gender into future large-scale community-based renewable energy projects on small-island states. The Gender Action Plan developed by the Project details the specific objectives, outcomes and activities that will be followed to improve women's participation and enhance their social and economic empowerment within the project.

An important component of the project would be the social and community-based activities and engagement to be outlined within the Community and Social Engagement Benefits Plan. The Plan would highlight all of the community-based activities and projects to be undertaken by RSB and the proposed benefits of these activities to the St. Philip community. Gender will be appropriately considered within the scope of the design and implementation of these actions to ensure both women and men are able to benefit, as well as the youth.



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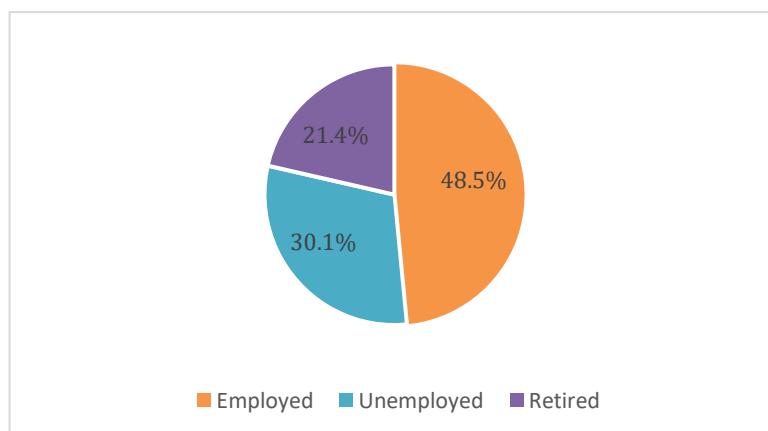
18 Annex

Table 7. Names of Roads and Communities Included in the Survey

Name of Road/Community	Number of respondents		Percentage		Total %
	Female	Male	Female	Male	
Padmore Village	19	17	12.3%	11.0%	23.2%
Circular Road	14	17	9.0%	11.0%	20.0%
Marchfield Village Road	12	12	7.7%	7.7%	15.5%
Harrow Land	10	5	6.5%	3.2%	9.7%
Neil Kirk Gardens	6	5	3.9%	3.2%	7.1%
Farm Road	6	3	3.9%	1.9%	5.8%
Harrow Road	1	6	0.6%	3.9%	4.5%
Crescent Drive	5	2	3.2%	1.3%	4.5%
Farm Gardens	4	1	2.6%	0.6%	3.2%
Farm Road Terrace	2	1	1.3%	0.6%	1.9%
Harrow Park	2	1	1.3%	0.6%	1.9%
Green Way / Farm Gardens	1	1	0.6%	0.6%	1.3%
Bushy Park Road	0	1	0.0%	0.6%	0.6%
Neil Kirk Road	0	1	0.0%	0.6%	0.6%
Total	82	73	52.9%	47.1%	100%

Table 8: Heads of Households

Head of Household	Number of Respondents		Percentage		Total %
	Female	Male	Female	Male	
Female-headed Household	56	53	36.1%	34.2%	70.3%
Joint responsibilities	14	16	9.0%	10.3%	19.4%
Male-headed Household	12	4	7.7%	2.6%	10.3%
Total	82	73	52.9%	47.1%	100%

**Figure 2 Employment Status of Other Households****Table 7: Community Attributes Valued by Respondents**

Attributes	Frequency of Response		Percentage		Total %
	Female	Male	Female	Male	
It is peaceful and quiet	60	40	38.7%	25.8%	64.5%
The people who live in the area are friendly	48	36	31.0%	23.2%	54.2%
Proximity to commercial, social and community services (e.g., businesses, healthcare, schools, church, recreational spaces, social clubs)	34	29	21.9%	18.7%	40.6%
It is safe and secure	33	28	21.3%	18.1%	39.4%
It is an attractive area	16	20	10.3%	12.9%	23.2%
It is breezy	1	1	0.6%	0.6%	1.3%
Nothing	0	1	0.0%	0.6%	0.6%

Renewstable Barbados Inc
Green Climate Fund- Gender Action Plan
July 2023

This gender action plan has been developed to support the implementation of gender specific actions which address the gender constraints and challenges identified under the Gender Assessment. This Gender Action Plan identifies specific activities, objectives, indicators and outcomes.

Impact Statement: To enhance the resilience of energy supply to female and male headed households, small farmers, business owners/ operators within the Farm Road, Padmore Village and Six Roads communities by improving sustainable energy generation in Barbados.				
Outcome: The renewable energy sector through the use of hydrogen energy improves its capacity to respond to gender issues by implementing at least five (5) gender responsive interventions by the end of the RSB project (Outputs 1, 2, 3, 4 and 7).				
Output 1: Gender specific gaps and requirements are identified and incorporated into the Renewstable Barbados Project				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost (\$USD)
Output 1.1 -Ensure findings of the Gender Assessment are incorporated into the Project's Results Management Framework	The RSB Project Results Management Framework includes specific gender activities, targets and indicators as outlined in the Project Management Plan (Yes/No)	Complete by year 1	RSB Project Management Team 1. ESG & Communications Lead	This cost is subsumed within the project's implementation ¹

¹ This activity will be led by the RSB Project Manager with support from the ESG and Communication Lead.

	Target 1.1.1- Integrate the recommendations of the Gender Assessment Study within the RSB Results Management Framework			
Output 1.2- Ensure that the project's Environmental & Social Management Plan (ESMP) promotes gender sensitive actions through the project's life cycle.	Develop/ Update the project's ESMP to address gender requirements and include specific gender actions and host sensitization sessions and workshops with various stakeholder groups on these gender issues Target 1.2.1- Develop/ Update the ESMP to address gender requirements and include gender specific actions Target 1.2.2- Conduct sensitization and awareness sessions with various stakeholder groups on and across the project on the gender actions specific to the ESMP (<1 sensitization session/workshop)	Complete by year 2	RSB Project Management Team 1. ESG & Communications Lead 2. Project Director	The cost associated with the development/ update of the ESMP is subsumed within the project's implementation \$5,000.00 (this cost covers the hosting of the related awareness sessions and workshops including rental of workshop facilities, equipment, etc)

Output 1.3-Ensure updates to the project's Standard Operating Procedures (SOPs) and (For example; EPC Contracts, Labour Management Plans, etc) address the requirements of gender as well as sexual exploitation, abuse and sexual harassment (SEAH), sexual and gender base violence (SGBV).	Update the project's SOP documents to reflect specific gender requirements and considerations. Target 1.3.1- No. of project specific documents and SOPs developed and revised to include gender considerations	Complete by year 1	RSB Project Management Team 1. Project Director 2. ESG & Communications Lead	This cost is subsumed within the project's implementation²
Output 1.5- Conduct training with the RSB Project staff on the updated SOPs that include gender considerations.	Facilitate training sessions with the RSB Project team and staff (inclusive of the RSB and EPC team and employees) on the gender inclusive updated SOPs and other related documents. Target 1.41- No. of training workshops/ information sessions hosted (<2) Target 1.4.2- No. of RSB staff who have participated in the training disaggregated by sex. (Yes/ No)	End of Project	RSB Project Management Team 1. Project Director 2. ESG & Communications Lead	This cost will be subsumed in Output 1.2 as it is envisioned that this activity will be done simultaneously with Output 1.2

² This activity will be led by the Project Director with support from the ESG and Communication Lead.

Output 1.6 -Organize experience sharing workshops with key government agencies and partners. By engaging government and non-government stakeholders and sharing the importance of integrating gender considerations into renewable energy projects opportunities may be created to influence policy development and decision-making processes.	Host information sharing workshop(s)/session (s) with Gov't stakeholders and partners on the importance of integrating gender into scalable renewable energy projects for Gov't consideration and use in future renewable energy projects. Target-1.5.1 -No. of information sessions/workshops hosted (<2) Target 1.5.2- % of stakeholders who have participated in the information sessions/workshops disaggregated by sex (Yes/ No)	End of Project	RSB Project Management Team & Government Ministries (Ministry of Energy and the Ministry of Agriculture)	\$5,000.00.00 (hosting of workshops and awareness sessions with Governmental agencies on gender integration in RE projects)
Output 2: Enhance capacity of the RSB project staff (inclusive of all project related contractors and employees) to integrate gender considerations/ requirements in renewable energy projects				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost (\$USD)
Output 2.1- Establish a MOU between HDF Caribbean and the UWI Cave Hill Campus and/or other tertiary level educational institutions (for example; BCC and SJPI) to support the training, skills	Develop a MOU between HDF Caribbean and tertiary level institutions in Barbados to support training, skills development and knowledge transfer in green energy hydrogen production and use.	Complete by year 1	RSB Project Management Team and key education institutions 1- ESG & Communications Lead	This cost is subsumed within the project's implementation ³

³ The development of the MOU will be led by the RSB Project's ESG and Communications Lead

development and capacity building of all RSB Project staff in gender sensitivity, SEAH and SGBV training	Target 2.1.1- One (1) signed MOU between HDF Caribbean and tertiary level institutions in Barbados		2- UWI Cave Hill Campus (IGDS & Sustainable Energy Department) 3- Barbados Community College (BCC) 4- Samuel Jackman Prescod Institute of Technology (SJPI)	
Output 2.2- Develop a project specific SEAH and SGBV Training Programme to reflect Gender Sensitivity in Workspaces for all RSB Project employees (inclusive of contractors, construction workers and other related employees)	Develop in partnership with the UWI IGDS Unit, gender sensitive training modules specific to SEAH and SGBV for the energy related construction sector. Target 2.2.1- No. of training modules/courses developed to support training of project staff (<2) training courses one in SEAH and one in SGBV	Complete by year 2	1. UWI Cave Hill Campus-IGDS Unit 2. ESG & Communications Lead 3. Project Director	\$40,000.00 (course development, creation of instructional videos training, training materials and facilities rental)
Output 2.3- Develop online self-paced mandatory SEAH and SGBV training to reflect Gender Sensitivity in Workspaces for all RSB Project employees (inclusive of contractors and employees)	Target 2.3.1- No. of RSB project staff that have completed the mandatory online SEAH and SGBV training (% of females; % of males captured and documented)	Complete by year 2	1. ESG & Communications Lead 2. Project Director 3. UWI Cave Hill Campus-IGDS Unit	

Output 3: Increase participation of women performing technical roles in the renewable energy sector				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost (\$USD)
Output 3.1 -Develop in collaboration with relevant training institutions National / CARICOM Vocational Qualification (N/CVQ) programmes in hydrogen energy production, storage and use	Target 3.1.1 - No. of N/CVQs developed specific to hydrogen energy production, storage and use (<1)	End of Project	RSB Project Management Team and key Education Institution and Training Centres 1- ESG & Communications Lead 2- Technical Vocational Education & Training (TVET) Council 3- BCC 4- SJPI	\$18,000.00 (NVQ/CVQ development inclusive of external and internal verification cost and certification)
Output 3.2 -Design and implement in collaboration with relevant training institutions the RSB-Young Professional Green Hydrogen Energy Sector Internship Programme encouraging female participation in the sector via technical roles	Target 3.2.1 - No. of women who applied or registered for the RSB- Young Professional Energy Sector Internship Programme (<10)	End of Project	RSB Project Management Team and key Education Institutions 1- ESG & Communications Lead 2- BCC 3- SJPI	This cost is subsumed under the project implementation ⁴
Output 3.3 - Design and implement in collaboration with relevant training institutions the RSB-Young	Target 3.3.1 - No. of female interns participating in the RSB- Young Professional	End of Project	RSB Project Management Team and key Education Institutions	\$16,200.00 (stipend co-financing between RSB (inclusive of HDF and Rubis))

⁴ This activity will be led by the ESG and Communications Lead.

Professional Green Hydrogen Energy Sector Internship Programme encouraging female participation in the sector via technical roles	Energy Sector Internship Programme (6) ⁵		1- ESG & Communications Lead 2- BCC 3- SJPI 4- UWI-IGDS 5- UWI-Sustainable Energy	
Output 3.4- Collaborate with the Ministries of Education and Energy to host public education campaigns/ workshops which encourage the participation of girls and young women to aspire to leadership and technical roles in the renewable energy sector.	Target 3.4.1- No. of public engagement campaigns/ workshops hosted which encourage female participation in the RE sector (1)	Through-out the project	RSB Project Management Team and key Education Institutions 1- ESG & Communications Lead 2- Ministry of Energy 3- Ministry of Education 4- BCC 5- SJPI	\$5,000.00 (hosting of workshops and awareness sessions inclusive of rental of facilities, catering and workshops materials and facilitators)
Output 4: Conduct gender sensitive stakeholder consultations and awareness -raising programmes to ensure public knowledge and buy-in on the use of the RSB hydrogen energy solution and potential impacts.				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost (\$USD)
Output 4.1- Conduct gender sensitive stakeholder consultations to inform and solicit input from community members (i.e., heads of households, small farmers,	Target 4.1.1- No. of gender sensitive stakeholder consultations and awareness raising programmes hosted during the project implementation	At least quarterly until project ends	RSB Project Management Team 1- ESG & Communications Lead 2- Project Manager	\$12,000.00 (this cost covers the hosting of Town Hall Meeting, Community workshops and sensitization workshops)

⁵ 3 interns per year for 3 months each.

business owners and vulnerable populations) into the design, implementation and monitoring phases of the Project	(<4) community-based sensitization sessions, workshops or Town Hall Meetings hosted per year			
Output 4.2- Update the RSB Project webpage to host communication outputs including (press releases, new articles, stories and communication events) related to the RSB Project. In-addition to Communications/ Grievance Complaints Section which provides project's responses to community concerns as well as allow community members to formally issue any grievances and complaints that they may related to the project via a formal online GRM mechanism.	Target 4.2.1- No. of communication events published on the RSB webpage which provides opportunity for the sharing of information and updating of community members on the status of the project. (<3 communication stories, press release or related/ relevant updates captured per year and is captured on the RSB website) Target 4.2.2- No. of stories and other communication outputs developed which highlight the gendered impacts of the RSB project on the RE sector. (<3 communication stories, press release or related/ relevant updates captured per year which reflects gender specific considerations and is captured on the RSB website)	Throughout the project	RSB Project Management Team 1- ESG & Communications Lead 2- Project Director	This cost is subsumed within the existing Project Management cost ⁶

⁶ This activity is expected to be led by the ESG and Communications Lead.

	Target 4.2.3- Creation of an Online Complaints/ Grievance Mechanism as part of the RSB webpage which allows community members to issue complaints that may be disaggregated by sex with scope of real-time responses and feedback. ((1) Grievance/ Complaints Mechanism which is virtually accessible and provides an opportunity or community residents to raise concerns and issues.))			
Output 4.3- Develop a Community and Social Engagement Benefits Sharing Plan which takes on board gender considerations in the development of a community-based programme to benefit residents within St. Philip	Target 4.3.1-Number of women and men benefitting from the programme (Reports developed/ data collected captures the gender specific benefit or % male and % female beneficiaries from the community activities and projects)	Throughout the Project	RSB Project Management Team 1- ESG & Communications Lead 2- Project Director	This cost is subsumed within the existing Project Management cost ⁷

⁷ This activity is expected to be led by the ESG and Communications Lead.

with a specific focus on women and youth.	Target 4.3.2 -Number of community-based projects identified to benefit women and youth			
Output 5: Increase capacity of the RSB Project Management Team to respond to complaints from community members and other stakeholders.				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost (\$USD)
Output 5.1- Develop and implement) a community-based Grievance/ Complaints Mechanism (GRM) that is easily accessible by all stakeholder groups. The GRM will outline the project's grievance mechanisms across the various stakeholder groups, the procedures associated with issuance of such grievances and timelines for response.	Target 5.1- Develop as part of the ESMP a Grievance/ Complaints Mechanism plan for the different stakeholder groups. The plan will outline the procedures, actions and timelines for responses for the various responses being posed. (1) (Yes/ No) Target 5.2- Develop as part of the ESMP a community based GRM mechanism and Action Plan which facilitates the issuance of complaints by community members for appropriate responses and feedback to be issued. (1) (Yes/ No)	Complete by year 1	RSB Project Management Team 1- ESG & Communications Lead 2- Project Director	This cost is subsumed within the existing contract of the ESG & Communications Lead

Output 6: Female and male heads of households, small business owners and other vulnerable populations have an increased awareness and understanding of the use and application of the RSB project in enhancing energy resilience.				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost (\$USD)
Output 6.1- Develop a community based handbooks/tool kits which are user friendly and increase the awareness and understanding of key stakeholders (female and male headed households, business owners and other vulnerable populations) on the benefits and impacts of hydrogen energy production.	Develop a community-based hand-book(s)/tool kit(s) on the RSB project and its hydrogen energy production and use in the energy and agricultural sector. Target 6.1.1.- (<1) handbooks/tool kits develop for specific community groups.	Complete by year 2	RSB Project Management Team 1- ESG & Communications Lead 2- Project Director	\$8,800.00 (this cost covers the development of the Handbook along with some printing of the documents for dissemination)
Output 6.2- Conduct education awareness sessions with key stakeholders (female and male headed households, business owners and other vulnerable populations) on the benefits and impacts of hydrogen energy production using the handbooks and tool-kits developed.	Conduct education awareness workshops on the RSB projects and its hydrogen use and impact in the energy and agricultural sector. Target 6.2.1- (<2) community-based education workshops/ information using the handbook/ toolkit developed.	End of Project	RSB Project Management Team 1- ESG & Communications Lead 2- Project Manager	This cost is subsumed in the public engagement and sensitization lines.
Output 7: Female and male heads of households, small farmers, business owners and other vulnerable populations have an increased understanding of how the demineralized water produced by the hydrogen energy plant or rain water harvesting can be used for irrigation and other ancillary uses				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost

Output 7.1- Undertake sensitization and awareness sessions targeting community members, small farmers and business owners and operators and other vulnerable populations	No. of sensitization/ awareness workshops hosted Target 7.1.1- (<1) community level sensitization/ workshop sessions held targeting Female and male community members, small farmers, business owners and other vulnerable populations) (Where possible will include 50% females; 50% males) (Yes/ No)	End of Project	RSB Project Management Team 1- Project Director 2- ESG & Communications Lead	\$10,000.00 (this cost covers hosting of workshops and awareness sessions inclusive of rental of facilities, catering and workshops materials and facilitators)
Output 7.2- Provide training in the rainwater harvesting techniques targeting small farmers, community members, business owners and operators and other vulnerable populations	Target 7.2.1- (<1) training sessions conducted on rainwater harvesting and usage. Targeting small farmers, female and male community members, business owners and operators and other vulnerable populations (Yes/ No) Target 6.2.2- Provide training in rainwater harvesting to at least (<1) school in the Community. (Yes/ No)	End of Project	RSB Project Management Team 1- Project Director 2- ESG & Communications Lead	

Output 8: Small farmers and community members have an increased understanding of how hydrogen energy production can support sustainable agriculture including sustainable farming techniques				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost
Output 8- Undertake education and awareness raising programs with small farmers and industry stakeholders in the agricultural sector on the synergies between renewable energy and the sustainable agricultural sector	No. of workshops and education sessions hosted on the synergies between hydrogen energy production and sustainable agricultural and farming techniques Target 8.1.1 (<1) training sessions conducted on renewable energy technologies and sustainable agriculture and farming techniques. Targeting small farmers and agricultural stakeholders. Where possible will include 50% females; 50% males) (Yes/ No)	End of Project	RSB Project Management Team 1- Project Director 2- ESG & Communications Lead	This cost is subsumed in the community sensitization, training and engagement activities related to Output 7.1 and 7.2
Output 9: Enabling a gender responsive project management and governance environment for the duration of the Project				
Activities	Indicators and Targets	Timelines	Responsibilities	Cost
Output 9.1- Establish an Inclusive and Participatory Steering Committee (PSC) that includes the Institute for Gender and Development Studies (IDGS) at the UWI Cave Hill Campus or Bureau of Gender Affairs (BGA) or a	Establish a PSC inclusive of a UWI IDGS or BGA representative Target 9.1.1- PSC established inclusive of the UWI IDGS or BGA or a Gender/ Social Inclusion Consultant	Complete by year 1	RSB Project Management Team 1- Project Director	No stand-alone cost identified (Staff time); However, the inclusion of the UWI IDGS or the Bureau of Gender Affairs (BGA) may require a participatory stipend. If a Gender/ Social Inclusion

Gender/ Social Inclusion Consultant to increase participation of women in decision-making				Consultant is preferred a cost may be incurred for time served.
Output 9.2 -Conduct gender sensitive training with the RSB Project Management Team to ensure gender dimensions are effectively considered throughout the duration of the project	Conduct at least one (1) gender sensitive training among Project Management Team and Staff per year. Where new staff are introduced, they will be required to undertake the training. Target 9.2.1- (<1) training per year for project staff.	End of project	RSB Project Management Team 1- ESG & Communications Lead 2- Project Director 3- UWI IGDS	This cost will be shared with Output 1.2 and Output 2.2 to ensure synergy and coordination between the project activities and keep cost down.
Output 9.3- Present annually to the PSC to report on the status and implementation of the Gender Action Plan (GAP)	Target 9.3.1- (<1) report developed and discussion held per year on the RSB GAP progress and implementation. Total (3) for project duration 2023-2025 (Yes/ No)	Annually	RSB Project Management Team 1- ESG & Communications Lead 2- Project Director	This cost is consumed within the existing contract of the ESG & Communications Lead